
CERN

Loss Maps SPS
Crab Cavity Phase Scan — Simulations

Outline

1. Overview
2. Horizontal crabs
   - Betatron loss maps
   - Off-momentum loss maps
3. Vertical crabs
   - Betatron loss maps
   - Off-momentum loss maps

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Overview

- All loss maps in this deck: **26 GeV**, TCP/TCSM at **5**σ.
- Crab cavities: total voltage **1 MV**, frequency **400.787294 MHz**. The kicking plane (Horizontal / Vertical) is the top-level split below.
- Working point $Q_x = 20.13$, $Q_y = 20.18$; chromaticity $\xi_x = 0.405$, $\xi_y = 0.35$; normalised emittances $\epsilon_x = \epsilon_y = 2$ μm.
- Two loss-map types: **Betatron** (single-plane blow-up, excited Horizontally or Vertically) and **Off-momentum** (RF frequency sweep, DPpos or DPneg).
- Crab cavity phase scanned from -179° to 179° in steps of 30° (13 values), with and without machine errors – unlike the MD, no “slow” loss map was simulated.

Overview — scan dimensions

Dimension	Values
Crab cavity kicking plane	Horizontal, Vertical
Loss map type	Betatron, Off-momentum
Betatron excitation plane	Horizontal, Vertical
Off-momentum sweep sign	DPpos, DPneg
Errors	Without errors, With errors
Crab cavity phase	-179° to 179° in steps of 30° (13 values)
Total loss maps	208

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Horizontal crabs — Betatron loss maps

- Horizontal
- Vertical

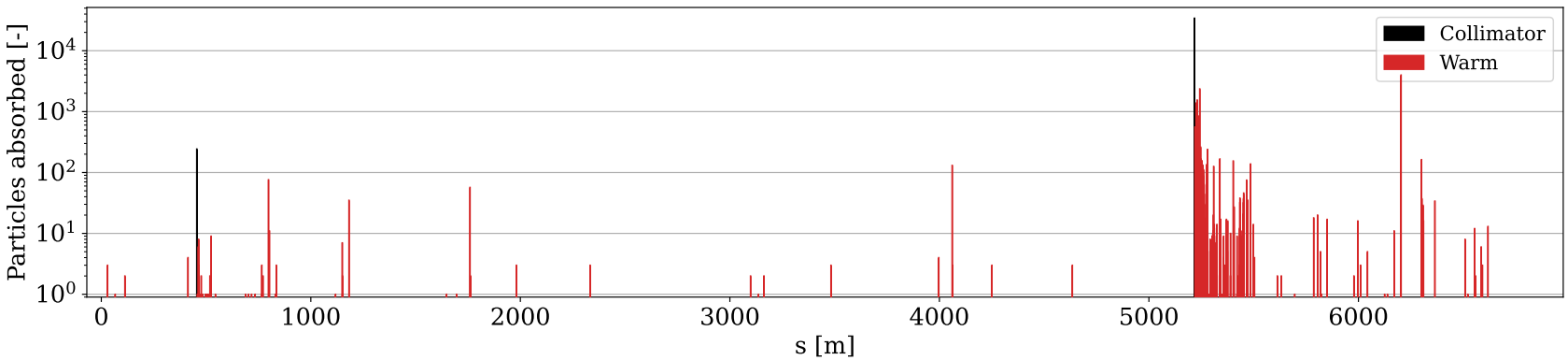
Horizontal crabs — Betatron loss maps — Horizontal

- Without errors
- With errors

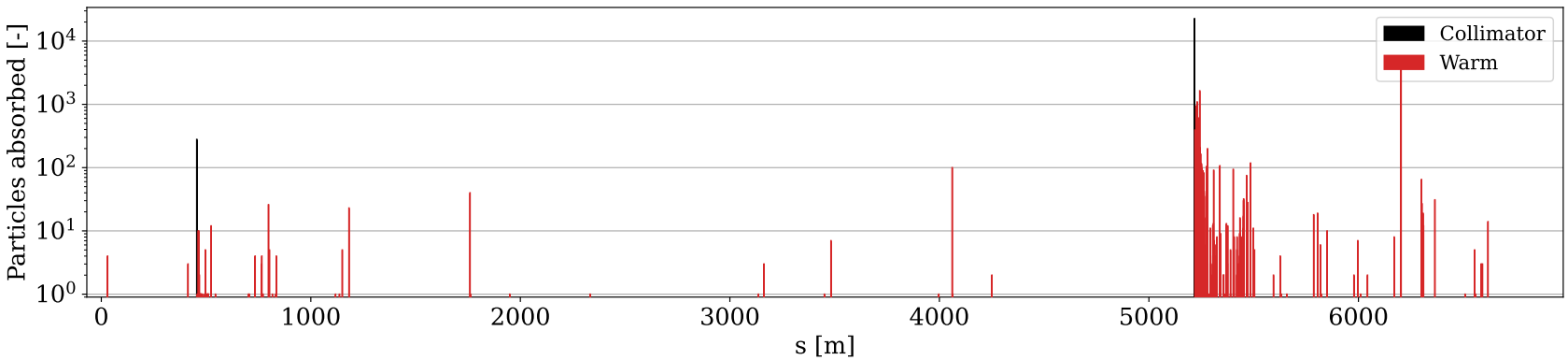
Horizontal crabs — Betatron loss maps — Horizontal — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

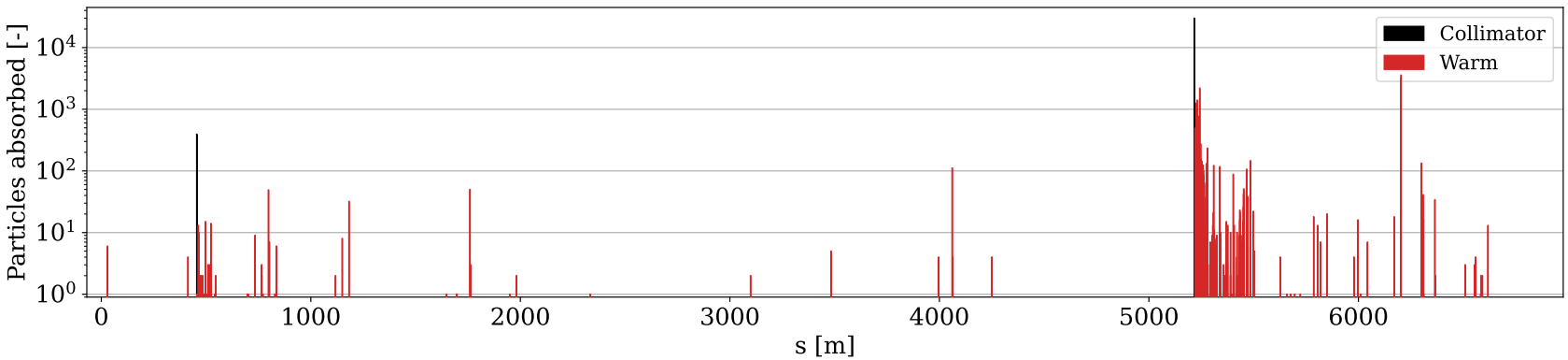
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase $-179°$



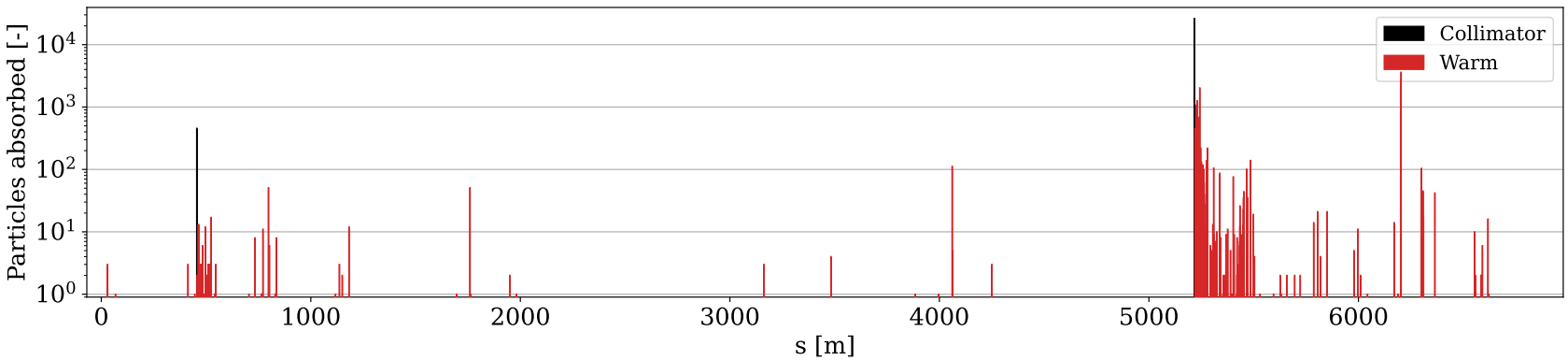
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase $-150°$



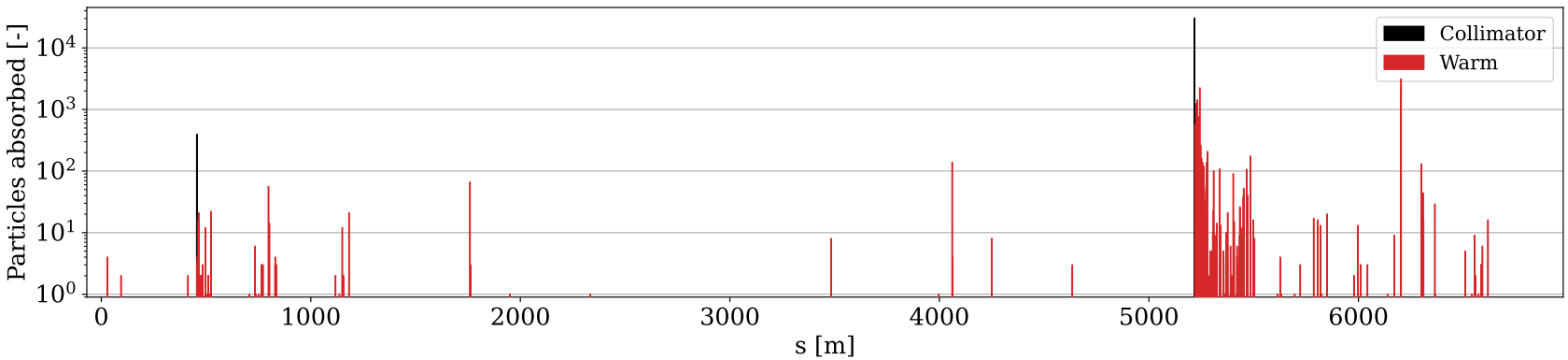
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase $-120°$



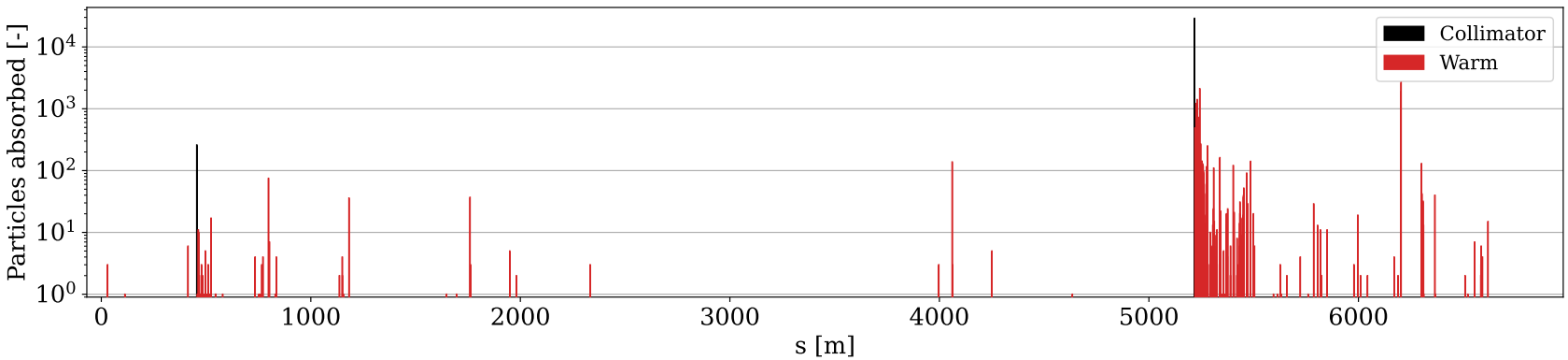
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase −90°



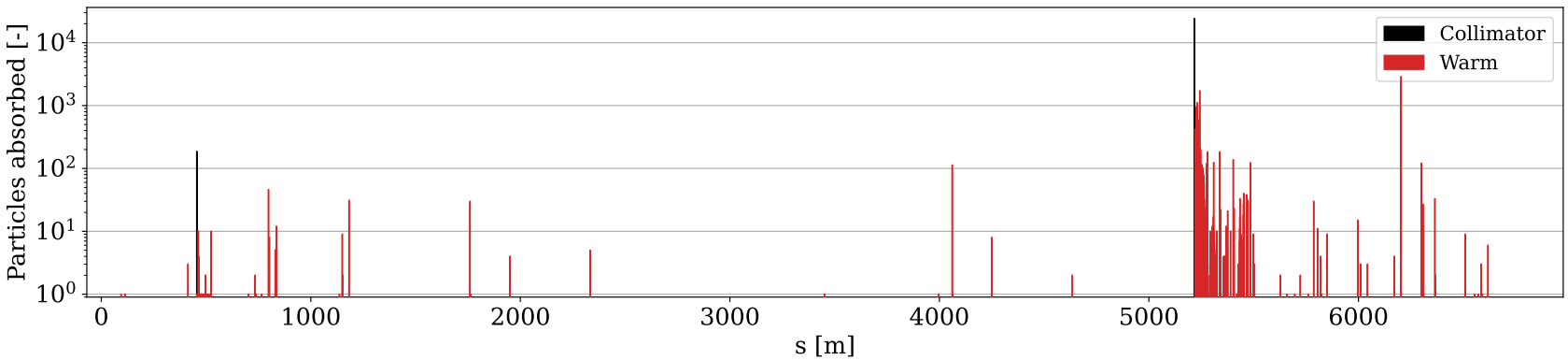
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase −60°



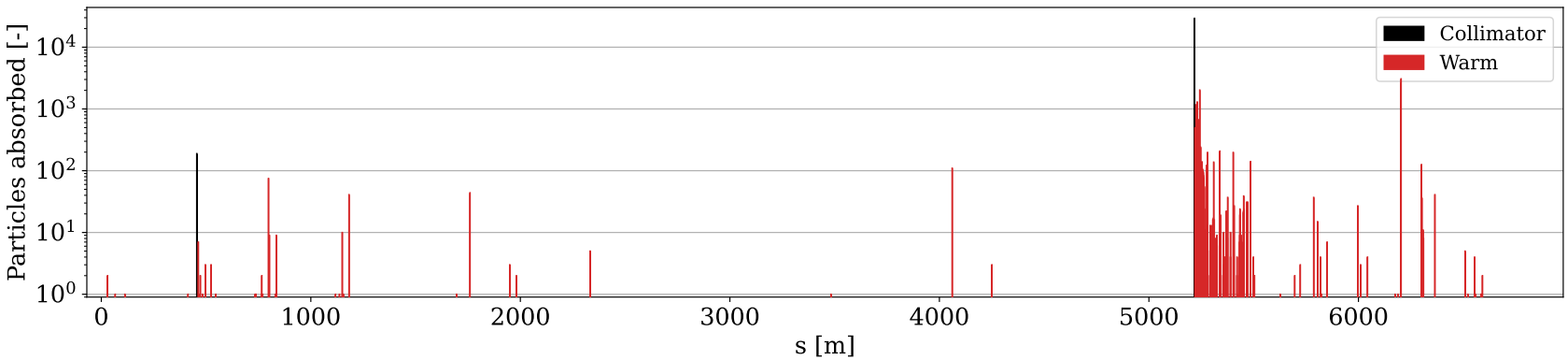
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase $-30°$



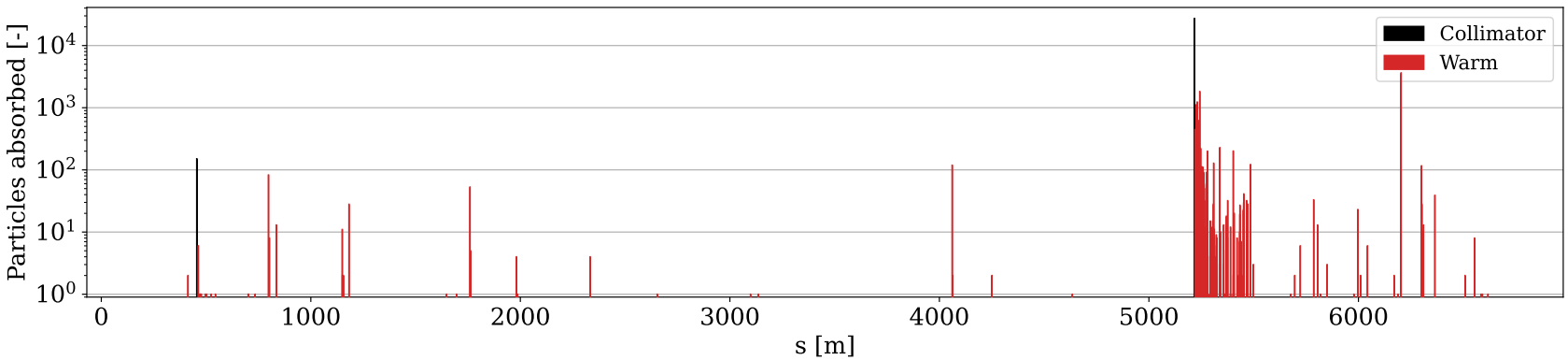
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 0°



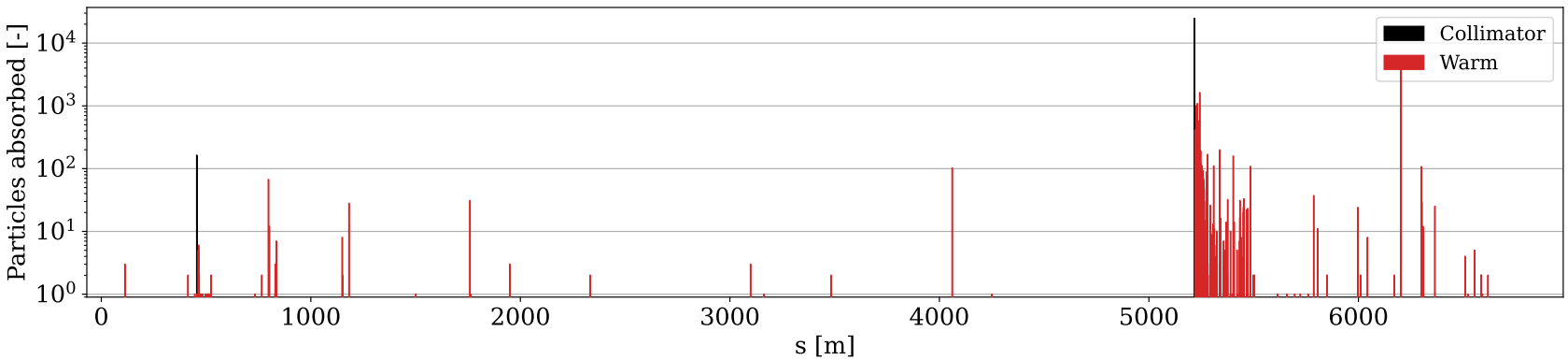
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 30°



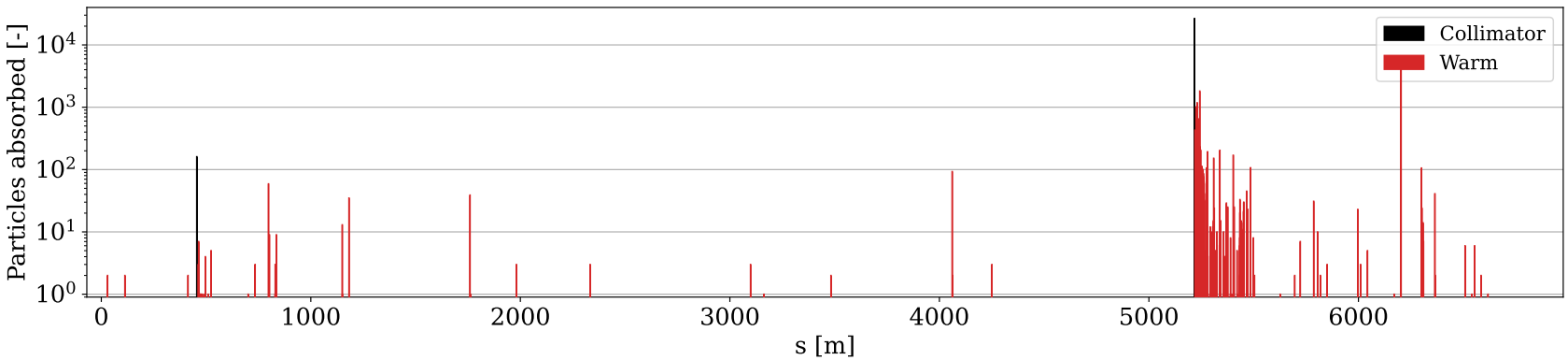
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 60°



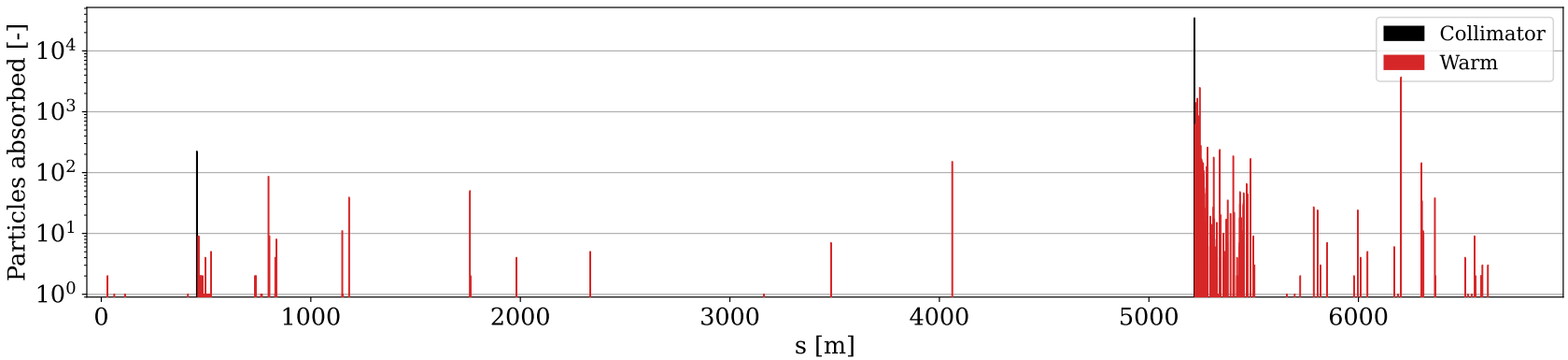
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 90°



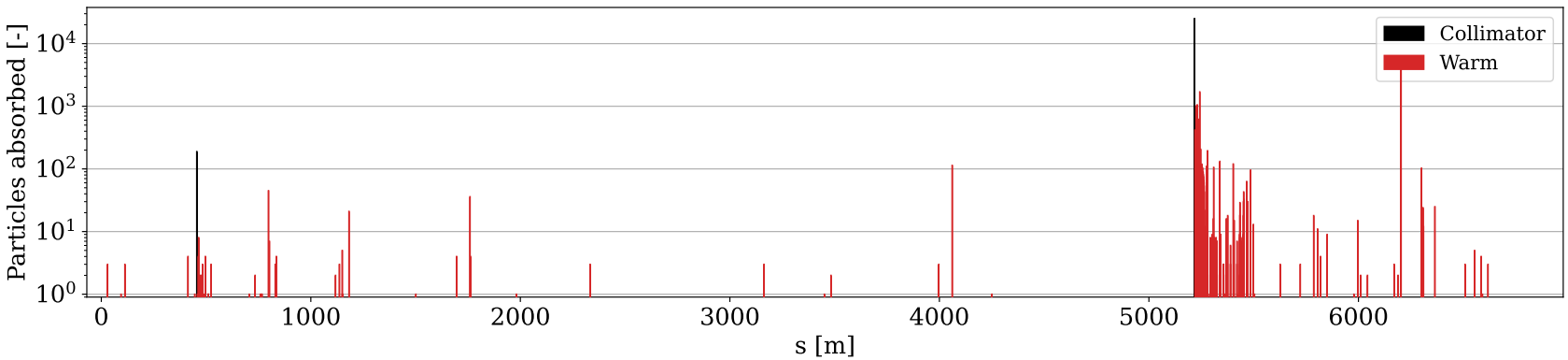
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 120°



Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 150°



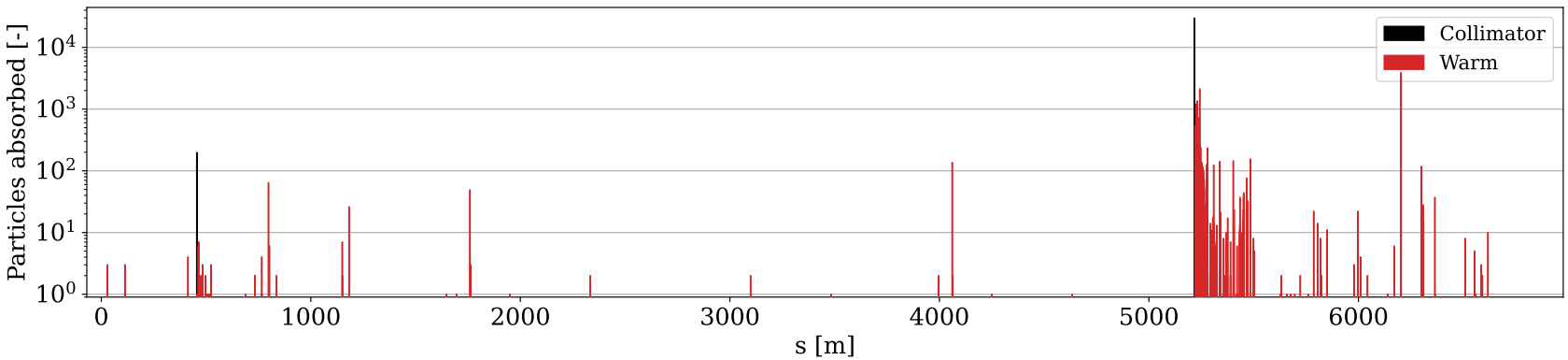
Horizontal crabs — Betatron loss maps — Horizontal — Without errors — Phase 179°



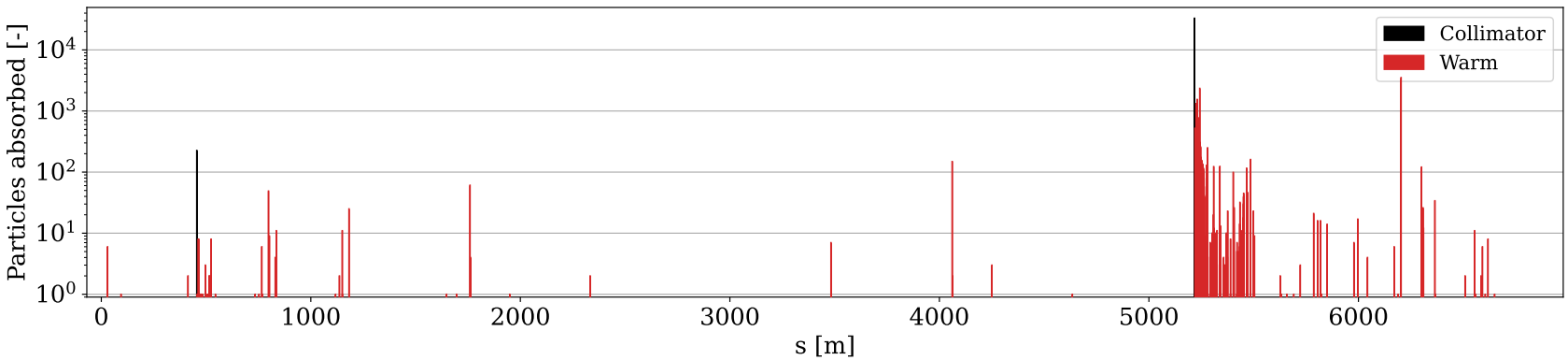
Horizontal crabs — Betatron loss maps — Horizontal — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

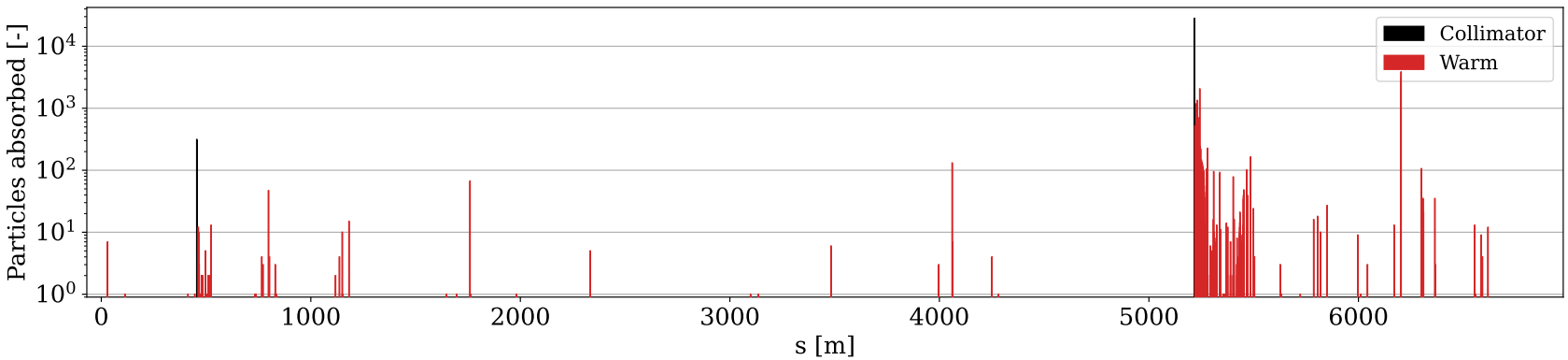
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $-179°$



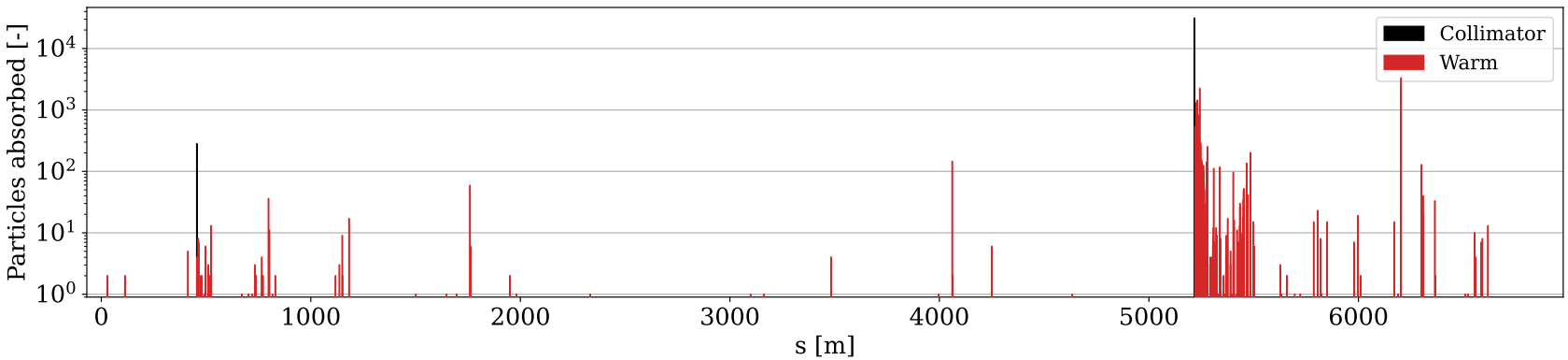
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $-150°$



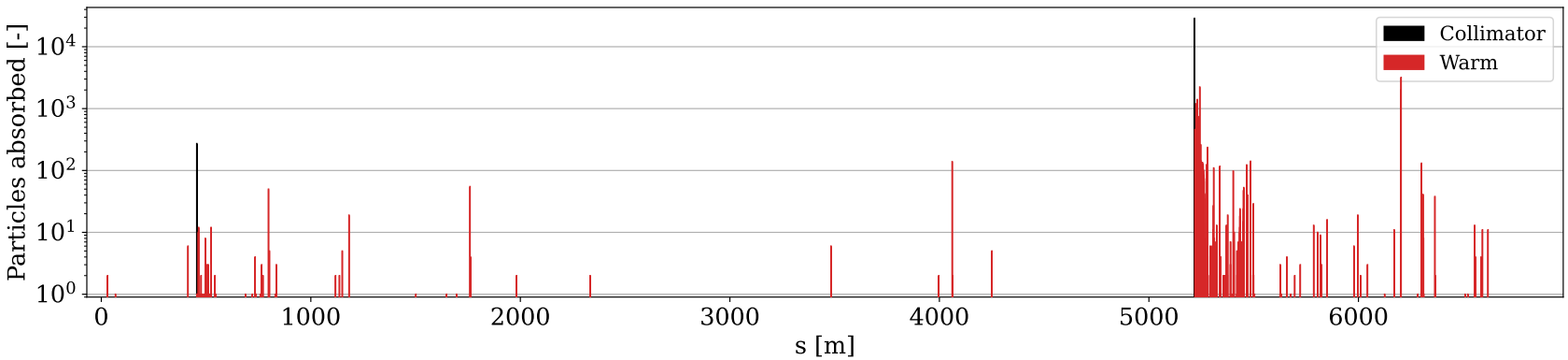
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $-120°$



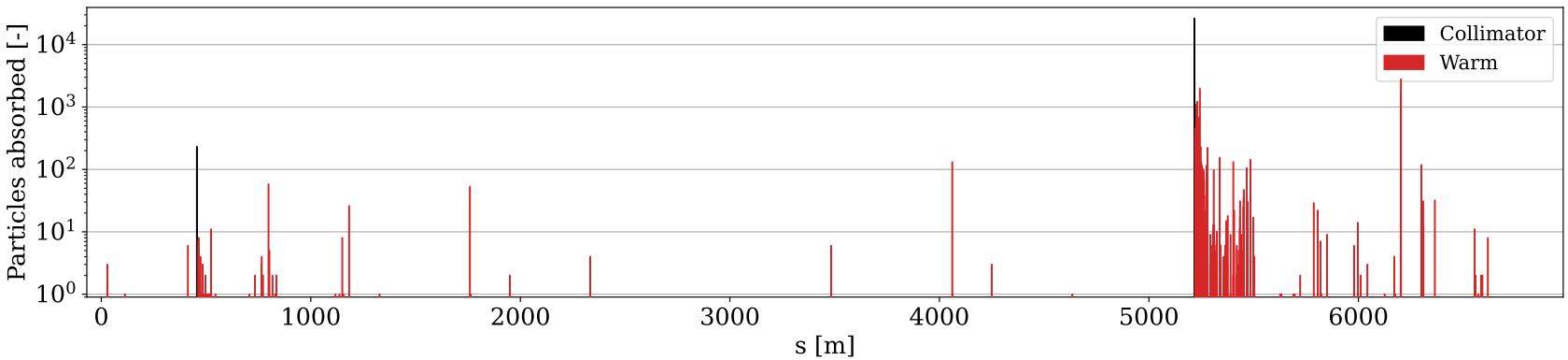
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase −90°



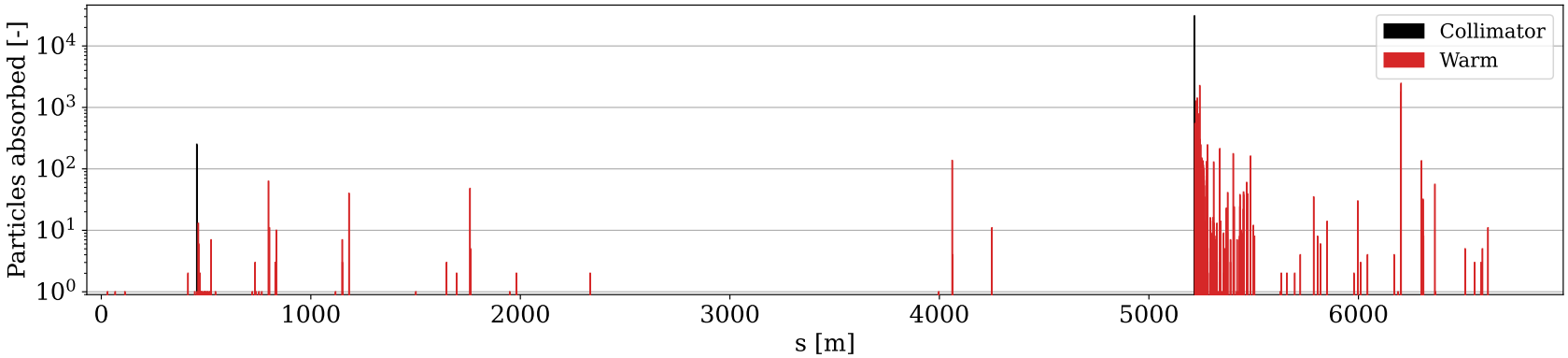
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $-60°$



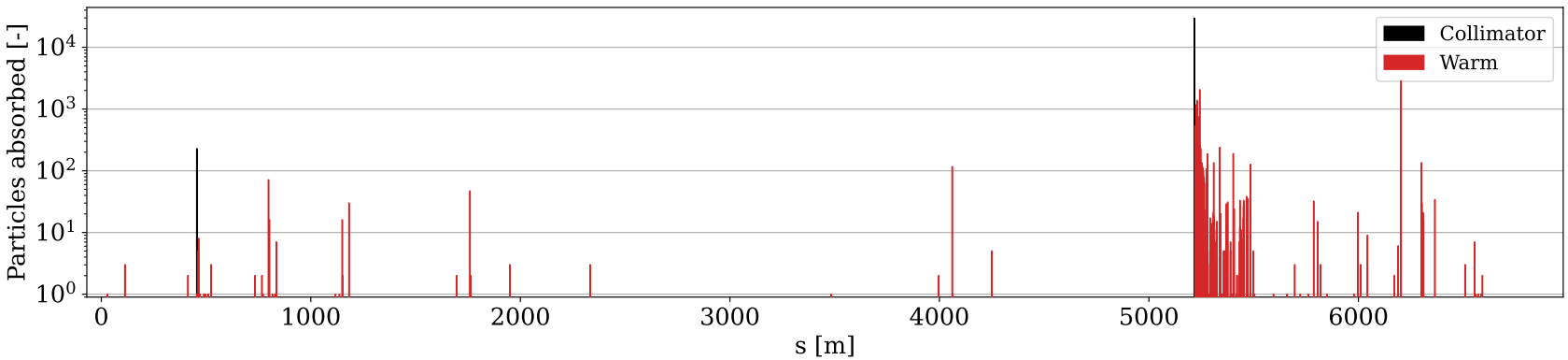
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $-30°$



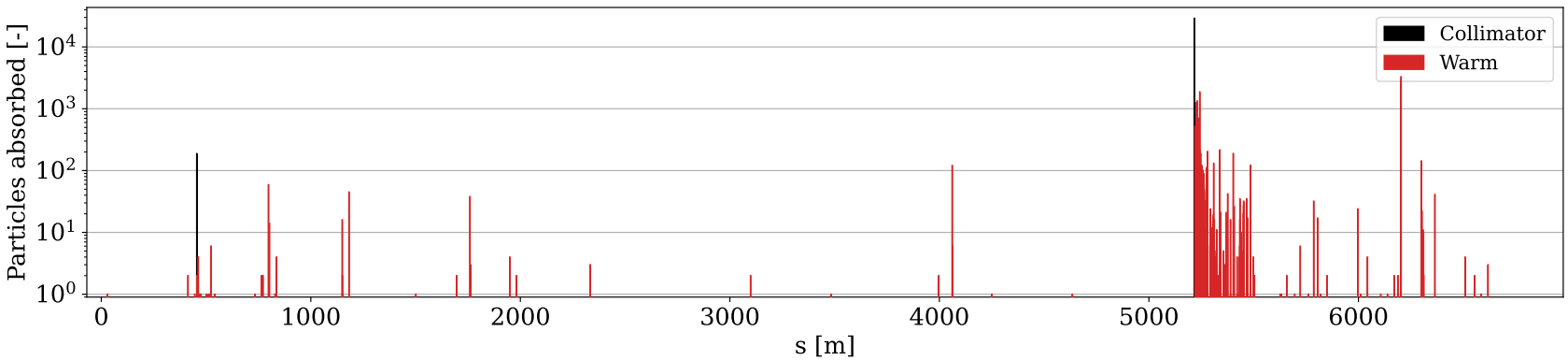
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase $0°$



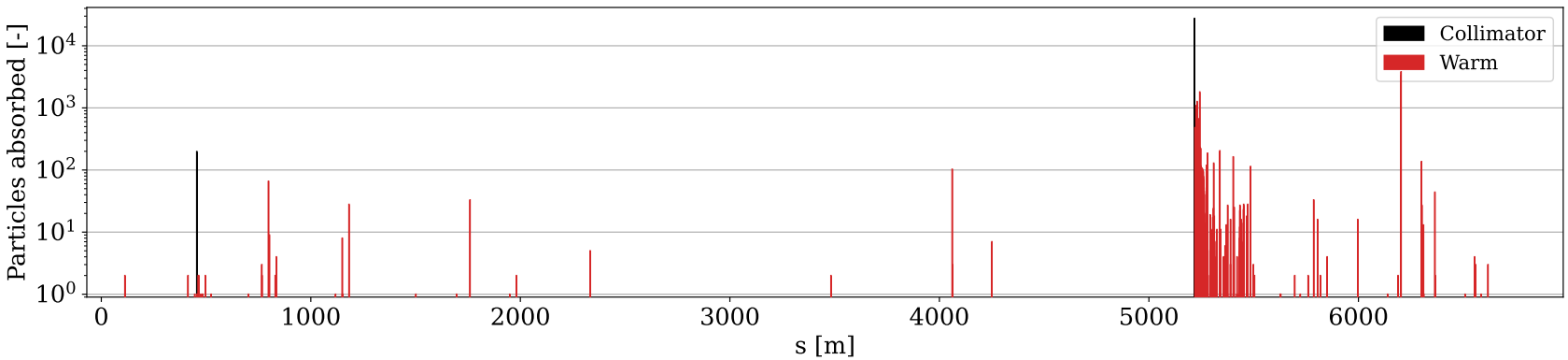
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 30°



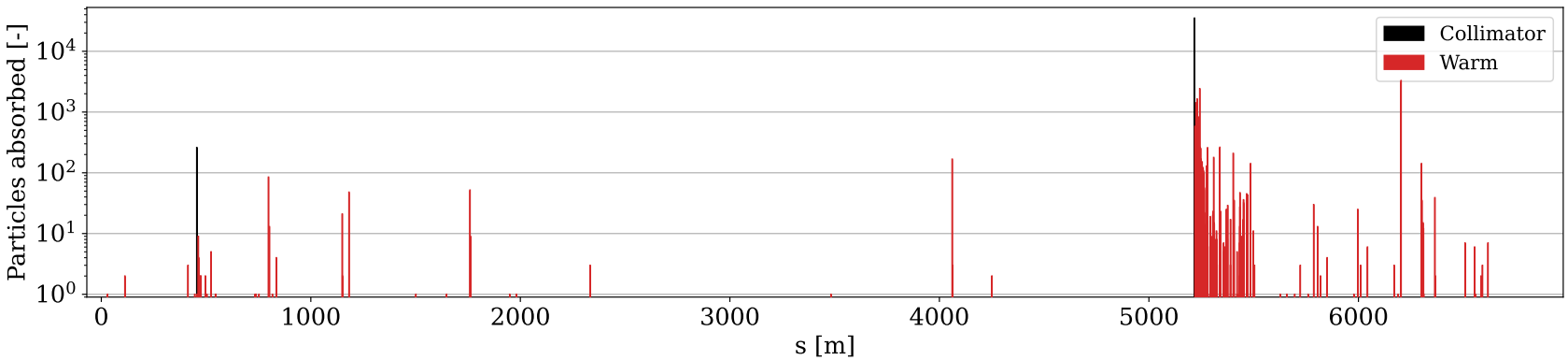
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 60°



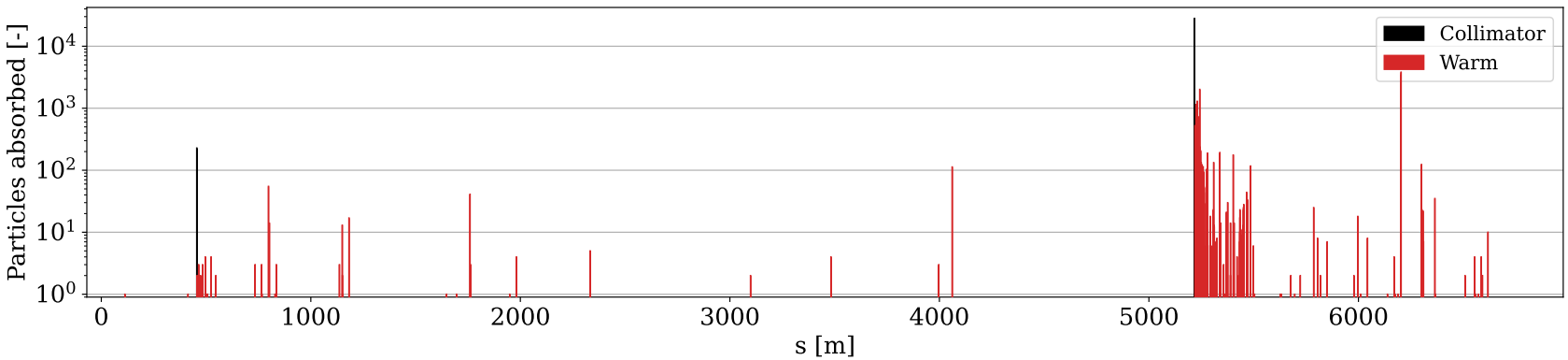
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 90°



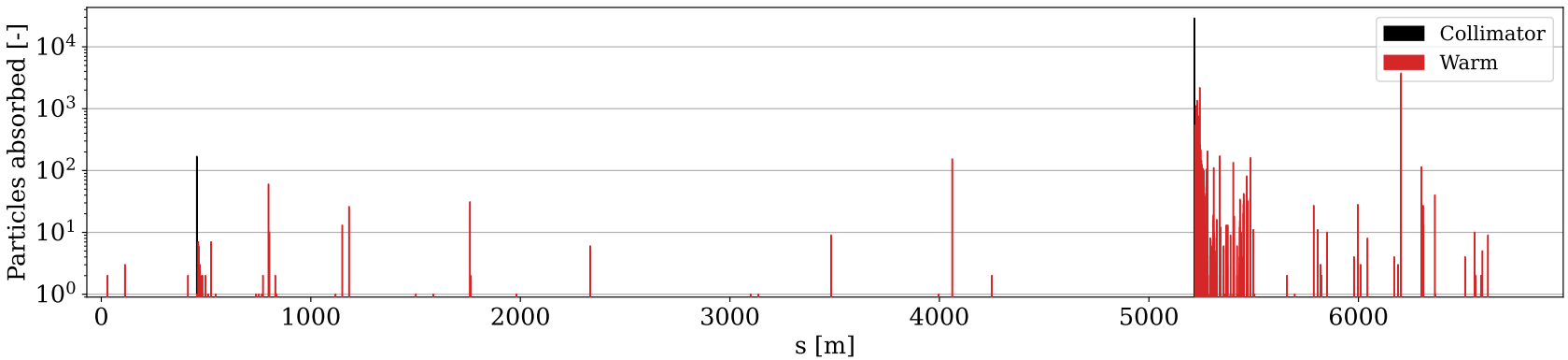
Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 120°



Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 150°



Horizontal crabs — Betatron loss maps — Horizontal — With errors — Phase 179°



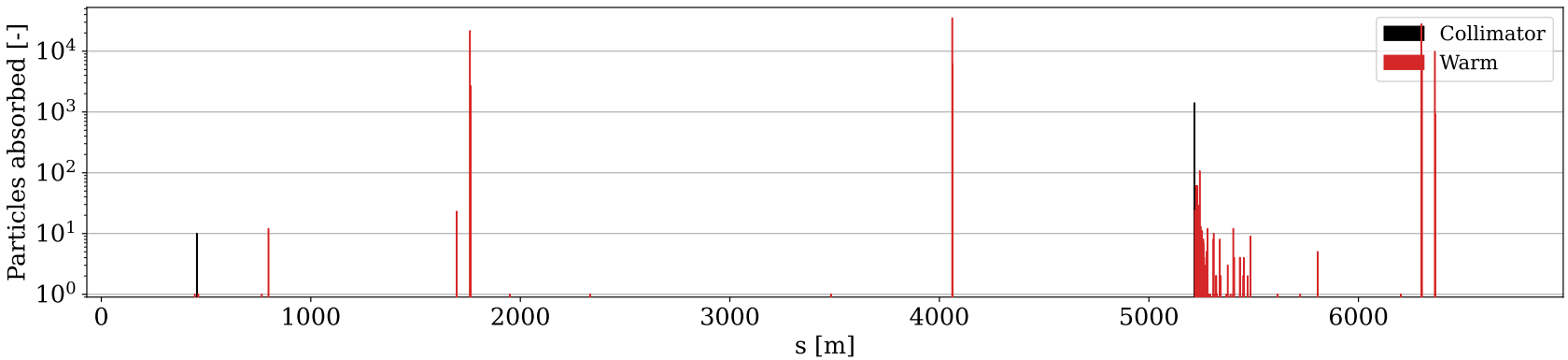
Horizontal crabs — Betatron loss maps — Vertical

- Without errors
- With errors

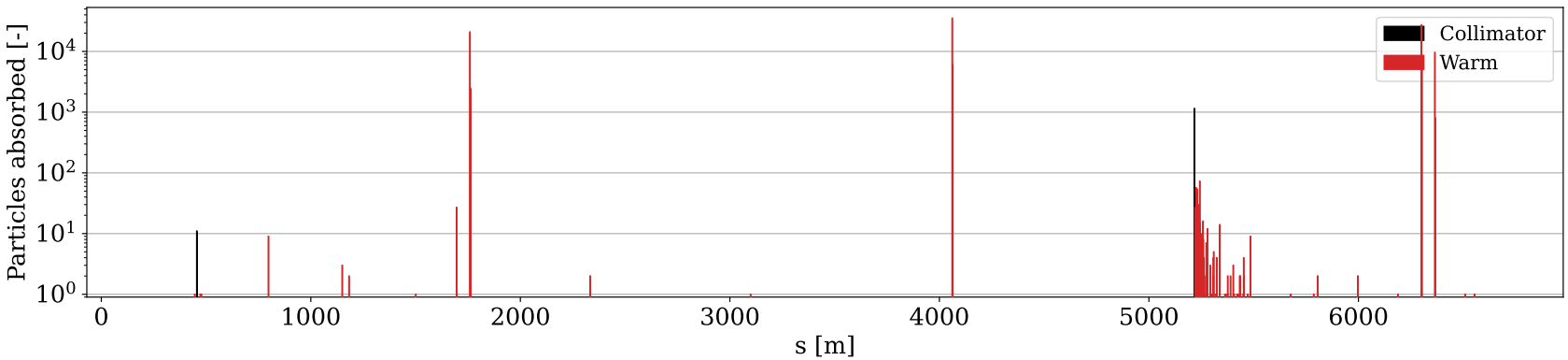
Horizontal crabs — Betatron loss maps — Vertical — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

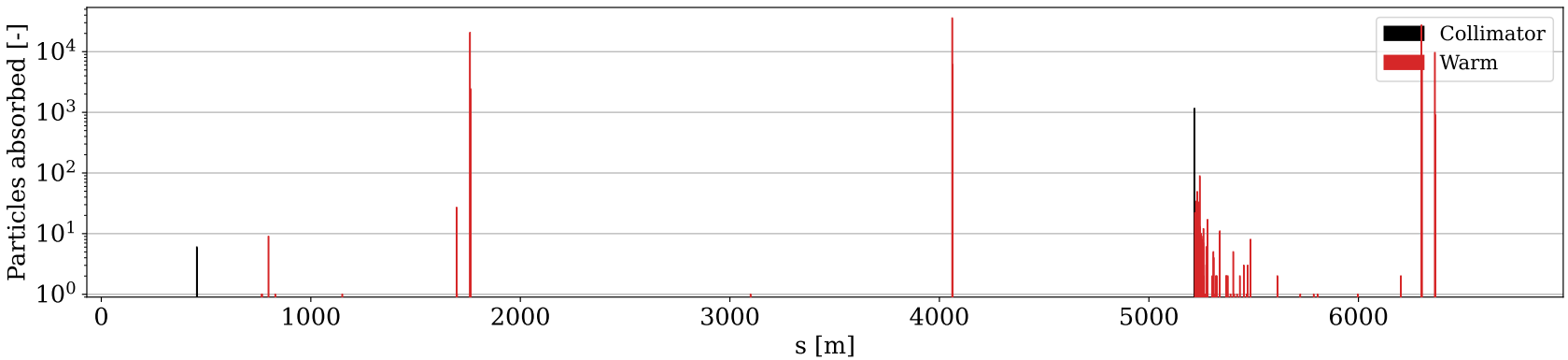
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $-179°$



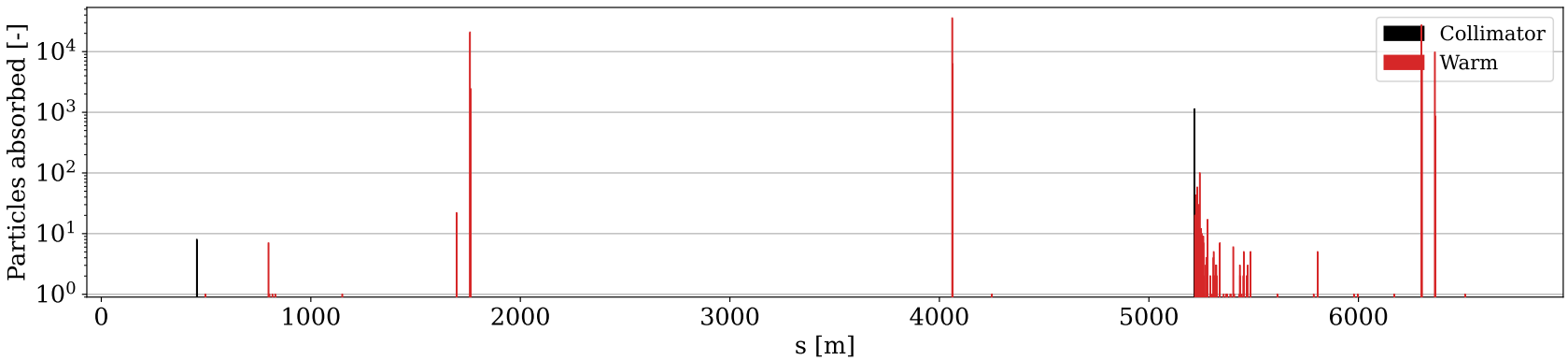
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $-150°$



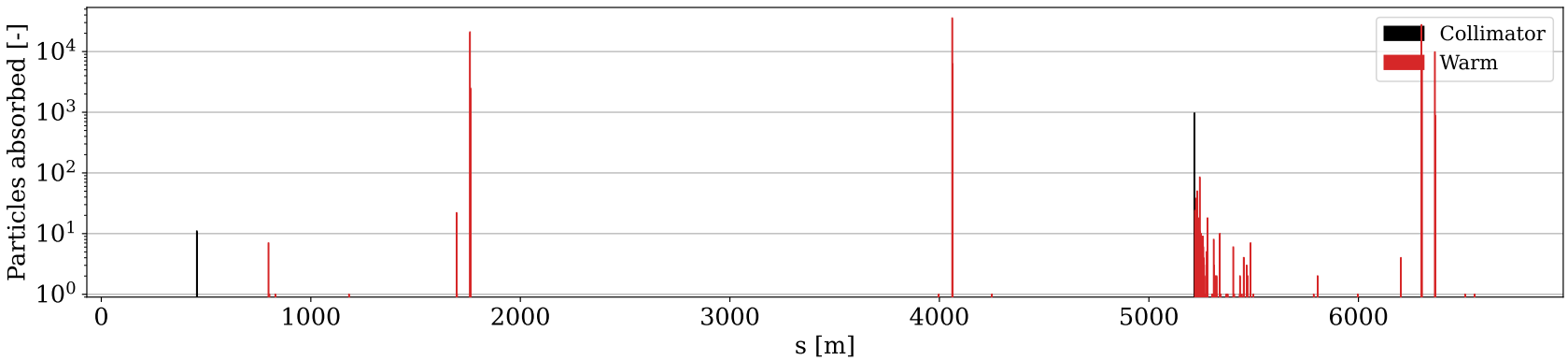
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $-120°$



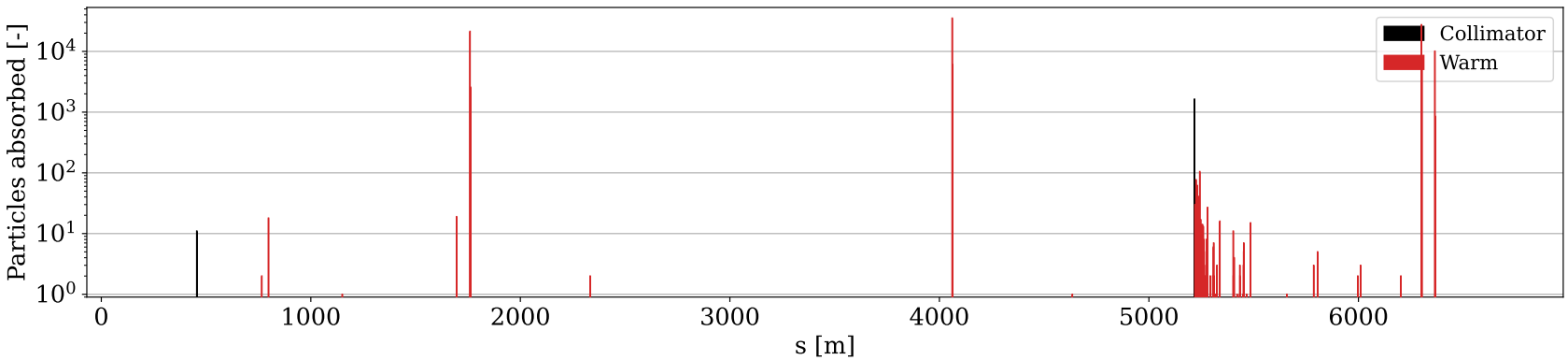
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase −90°



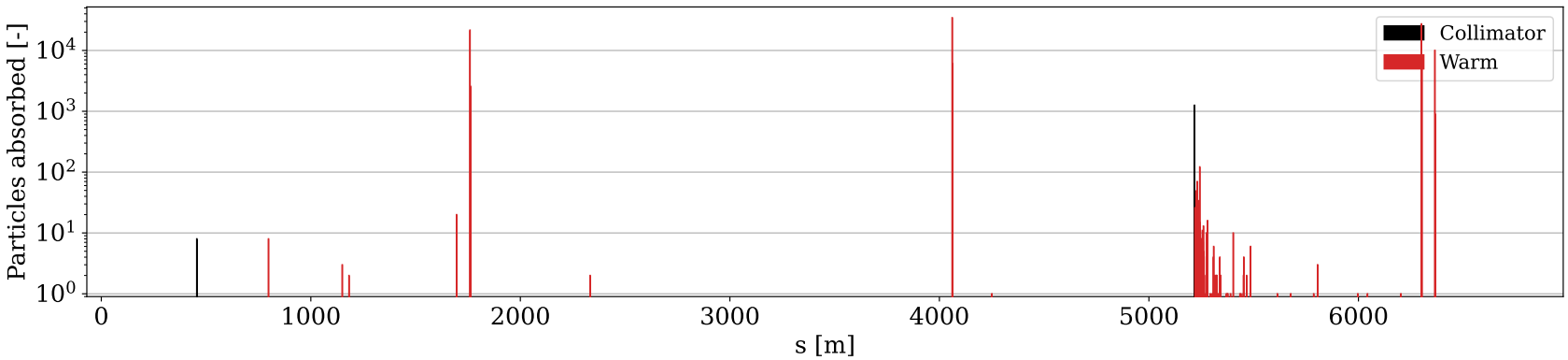
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $-60°$



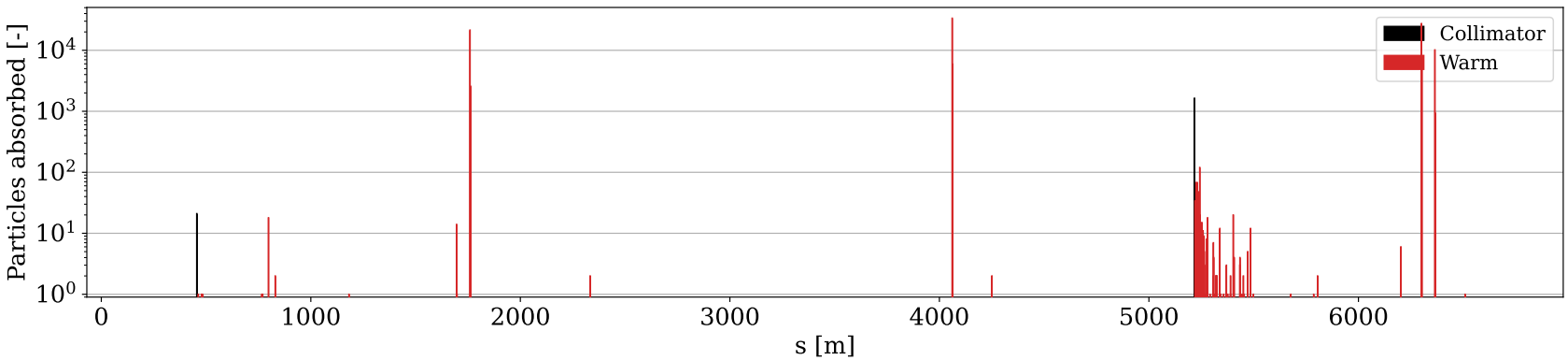
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $-30°$



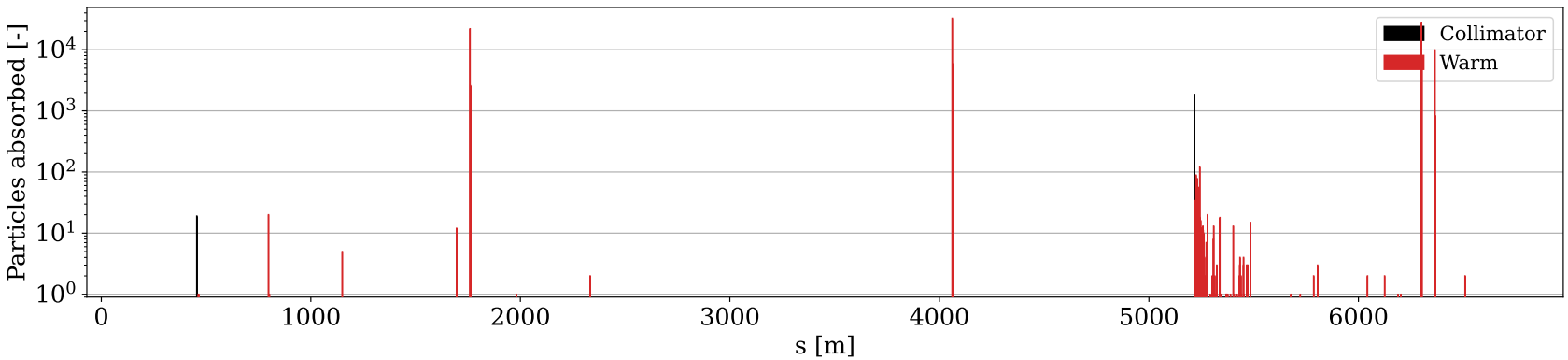
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase $0°$



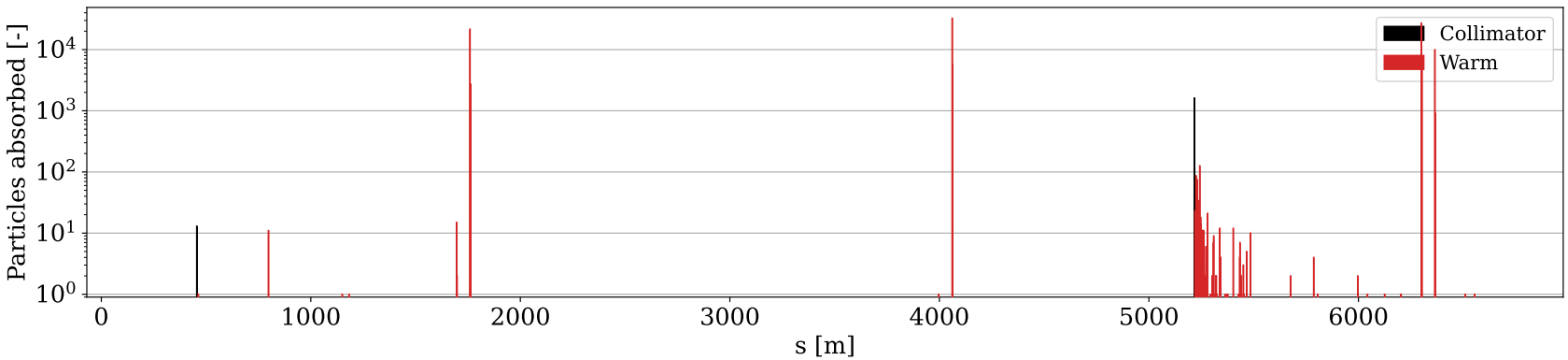
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 30°



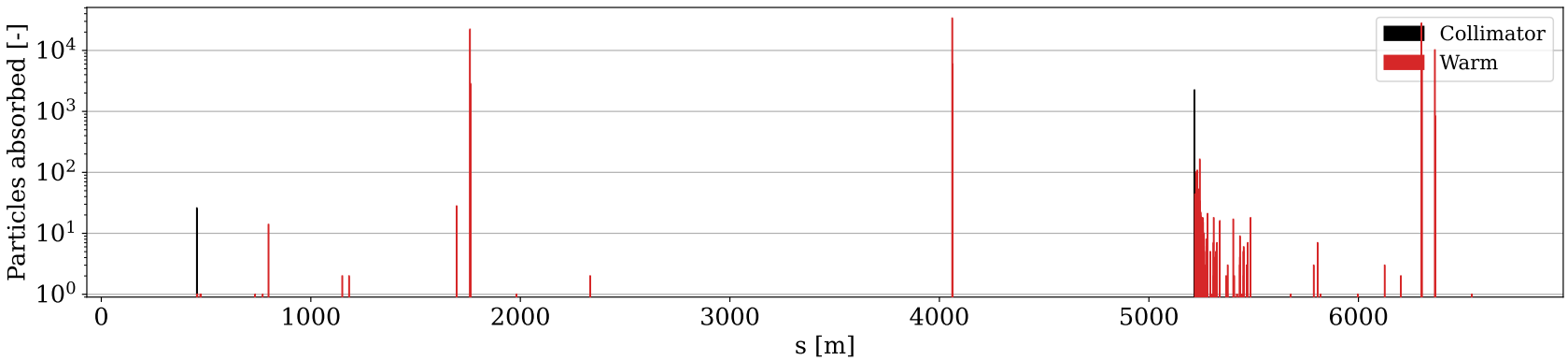
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 60°



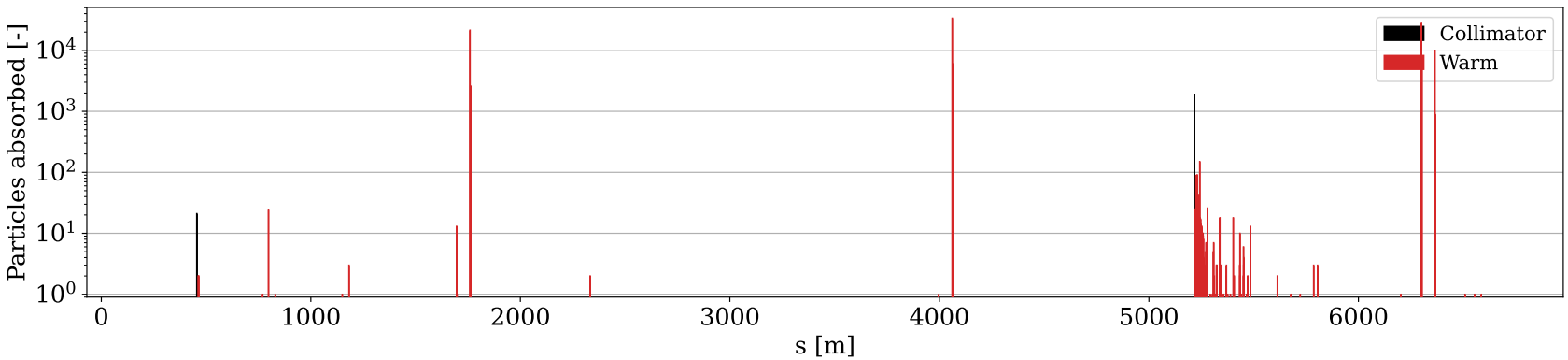
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 90°



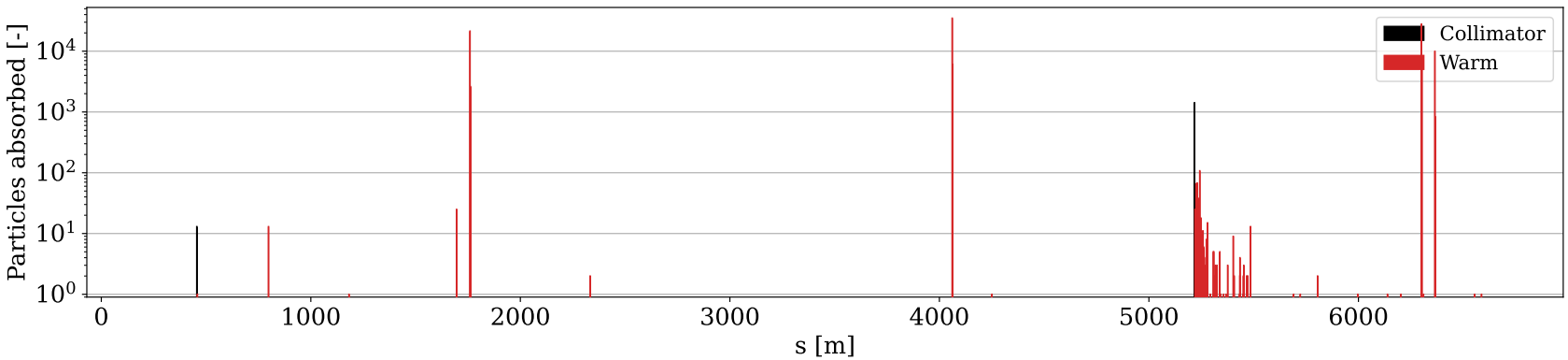
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 120°



Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 150°



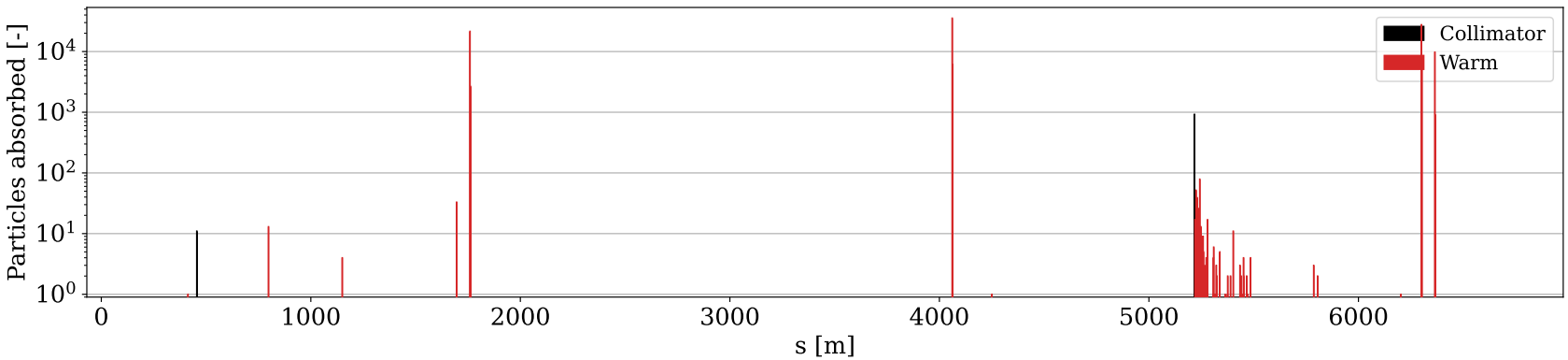
Horizontal crabs — Betatron loss maps — Vertical — Without errors — Phase 179°



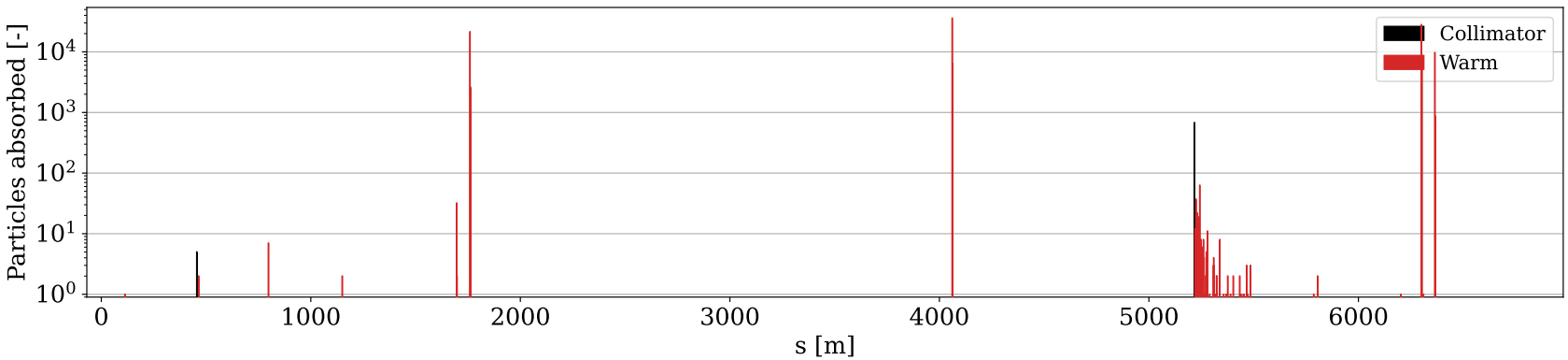
Horizontal crabs — Betatron loss maps — Vertical — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

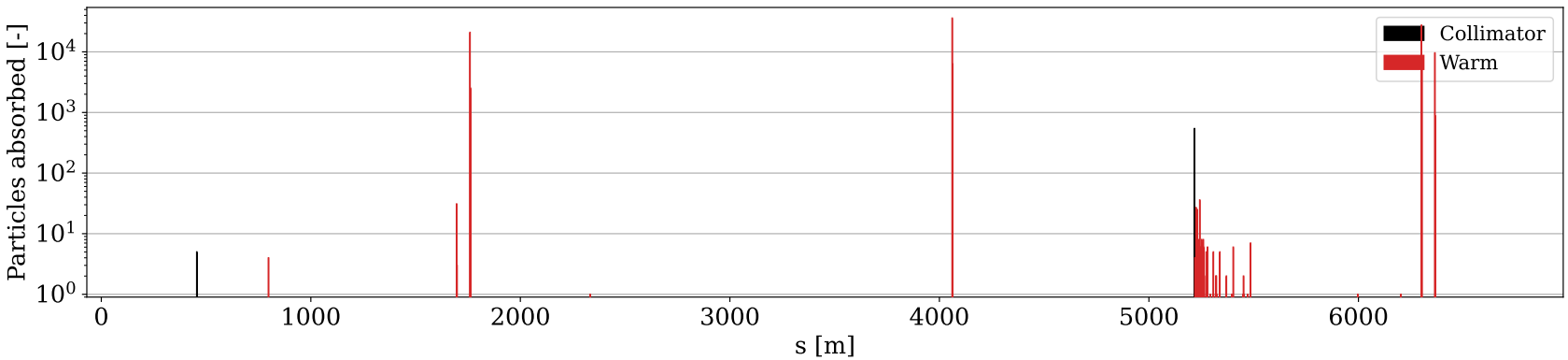
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-179°$



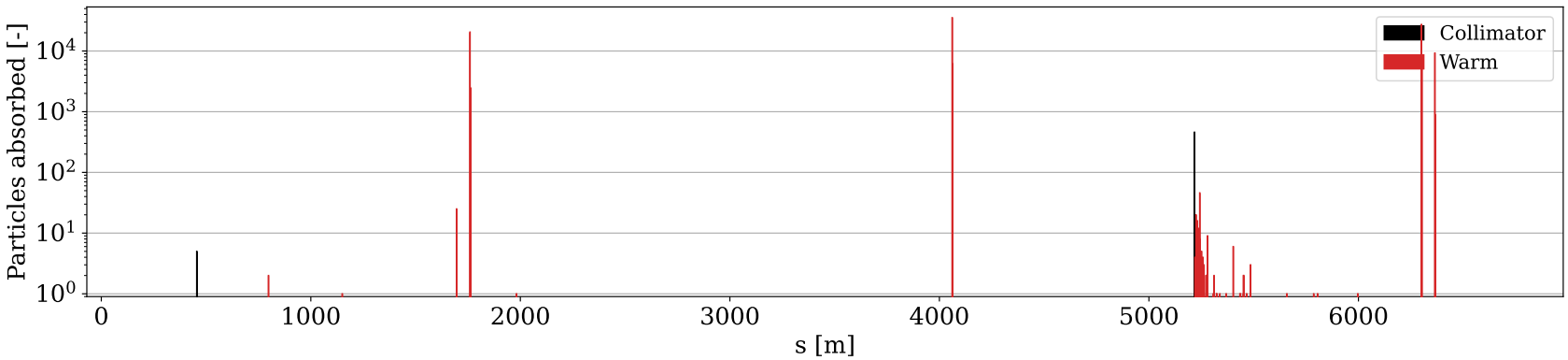
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-150°$



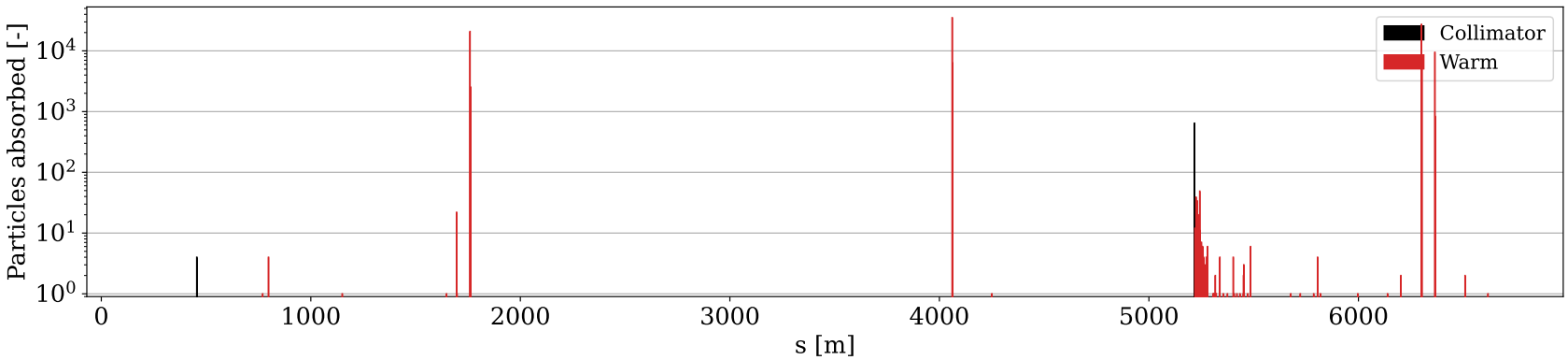
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-120°$



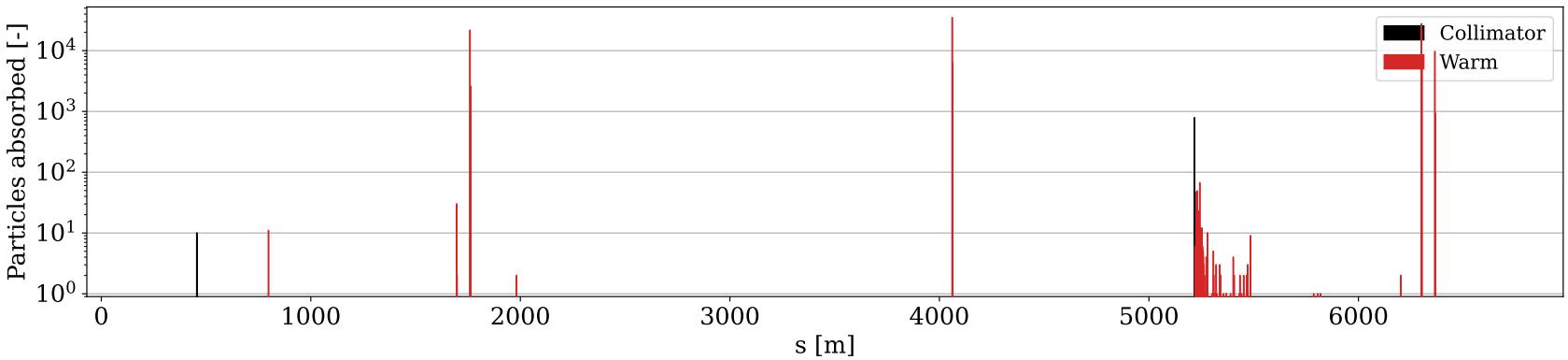
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-90°$



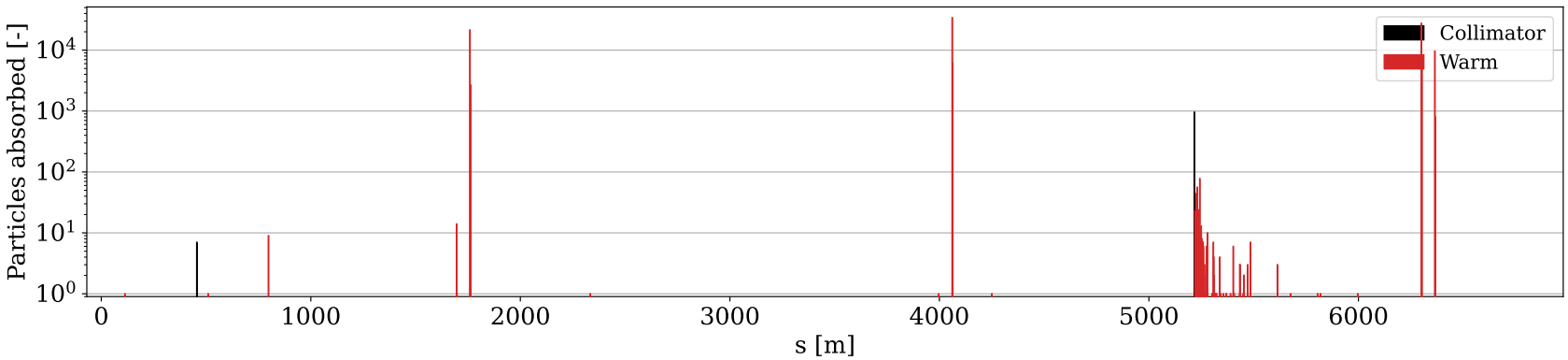
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-60°$



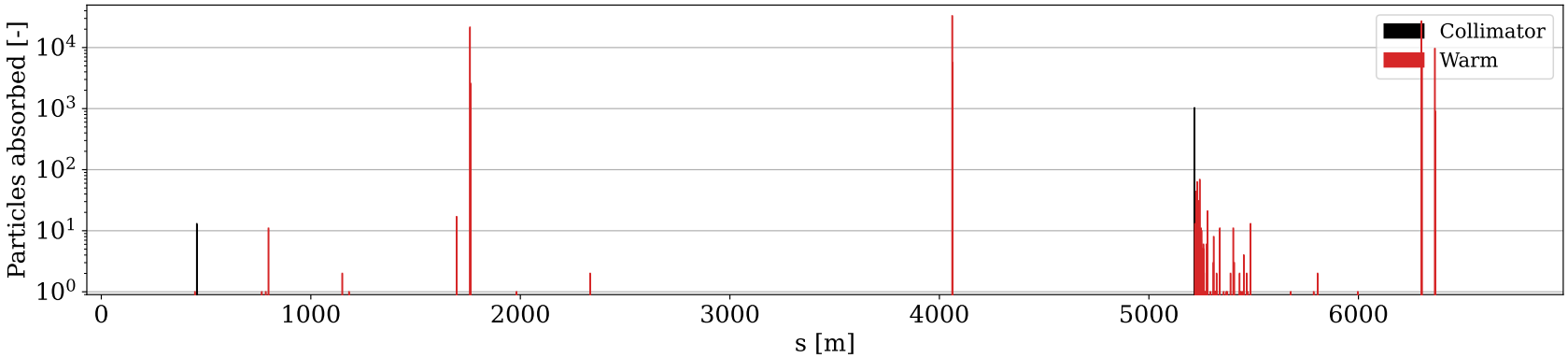
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $-30°$



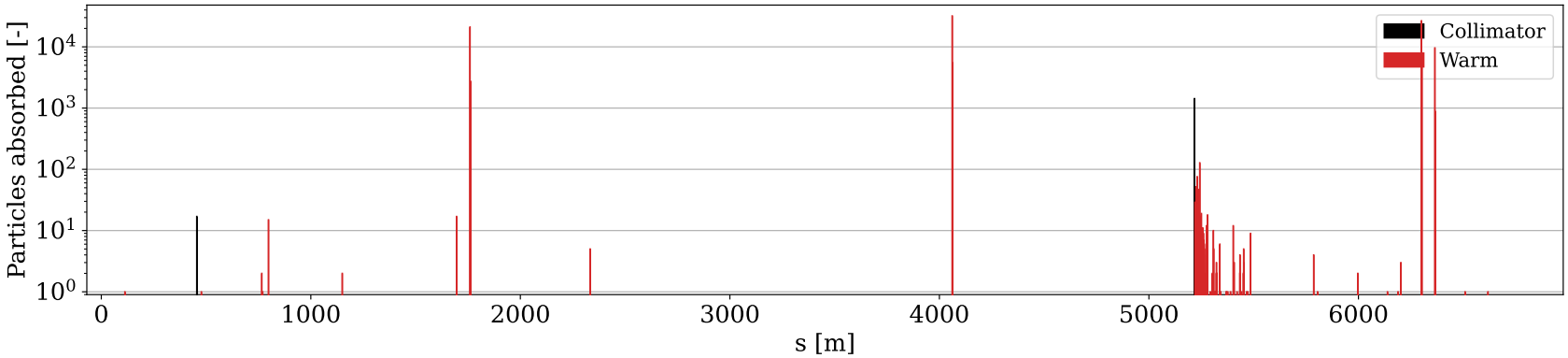
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $0°$



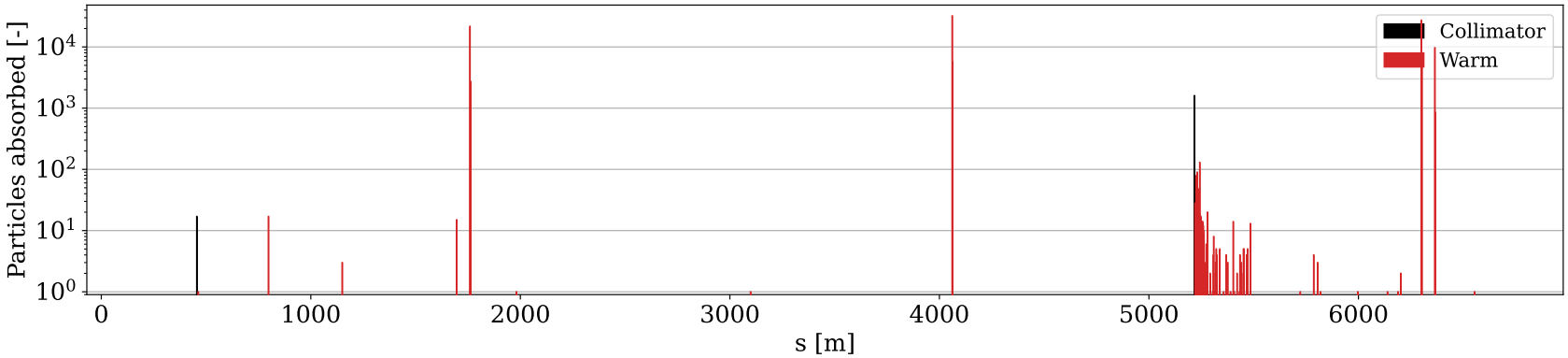
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $30°$



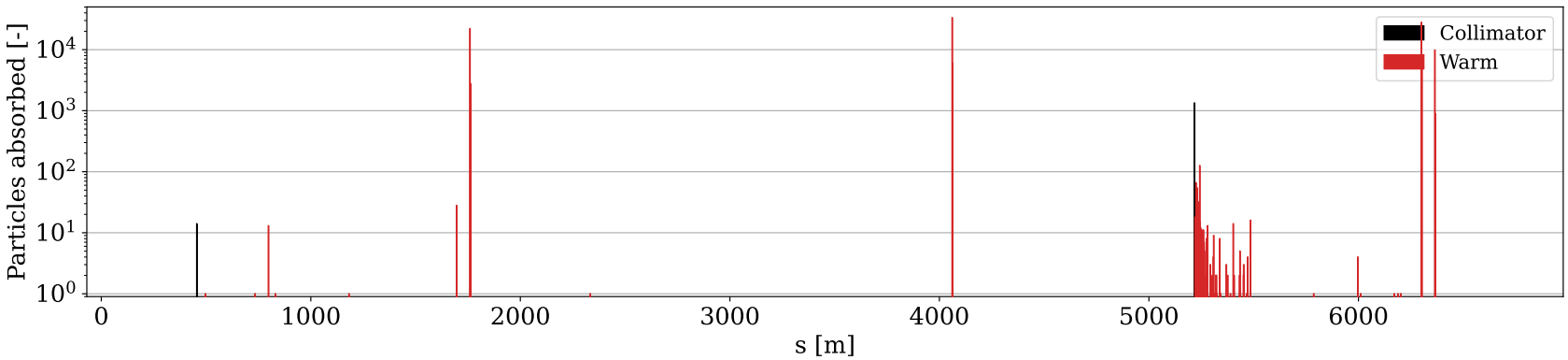
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $60°$



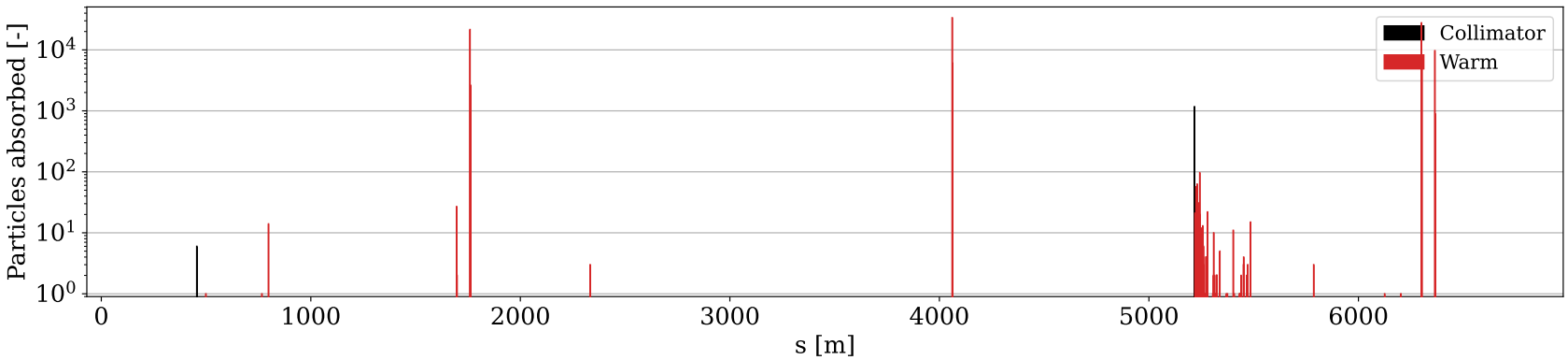
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $90°$



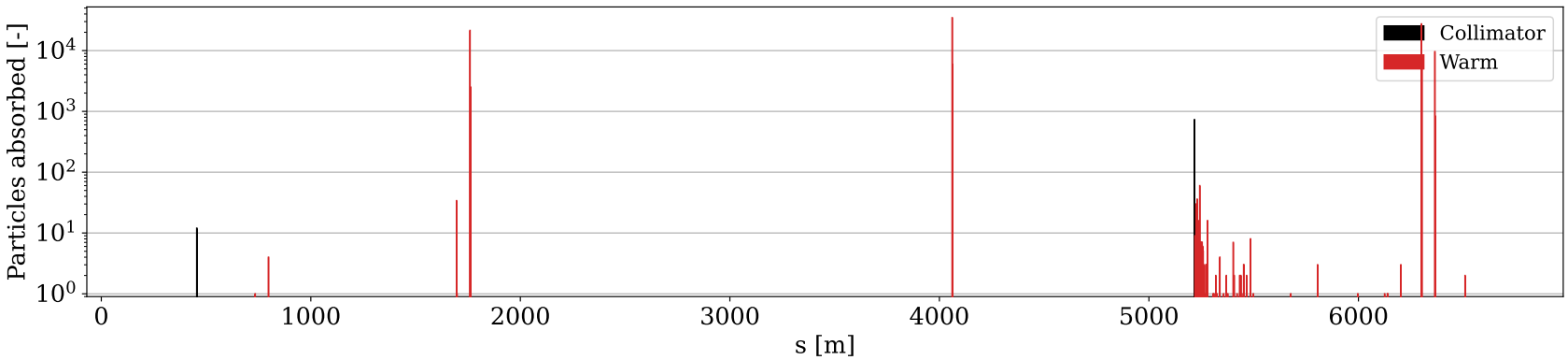
Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase 120°



Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase $150°$



Horizontal crabs — Betatron loss maps — Vertical — With errors — Phase 179°



Outline

1. Overview
2. Horizontal crabs
   - Betatron loss maps
   - Off-momentum loss maps
3. Vertical crabs
   - Betatron loss maps
   - Off-momentum loss maps

Horizontal crabs — Off-momentum loss maps

- DPpos
- DPneg

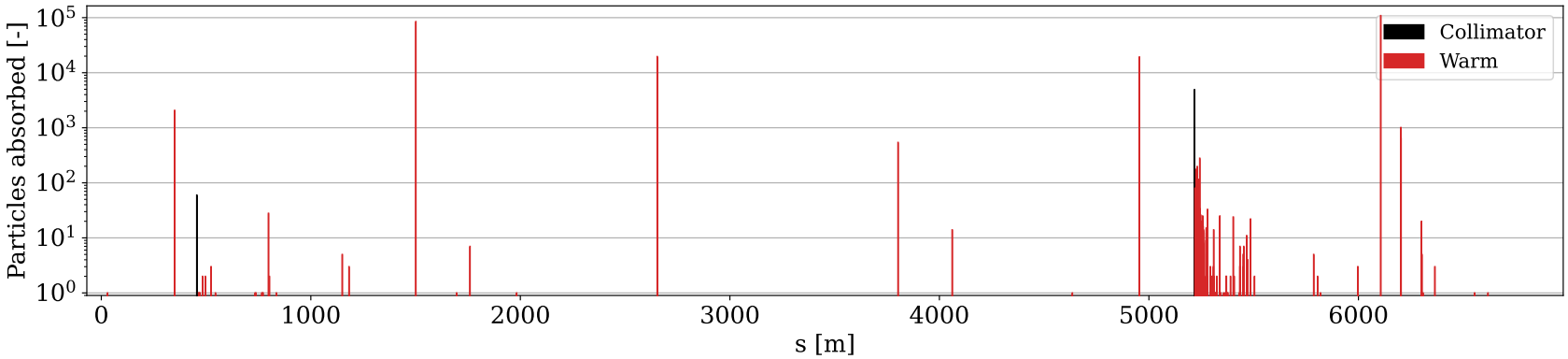
Horizontal crabs — Off-momentum loss maps — DPpos

- Without errors
- With errors

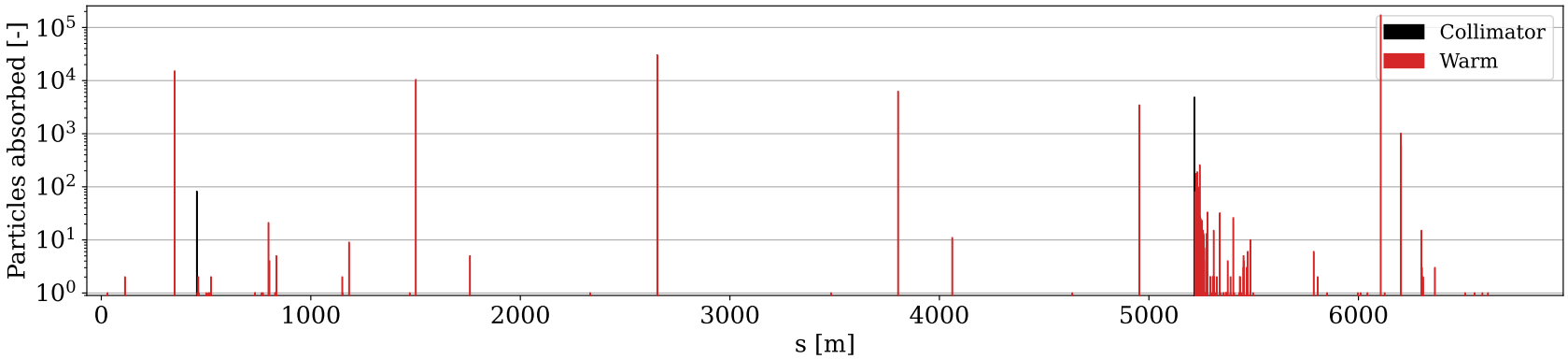
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

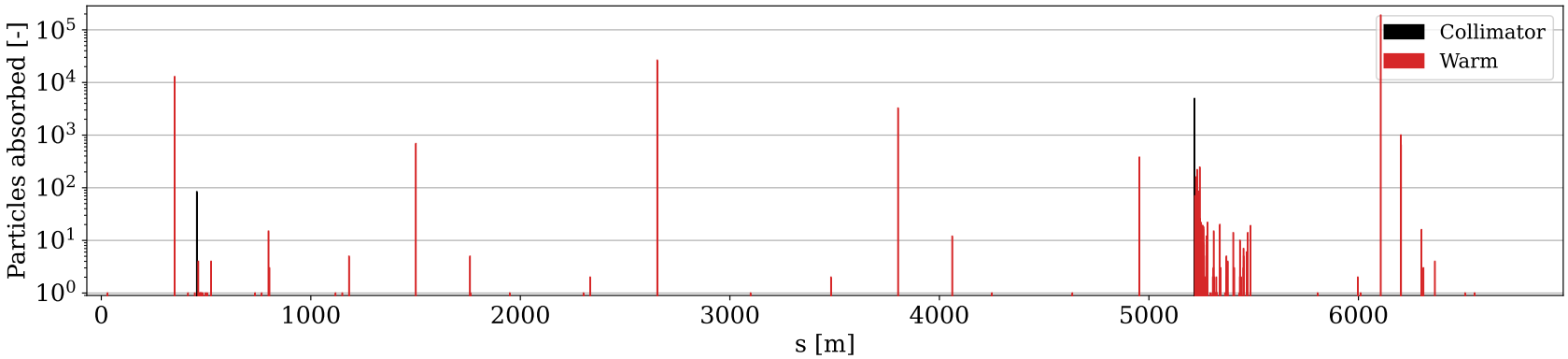
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-179°$



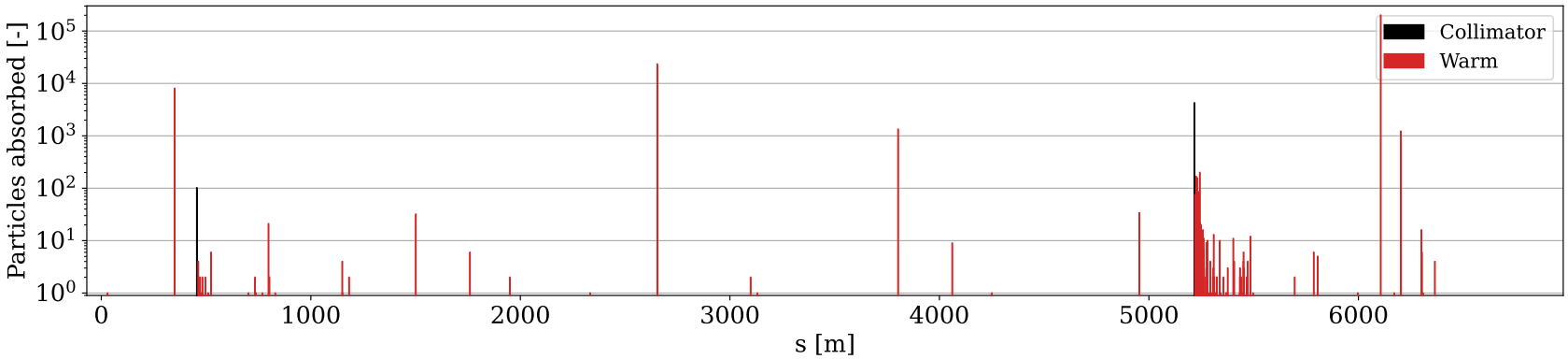
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-150°$



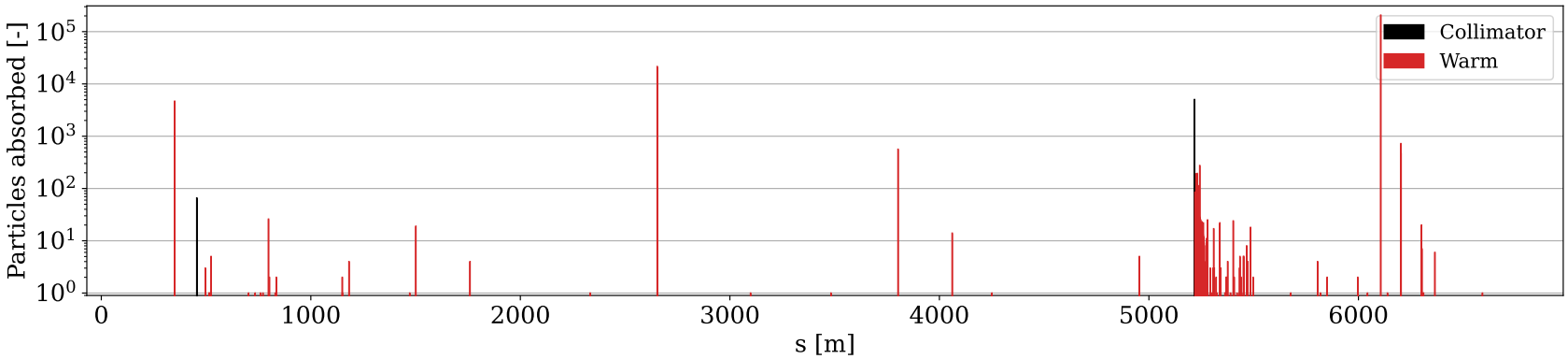
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-120°$



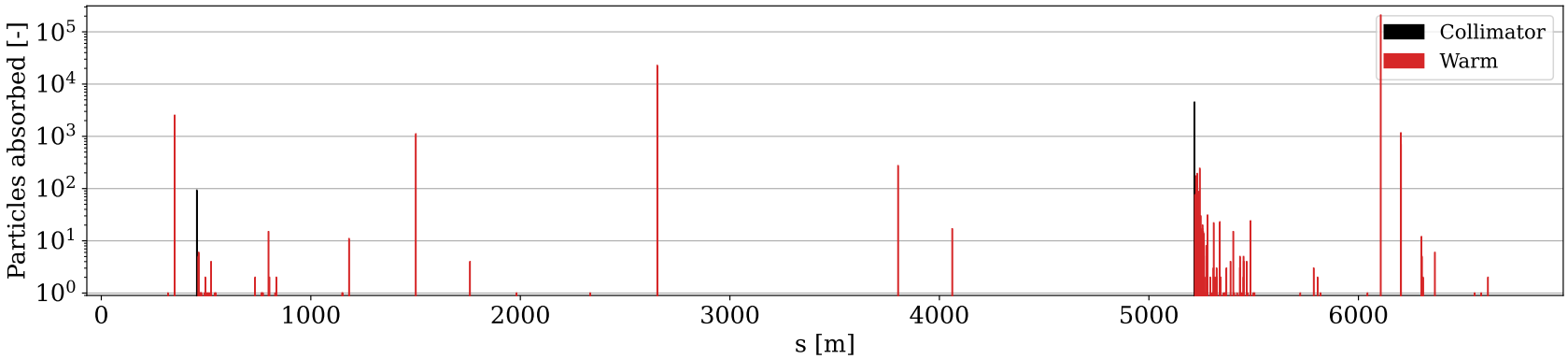
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-90°$



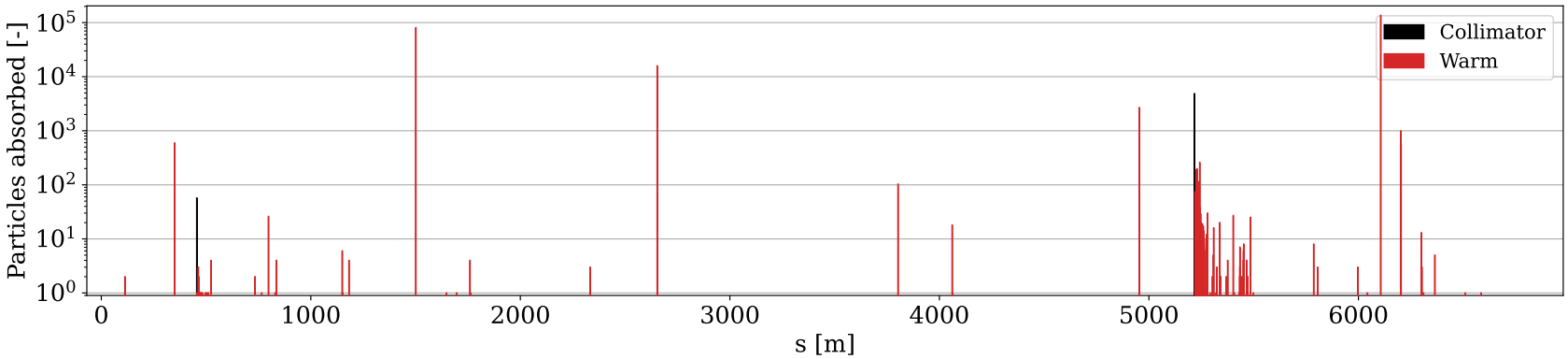
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-60°$



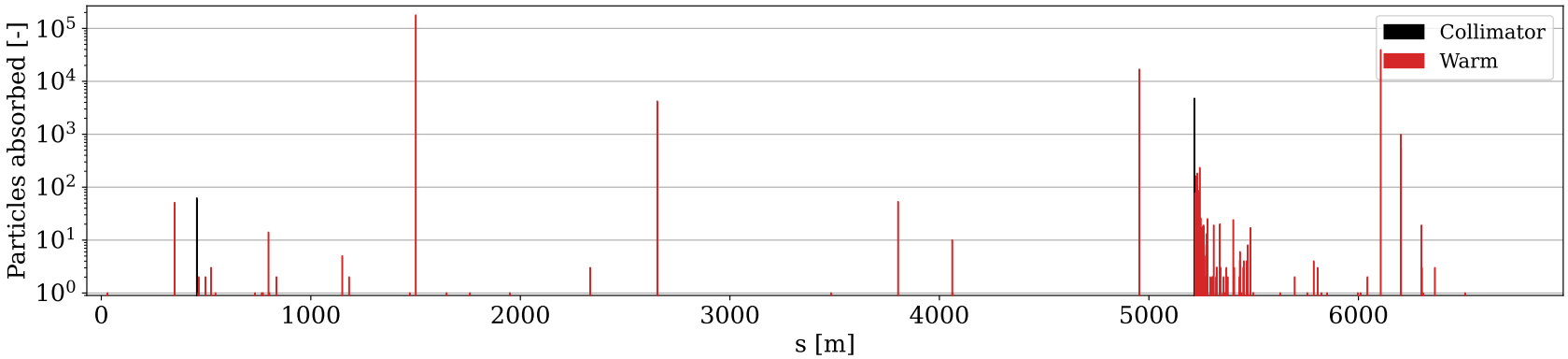
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-30°$



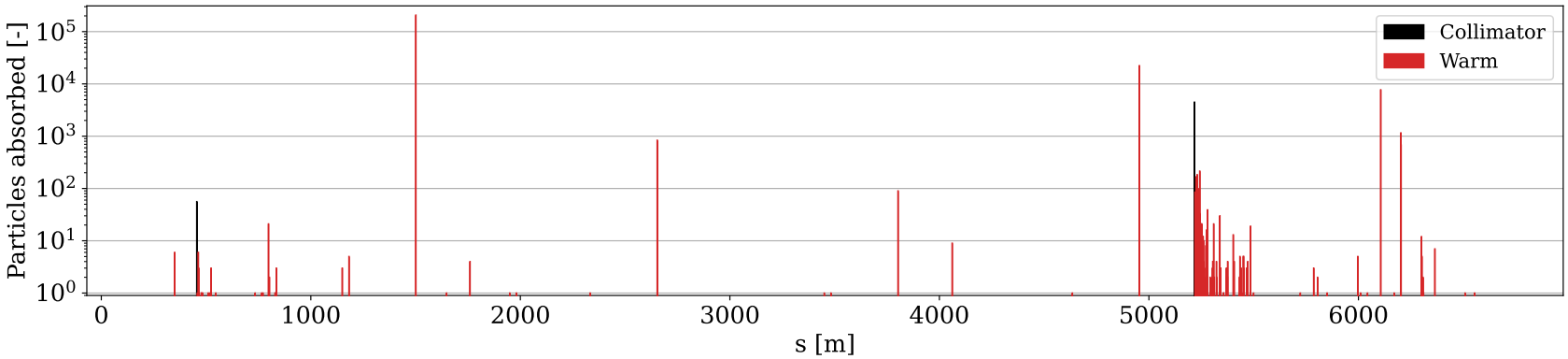
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $0°$



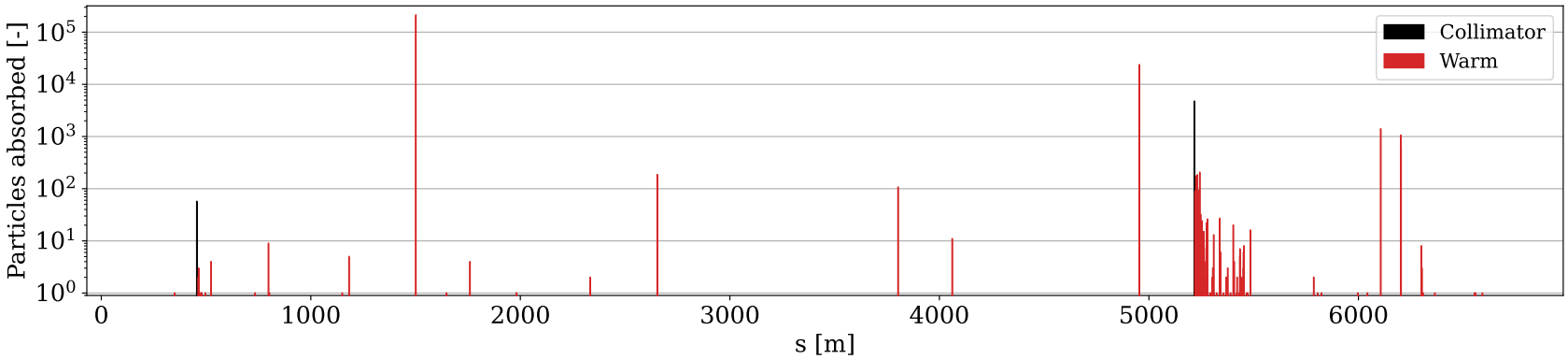
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase 30°



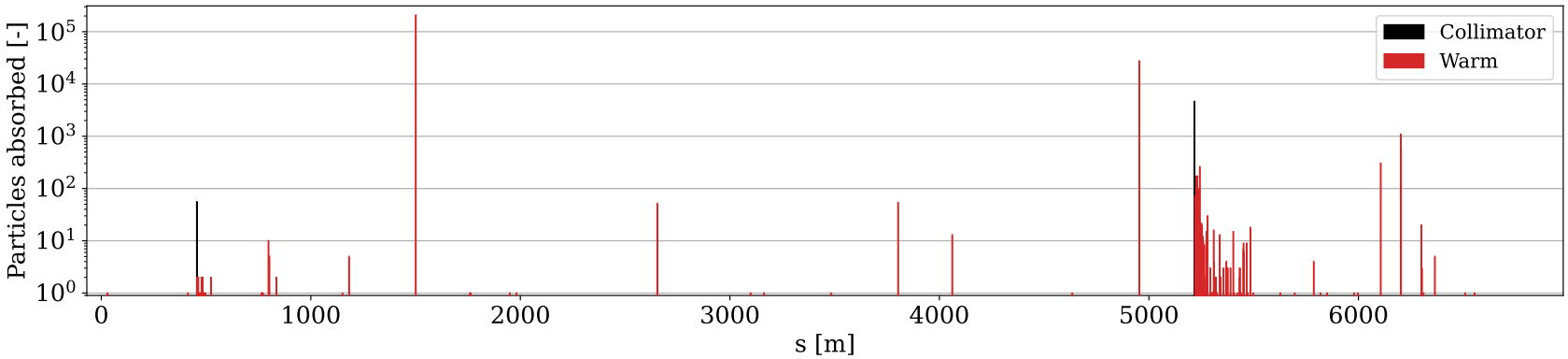
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $60°$



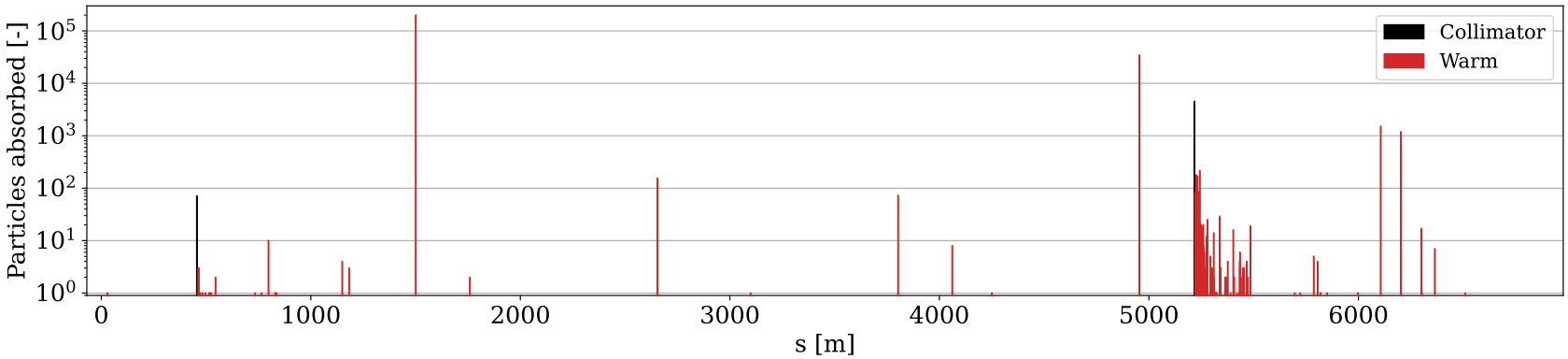
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase 90°



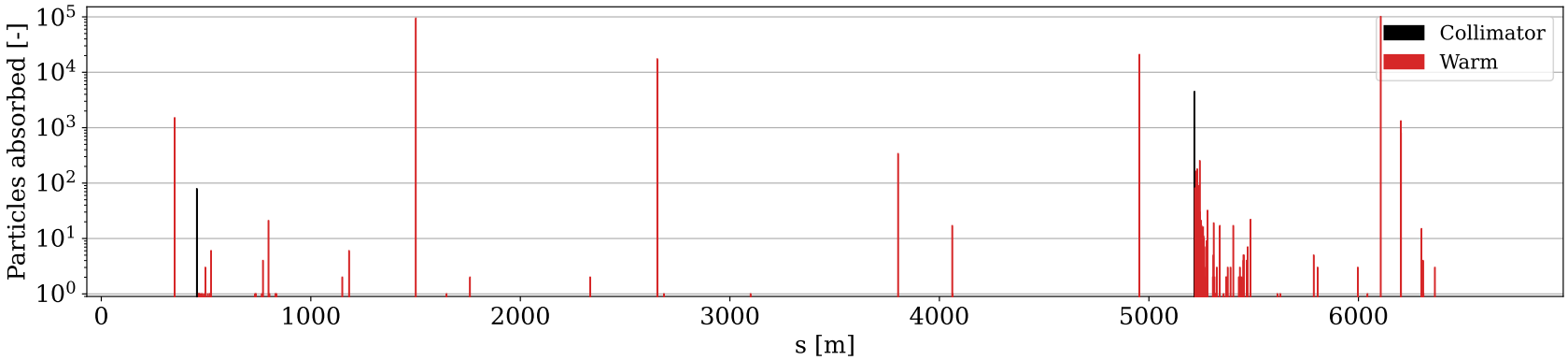
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase 120°



Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase $150°$



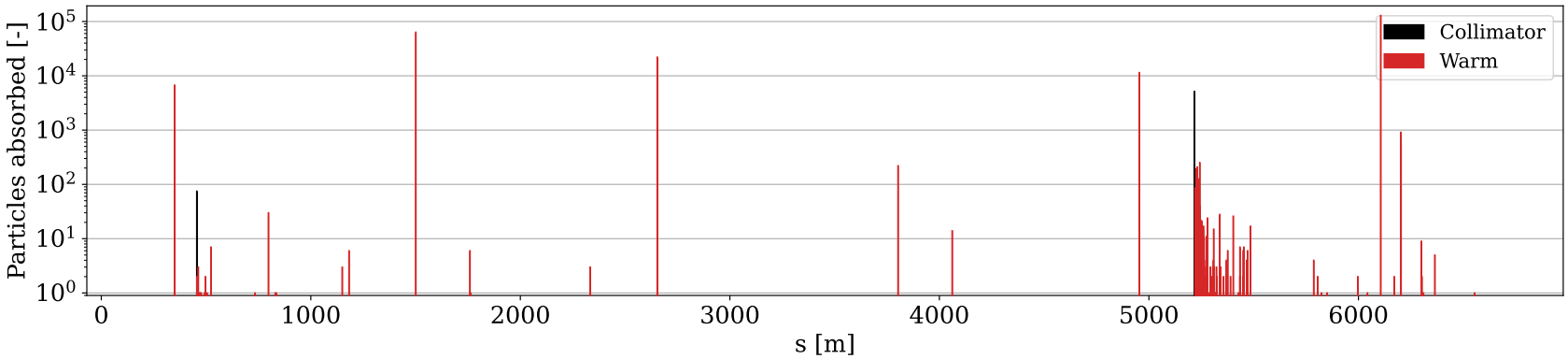
Horizontal crabs — Off-momentum loss maps — DPpos — Without errors — Phase 179°



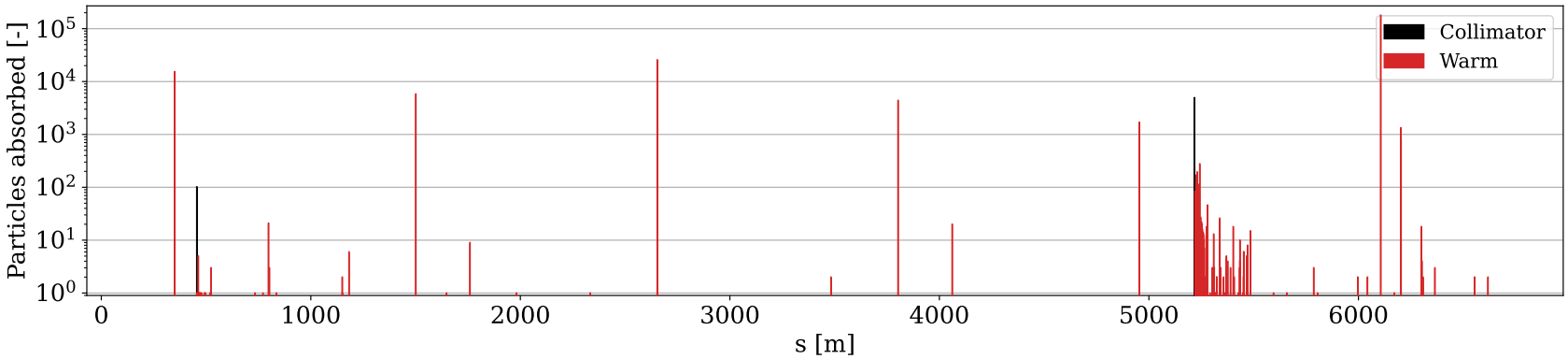
Horizontal crabs — Off-momentum loss maps — DPpos — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

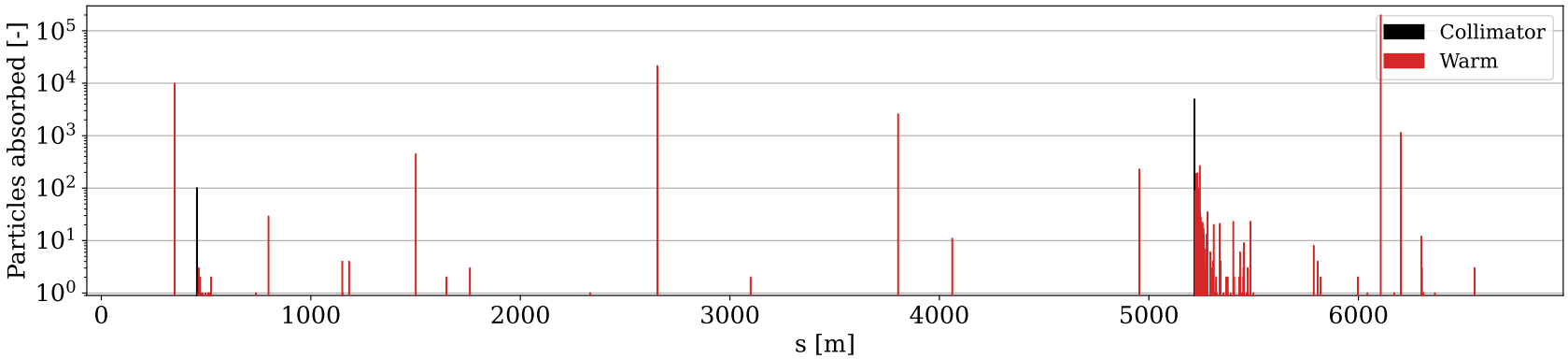
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase $-179°$



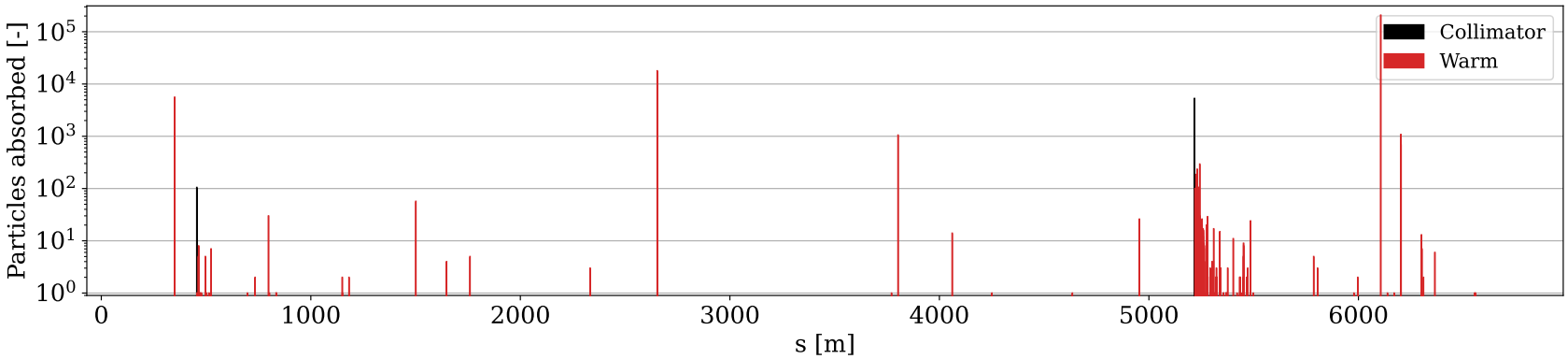
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase $-150°$



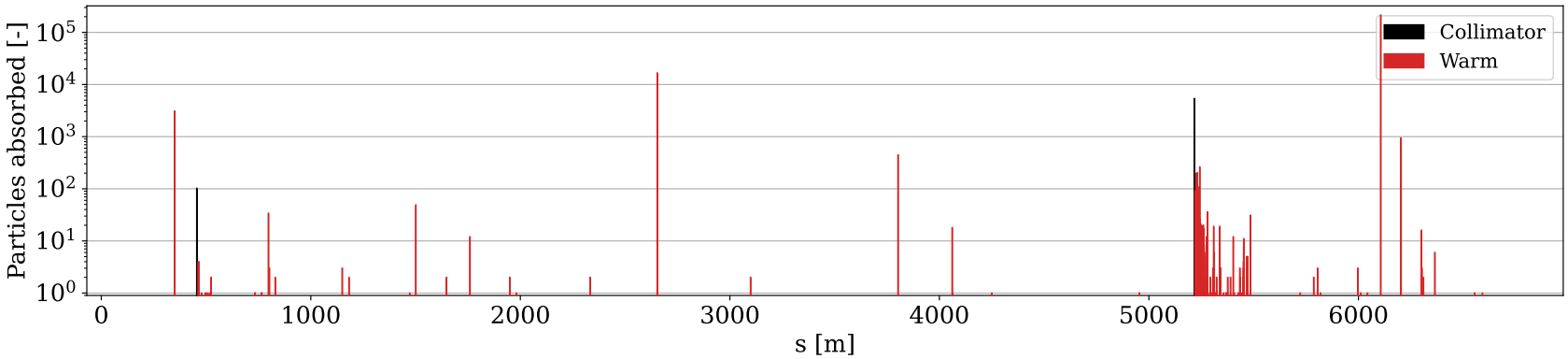
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase $-120°$



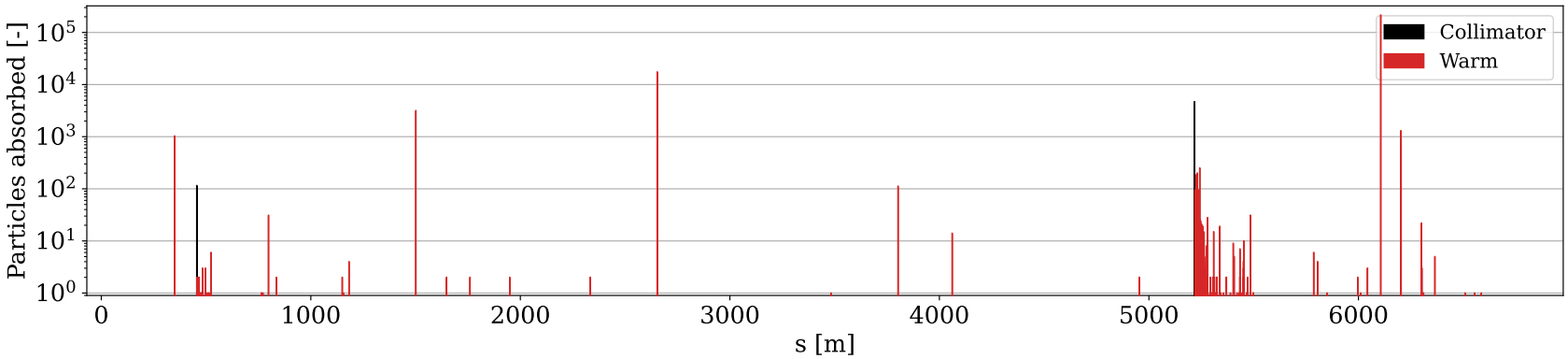
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase −90°



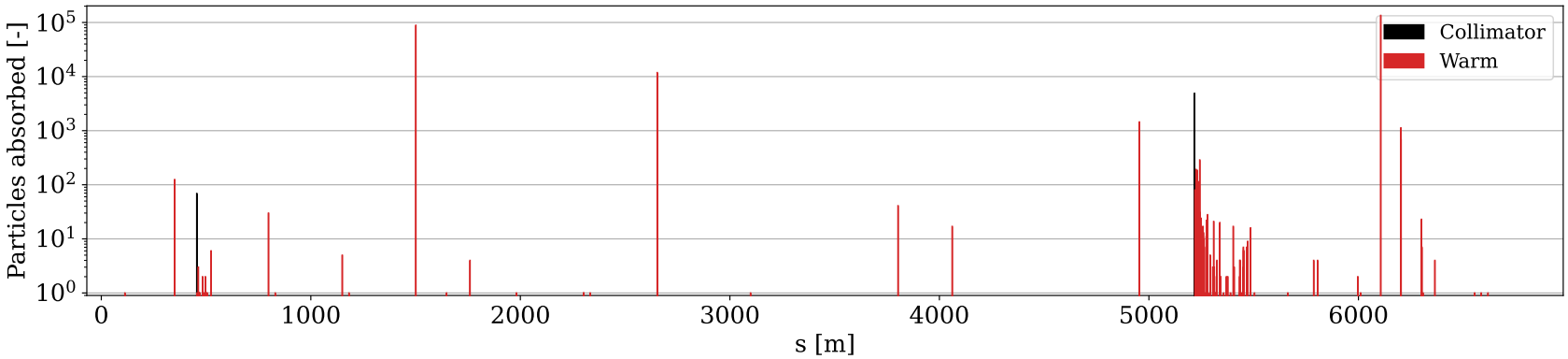
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase $-60°$



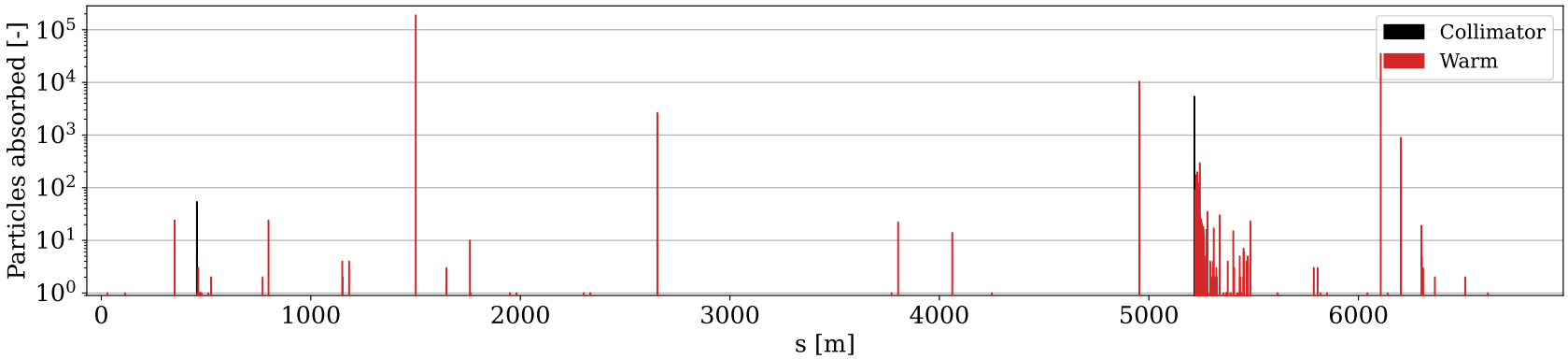
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase $-30°$



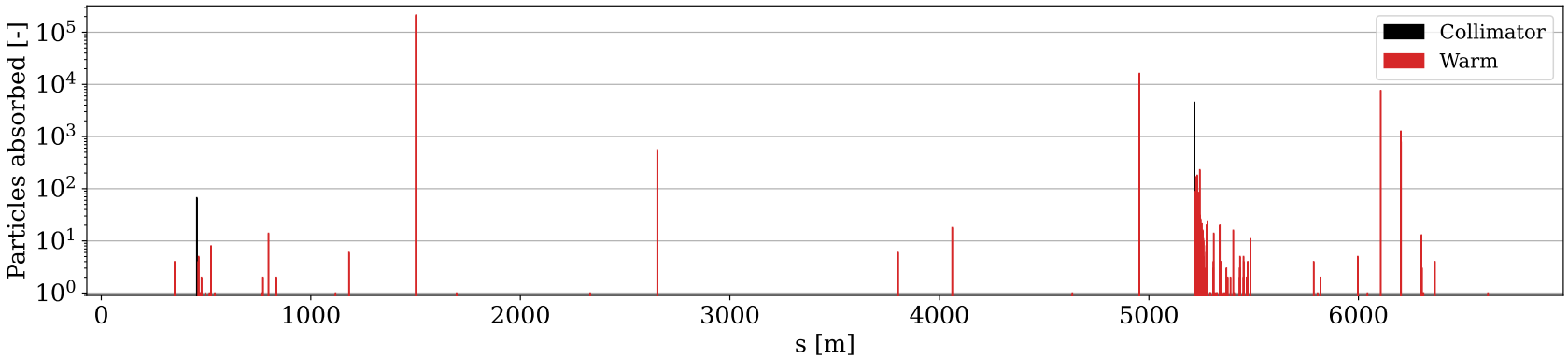
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 0°



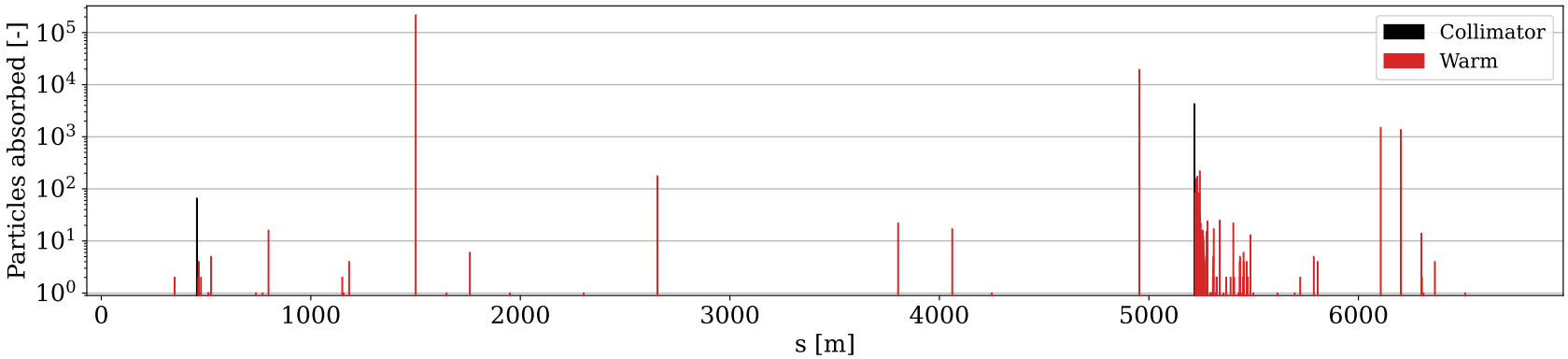
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 30°



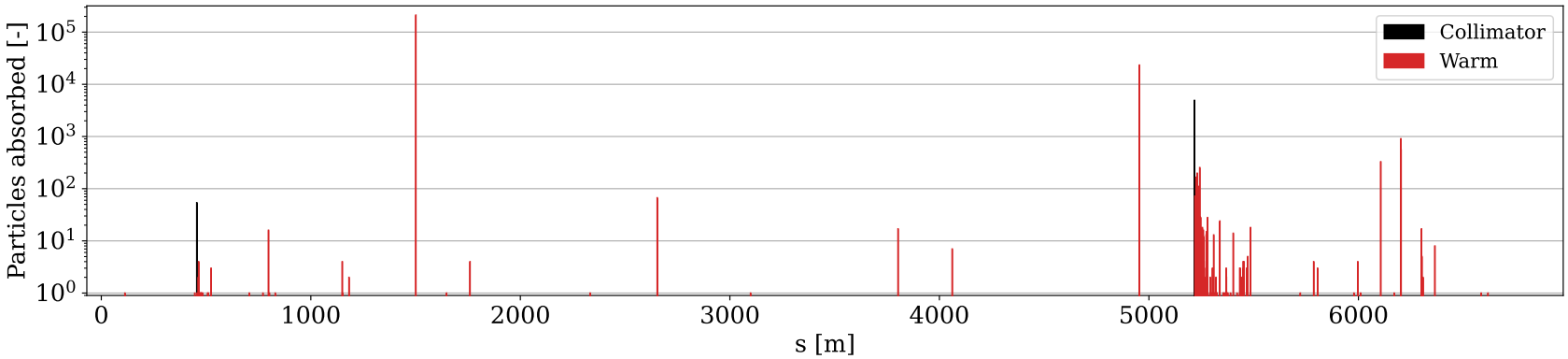
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 60°



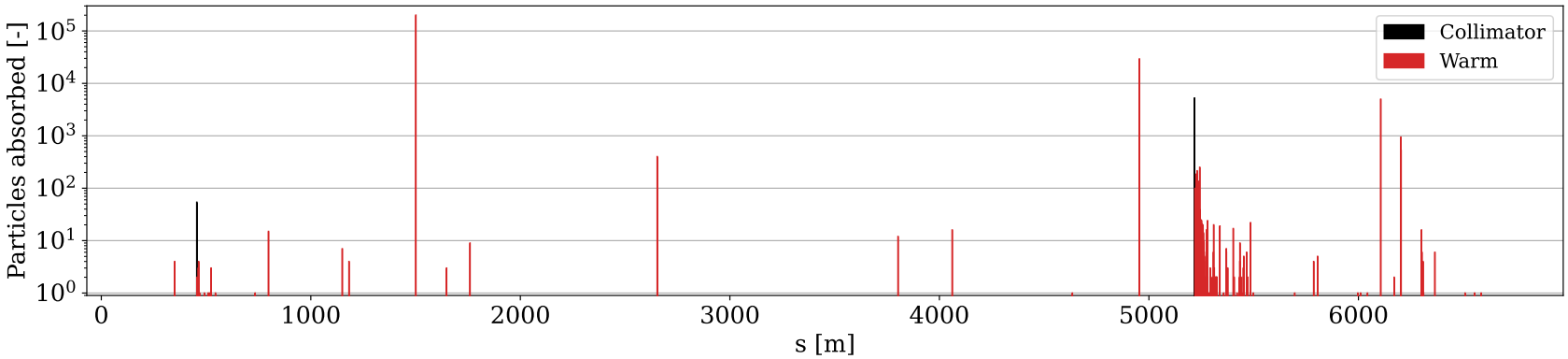
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 90°



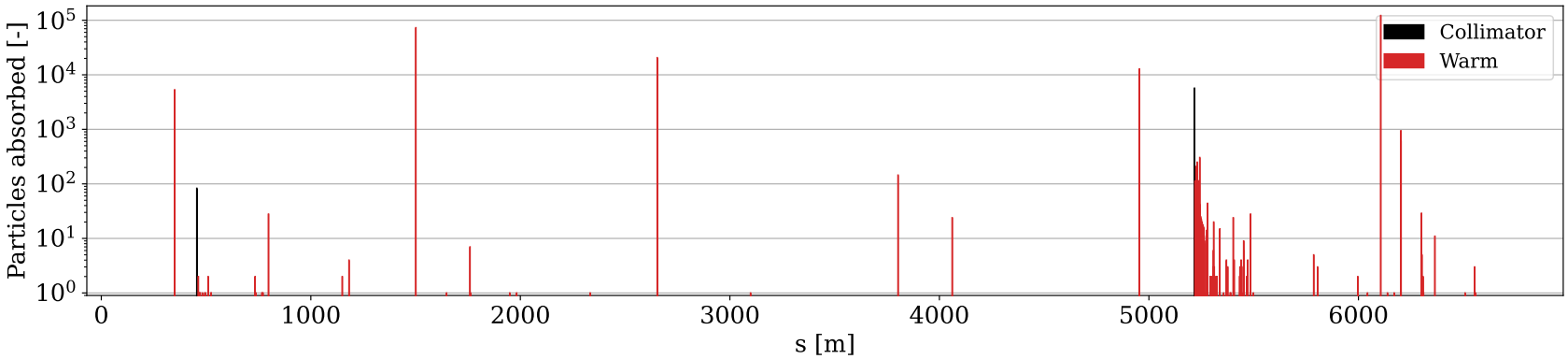
Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 120°



Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 150°



Horizontal crabs — Off-momentum loss maps — DPpos — With errors — Phase 179°



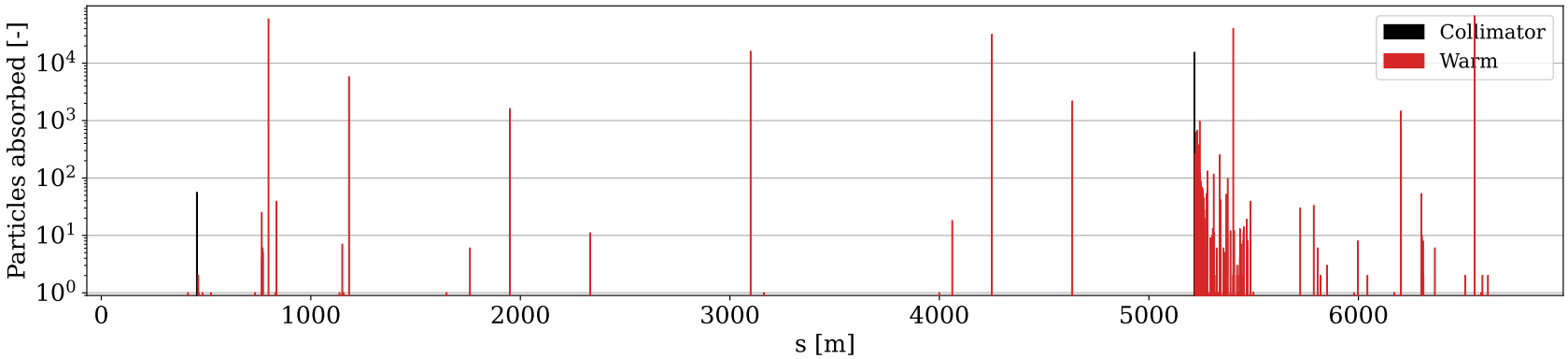
Horizontal crabs — Off-momentum loss maps — DPneg

- Without errors
- With errors

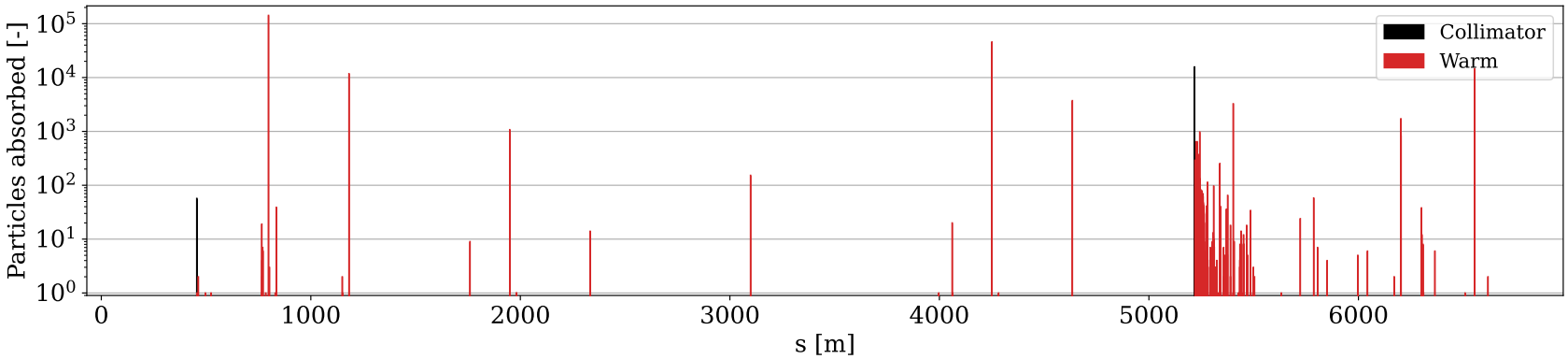
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

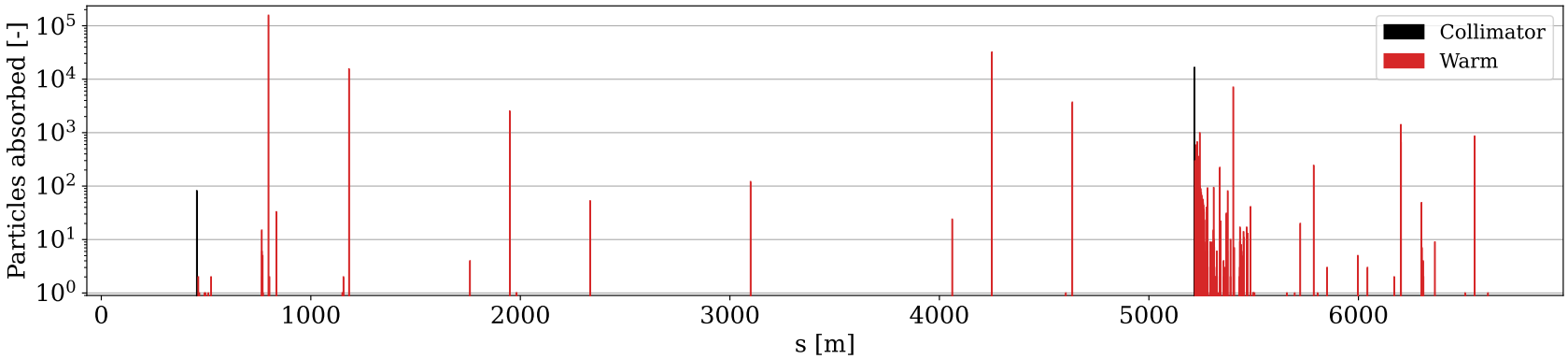
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-179°$



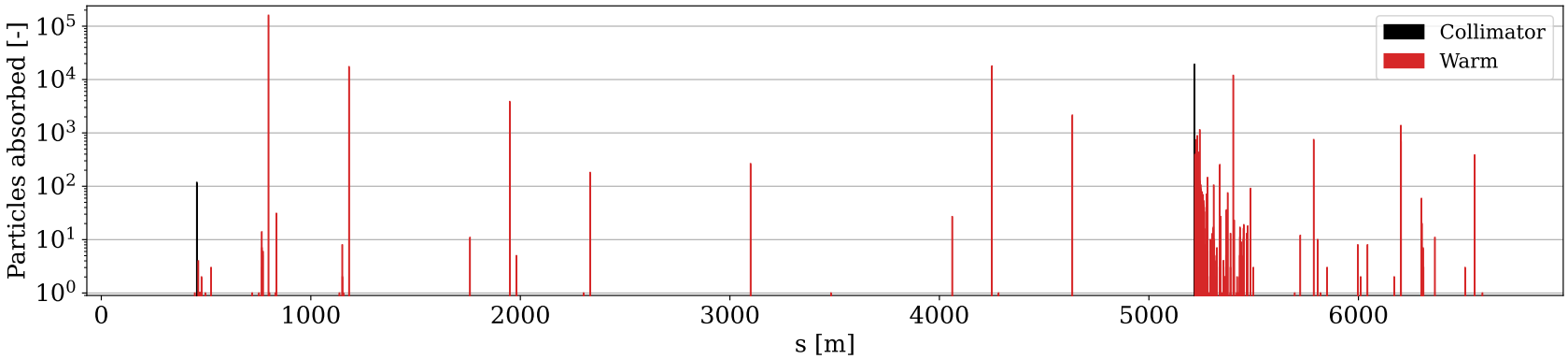
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-150°$



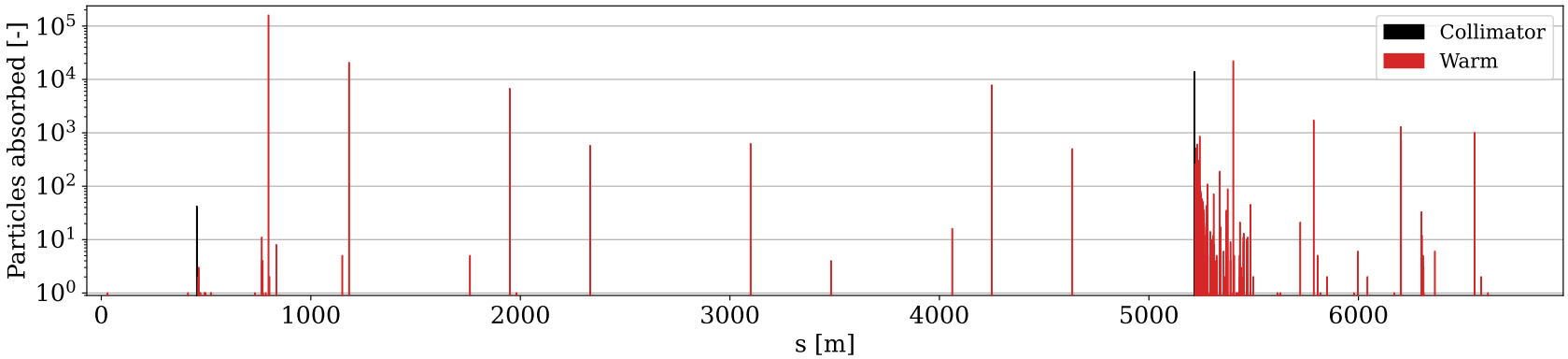
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-120°$



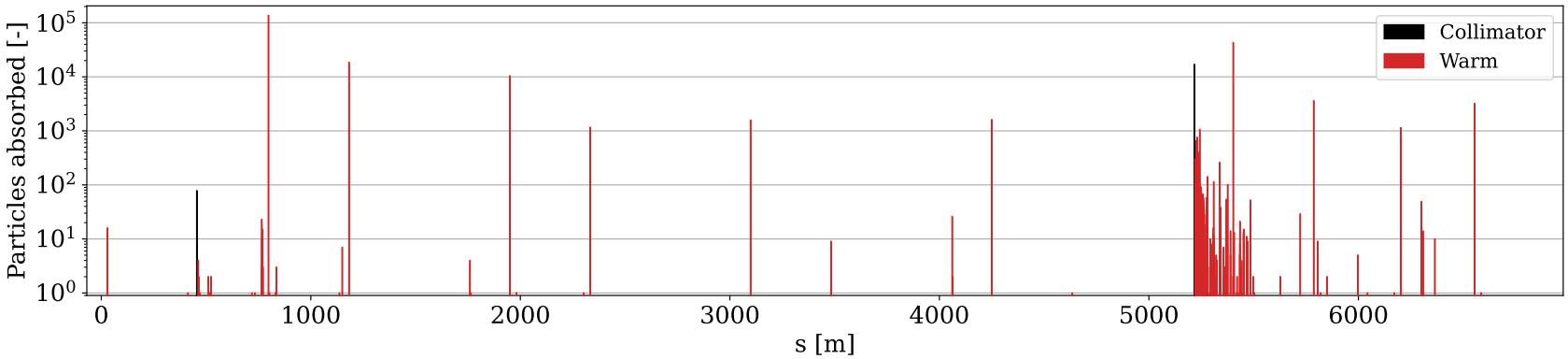
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-90°$



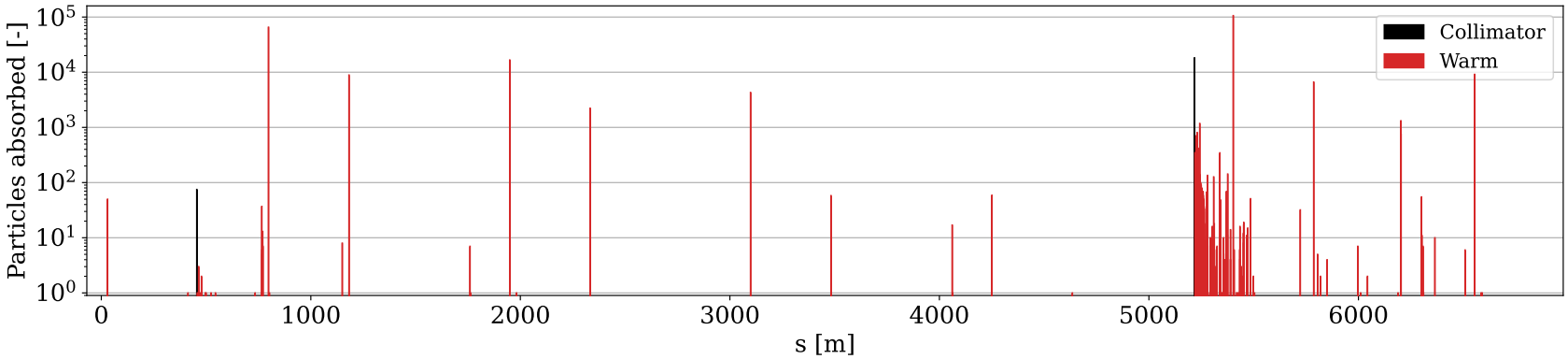
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-60°$



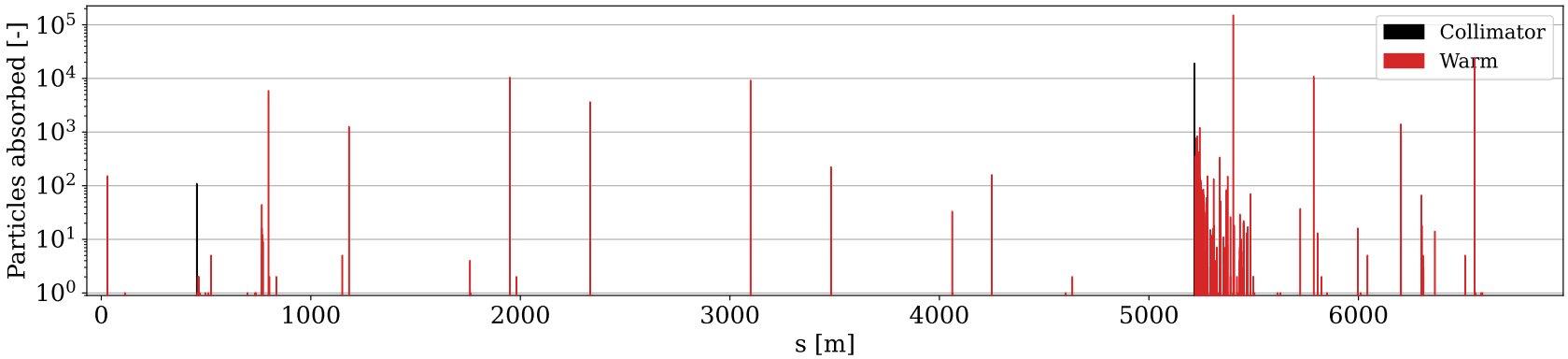
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-30°$



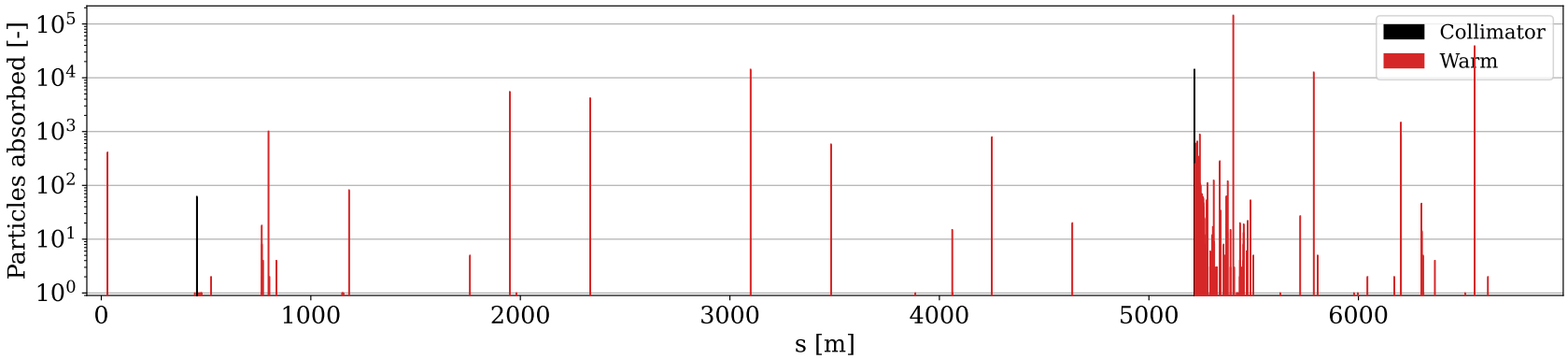
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $0°$



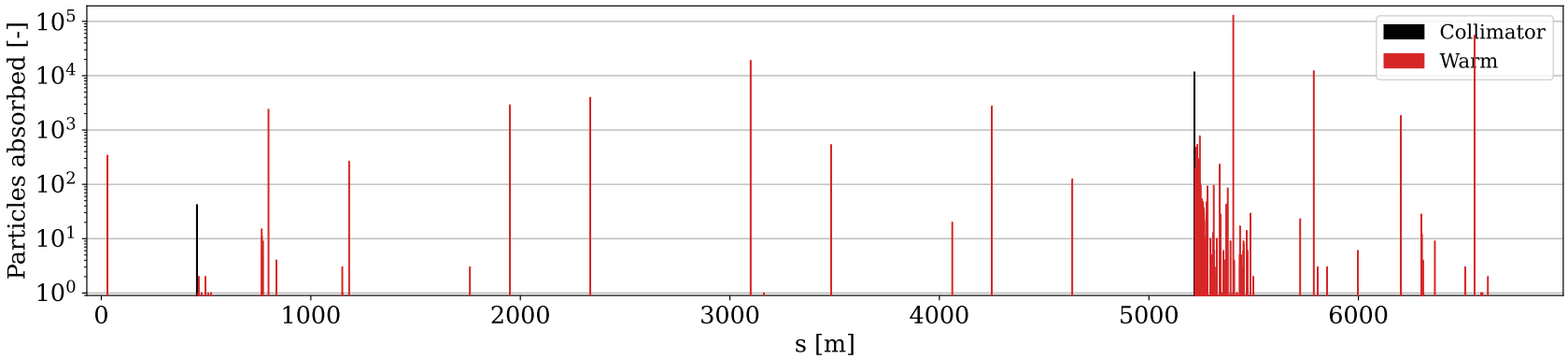
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase 30°



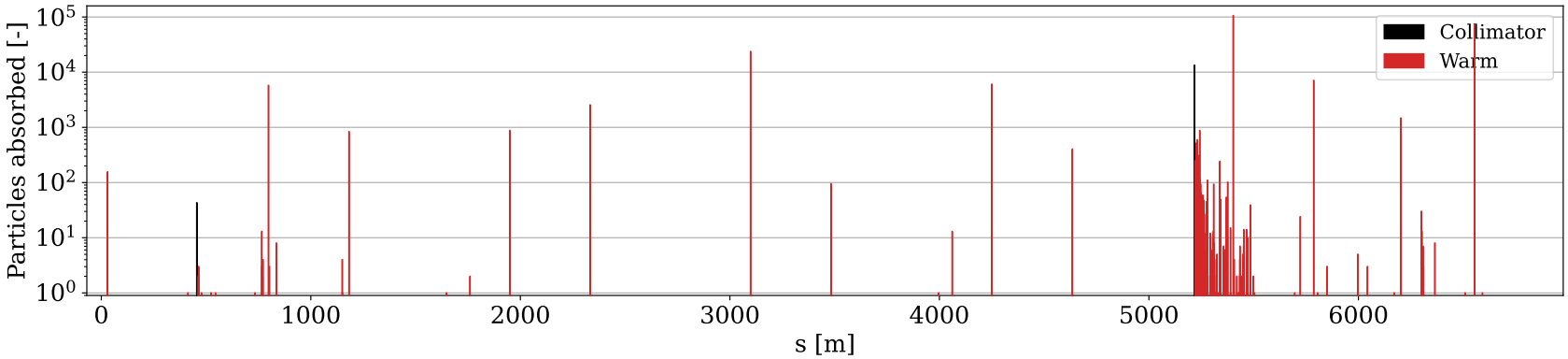
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $60°$



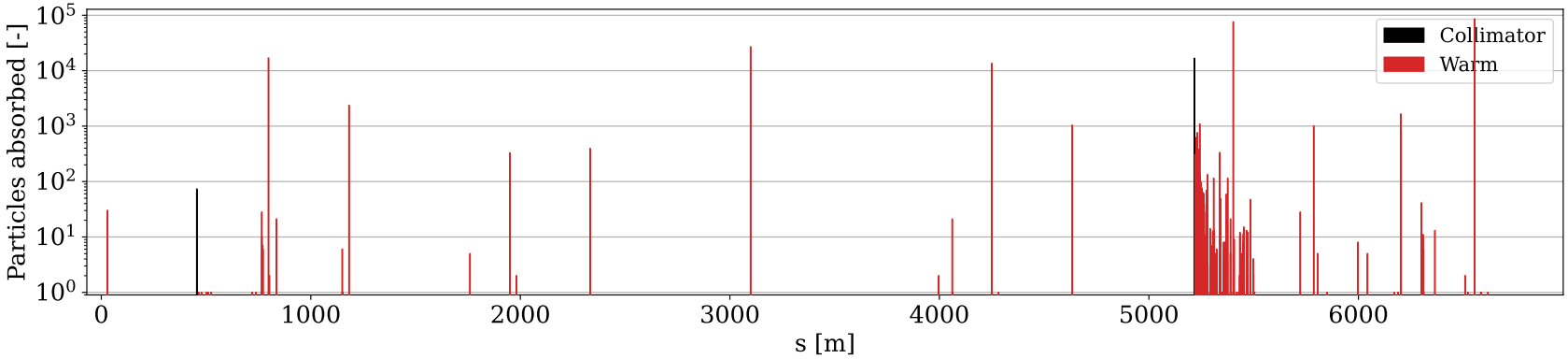
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase 90°



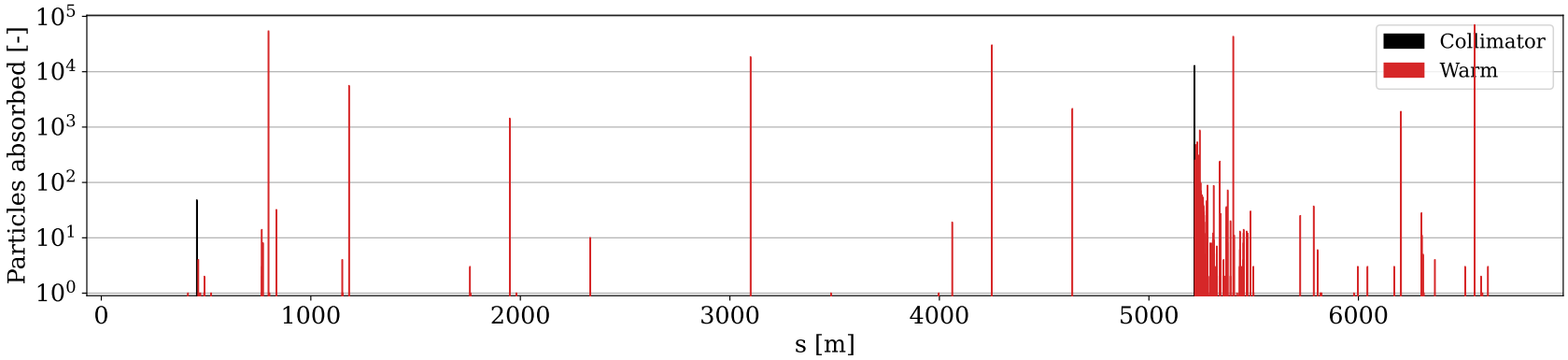
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase 120°



Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase $150°$



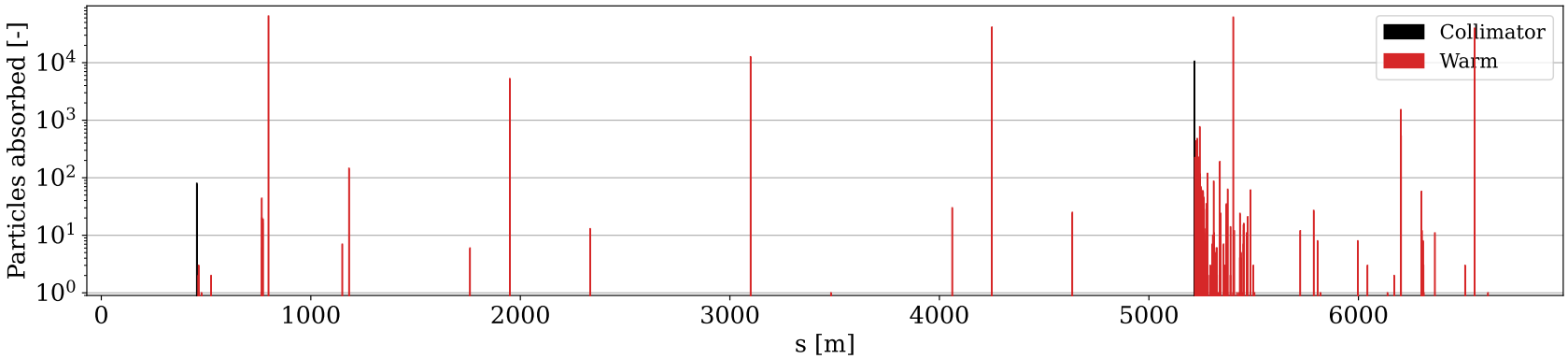
Horizontal crabs — Off-momentum loss maps — DPneg — Without errors — Phase 179°



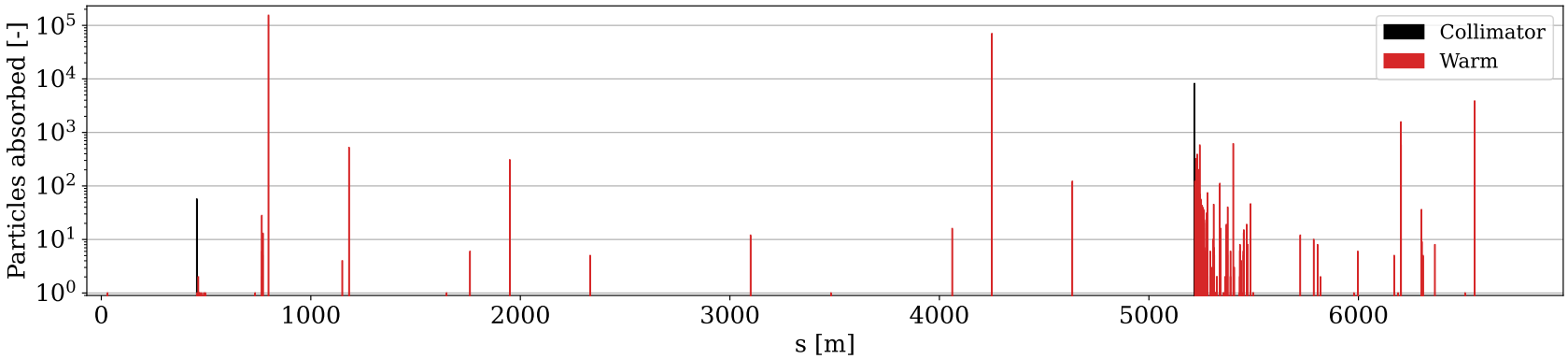
Horizontal crabs — Off-momentum loss maps — DPneg — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

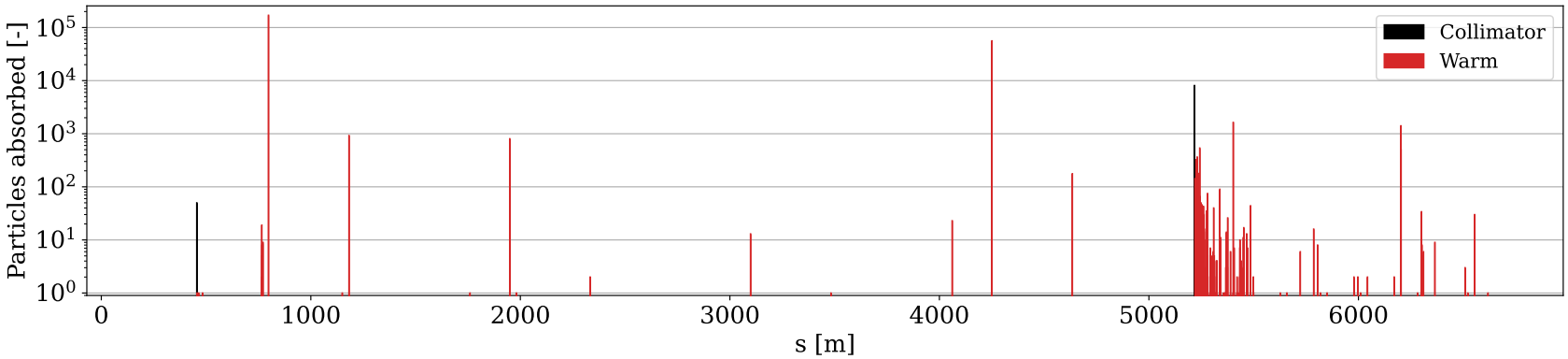
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase −179°



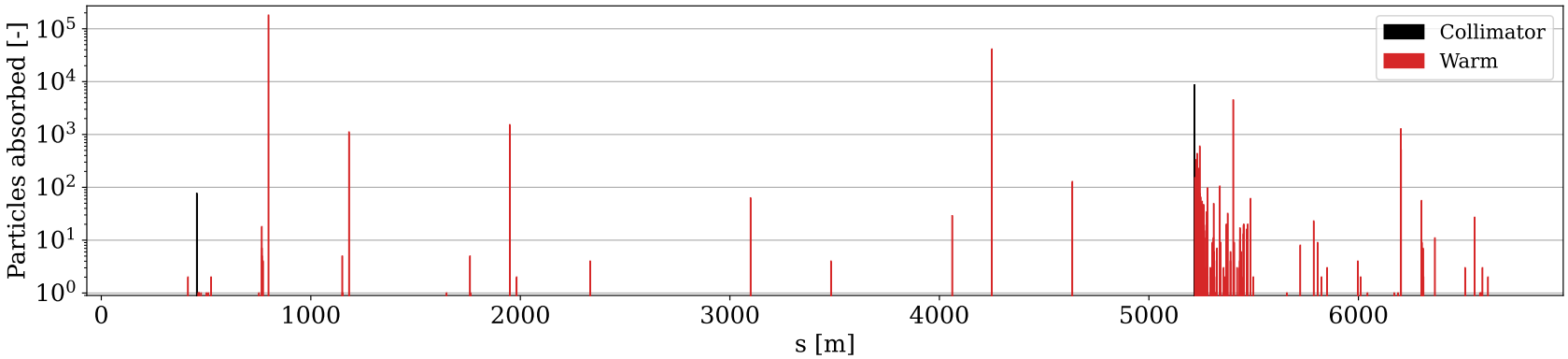
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase $-150°$



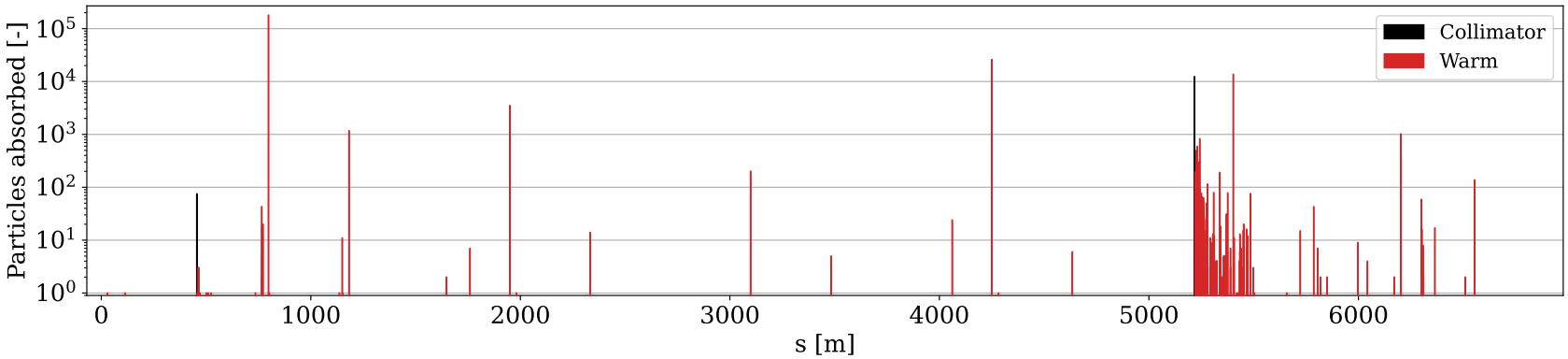
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase $-120°$



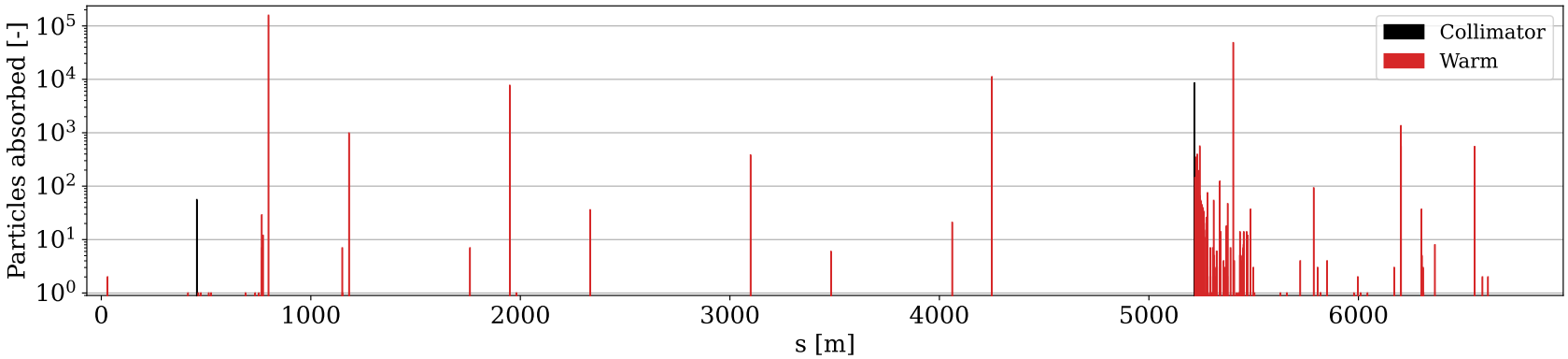
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase −90°



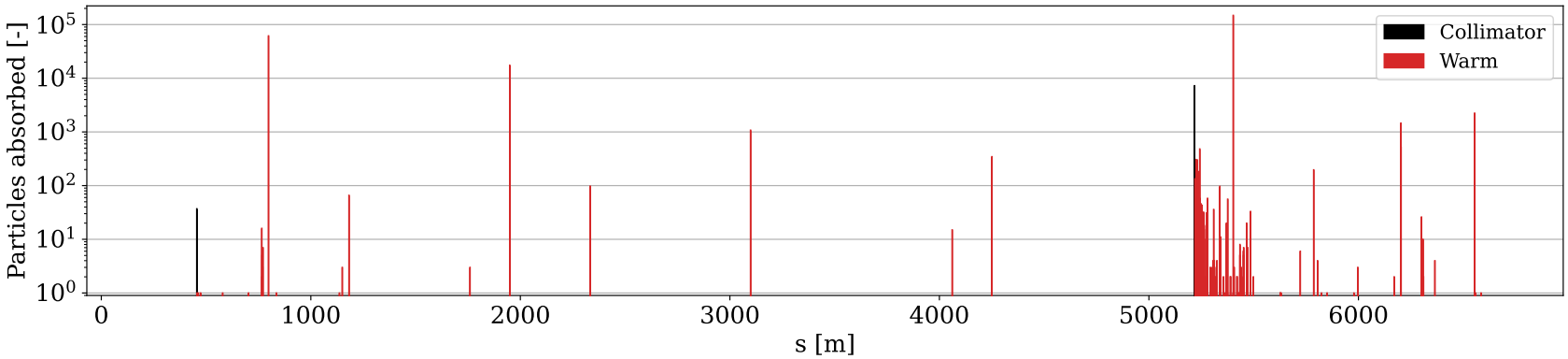
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase −60°



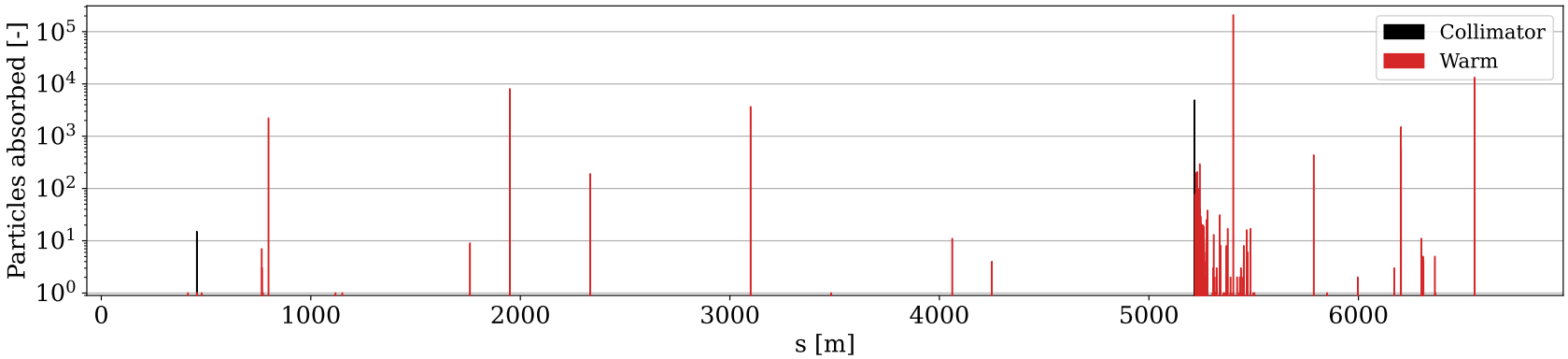
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase $-30°$



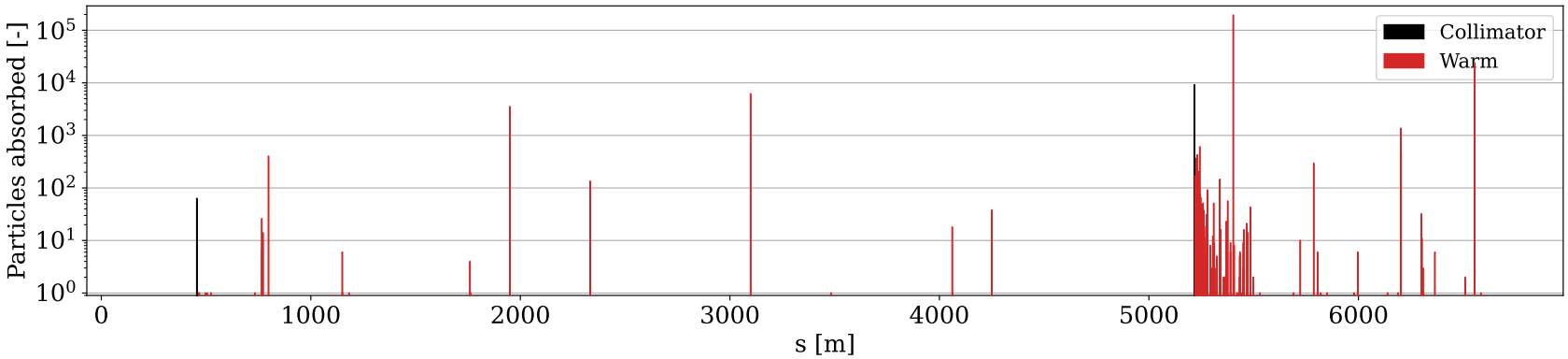
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 0°



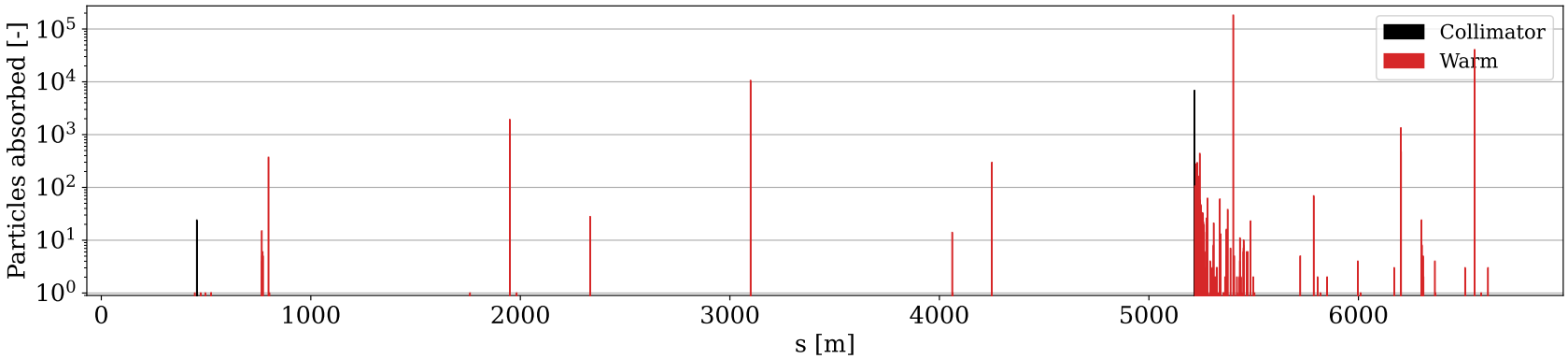
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 30°



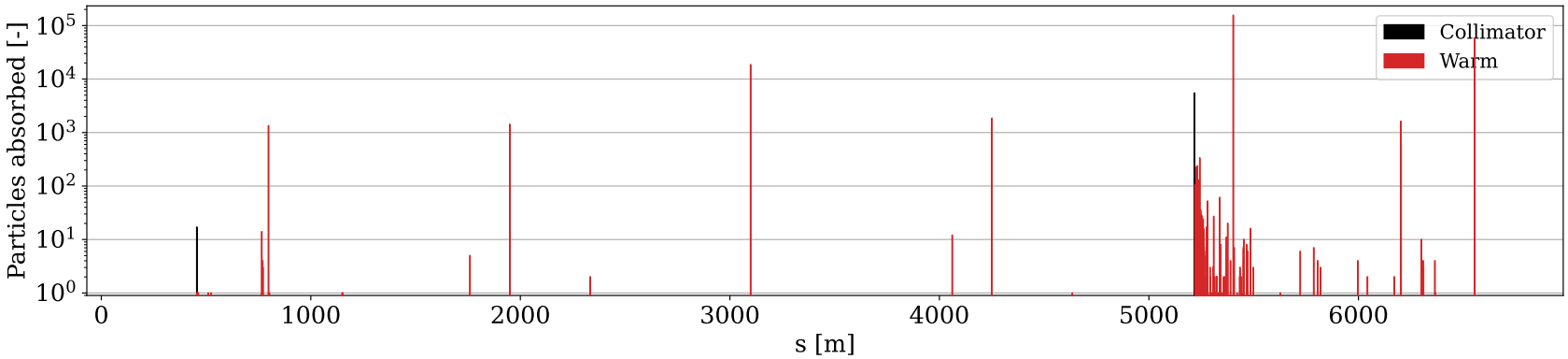
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 60°



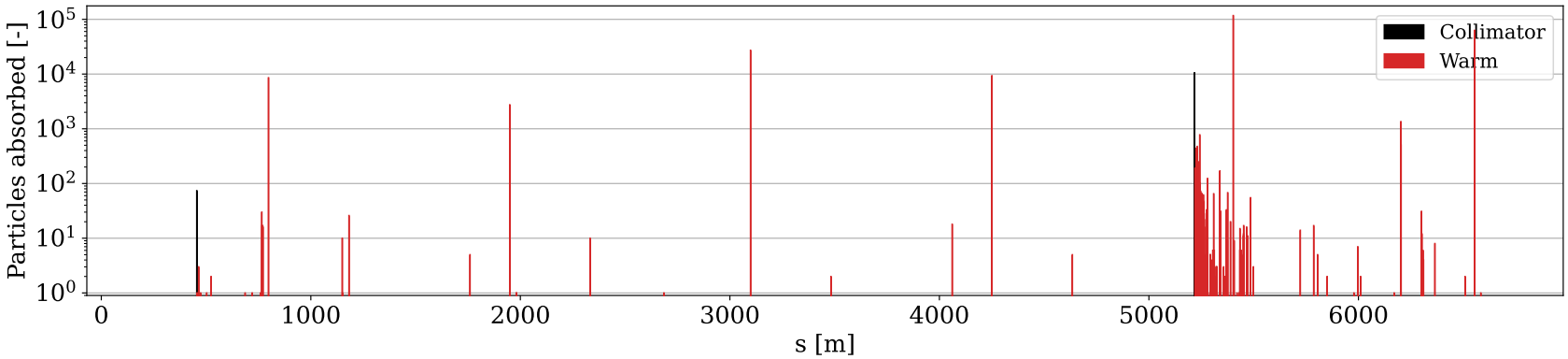
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 90°



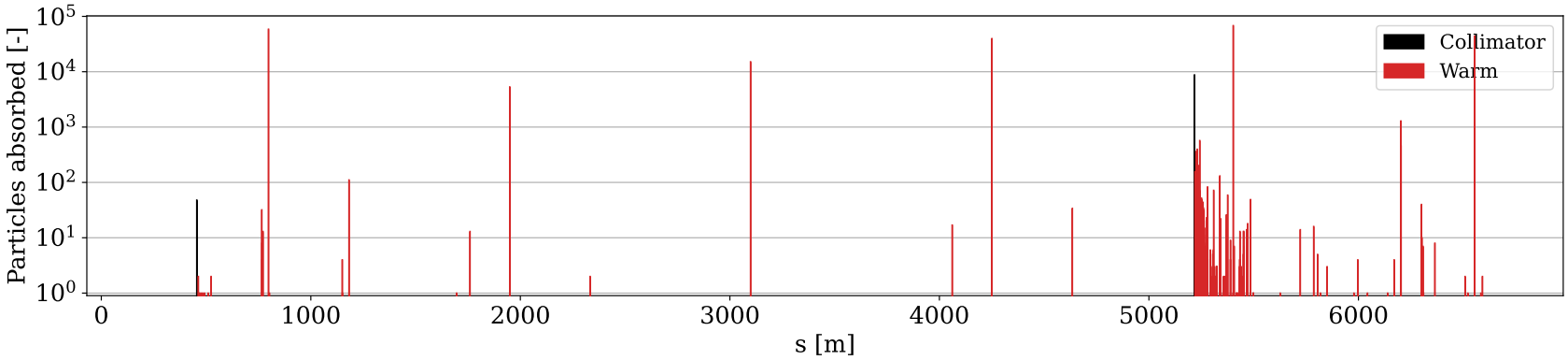
Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 120°



Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 150°



Horizontal crabs — Off-momentum loss maps — DPneg — With errors — Phase 179°



Outline

1. Overview
2. Horizontal crabs
   - Betatron loss maps
   - Off-momentum loss maps
3. Vertical crabs
   - Betatron loss maps
   - Off-momentum loss maps

Outline

1. Overview
2. Horizontal crabs
   - Betatron loss maps
   - Off-momentum loss maps
3. Vertical crabs
   - Betatron loss maps
   - Off-momentum loss maps

Vertical crabs — Betatron loss maps

- Horizontal
- Vertical

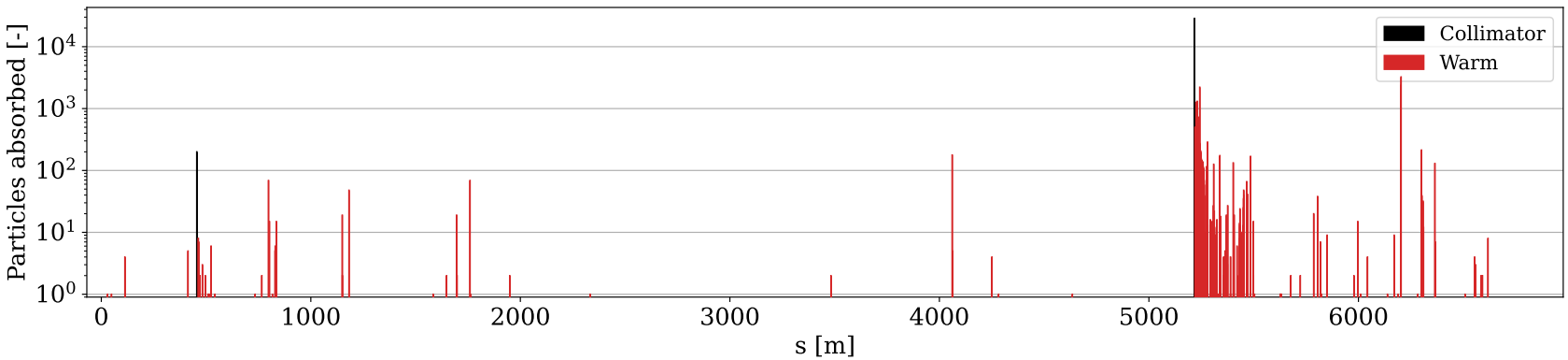
Vertical crabs — Betatron loss maps — Horizontal

- Without errors
- With errors

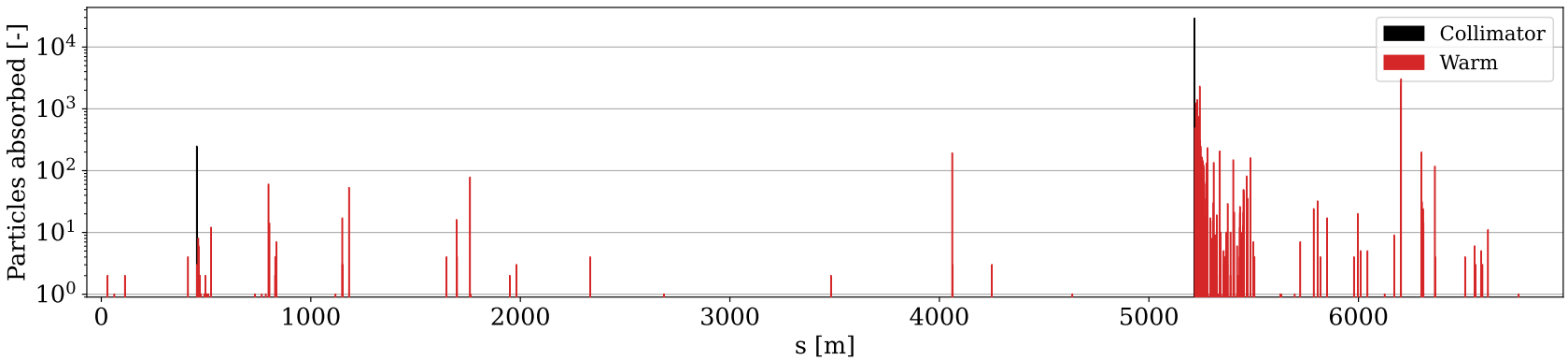
Vertical crabs — Betatron loss maps — Horizontal — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

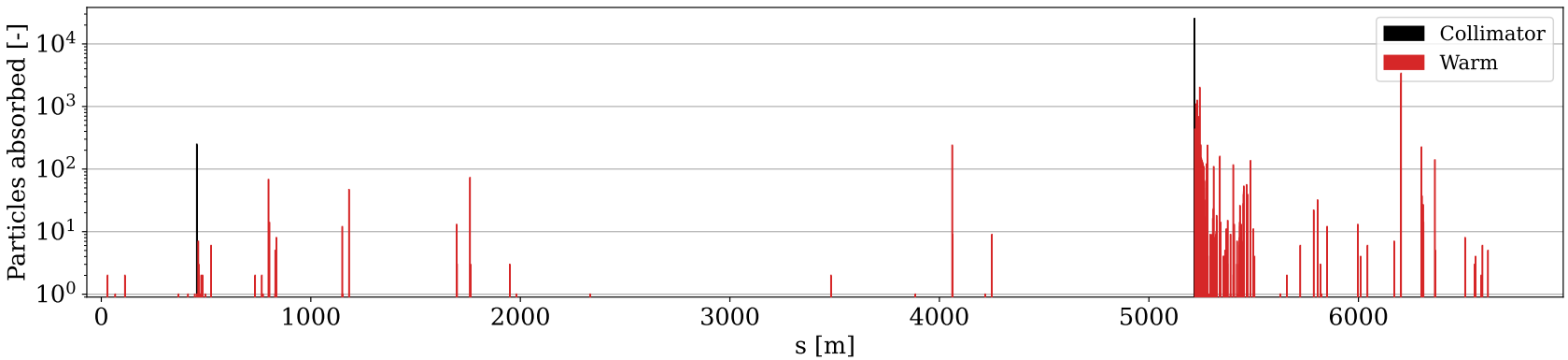
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase −179°



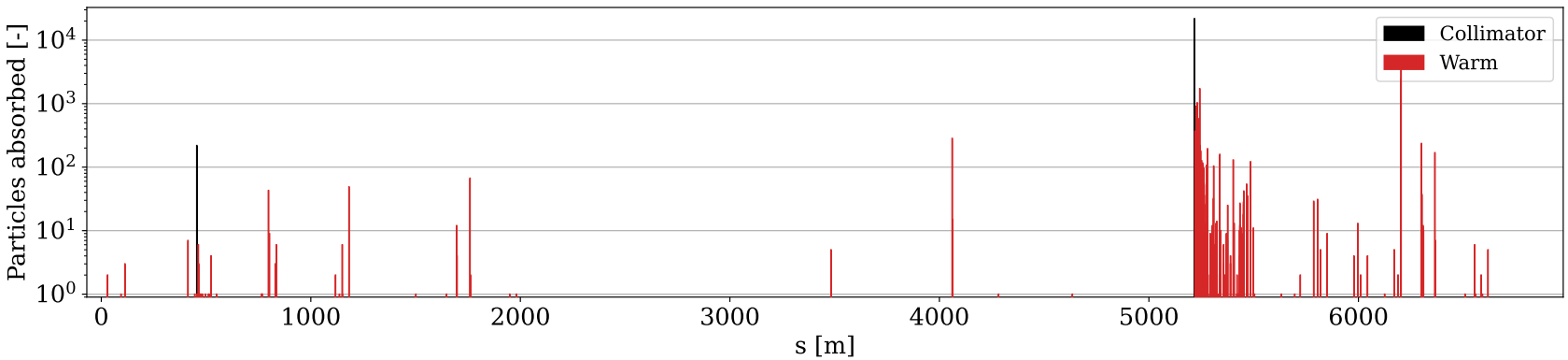
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase $-150°$



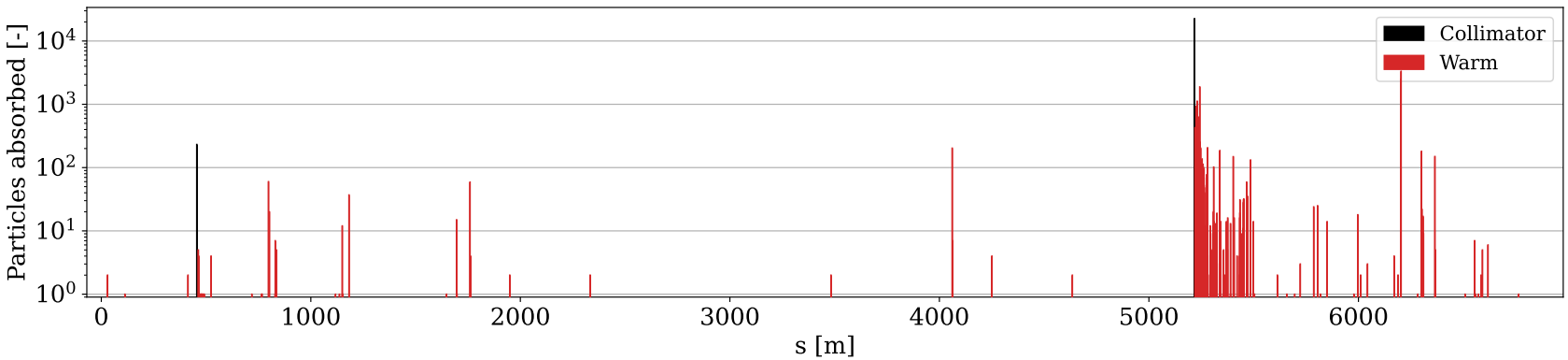
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase −120°



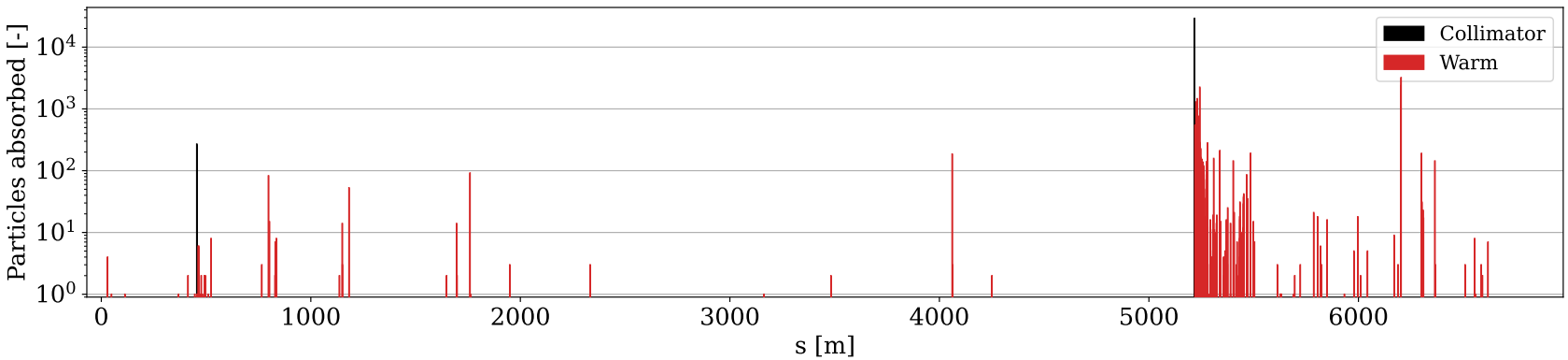
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase $-90°$



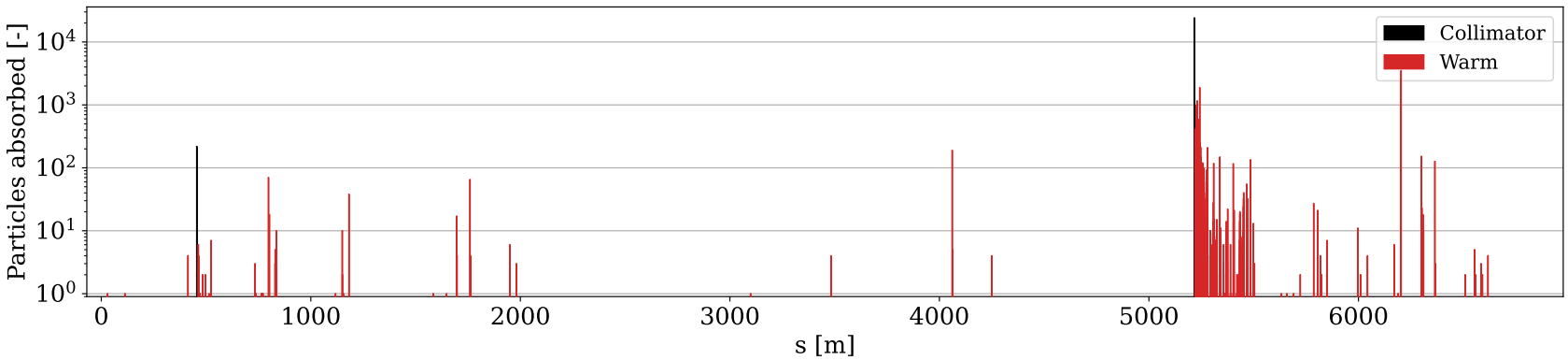
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase −60°



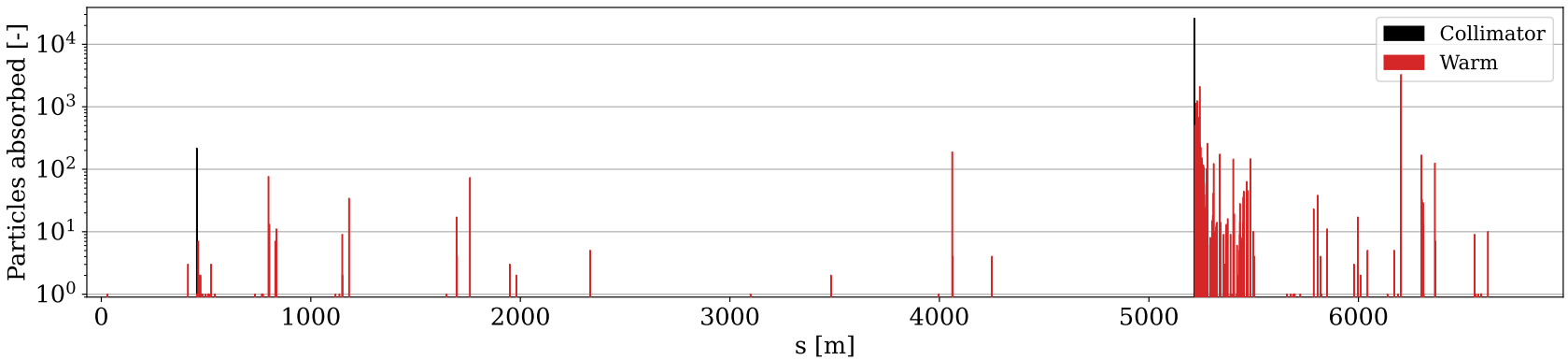
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase $-30°$



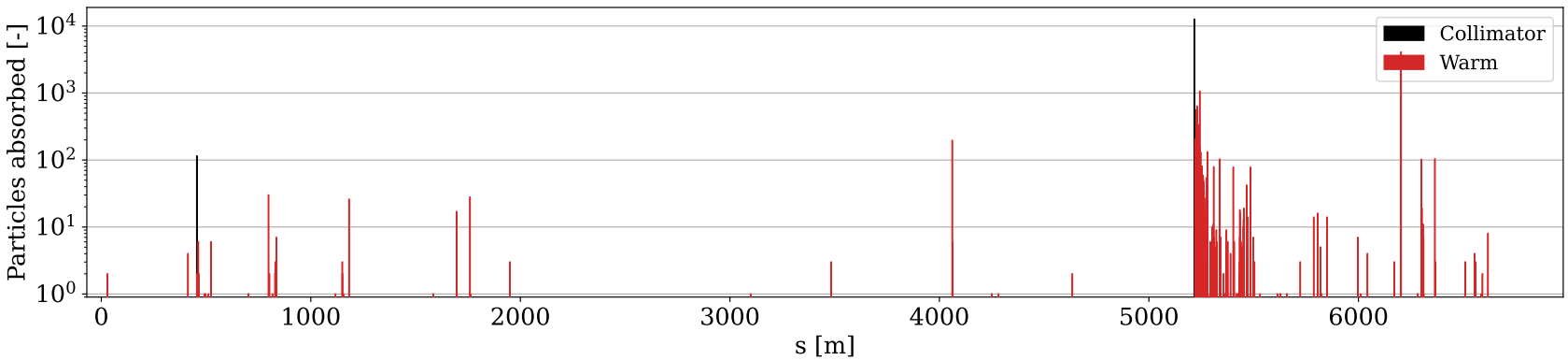
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase $0°$



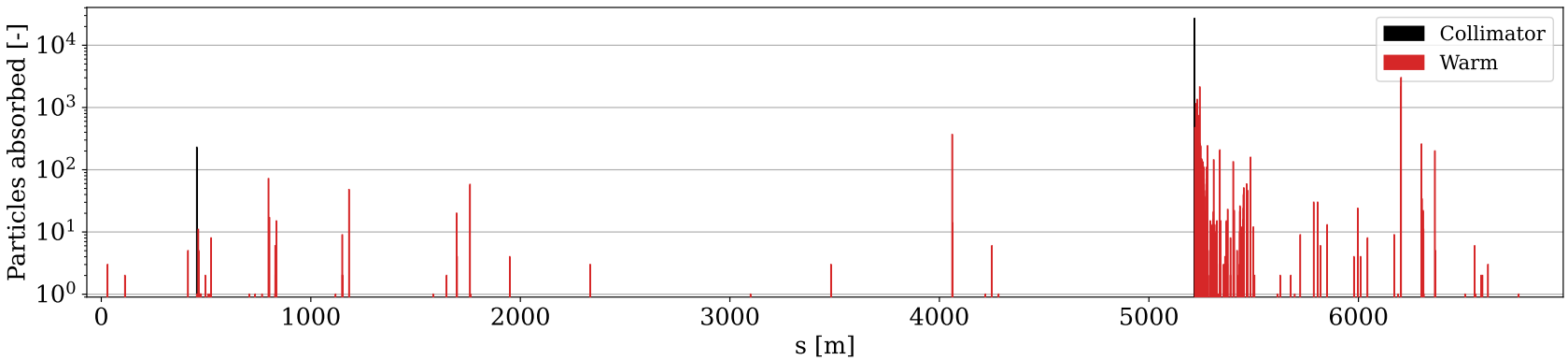
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 30°



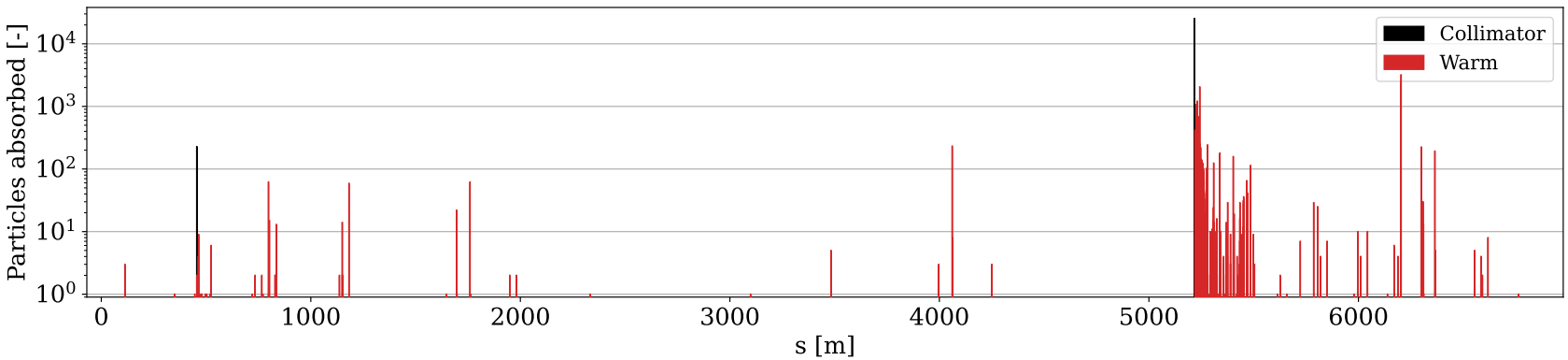
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 60°



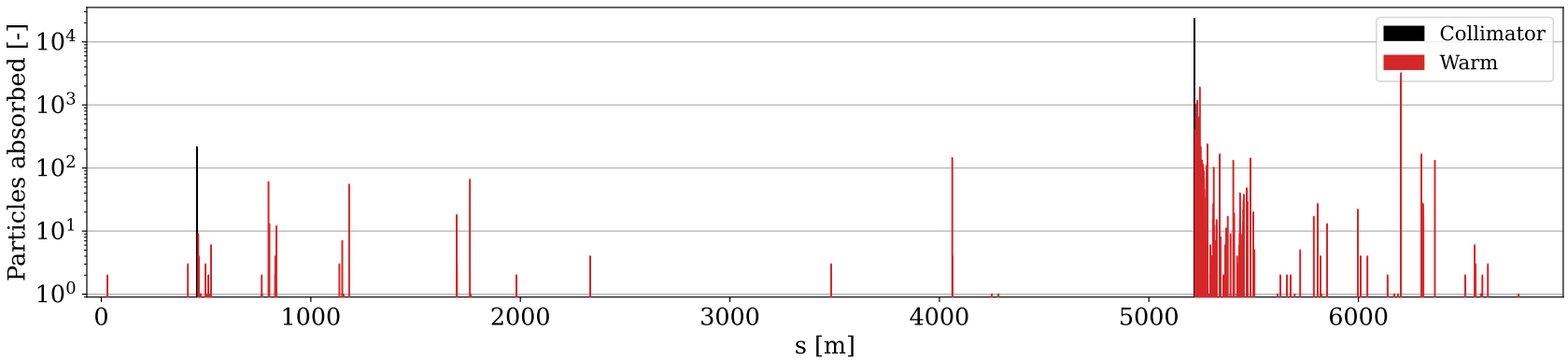
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 90°



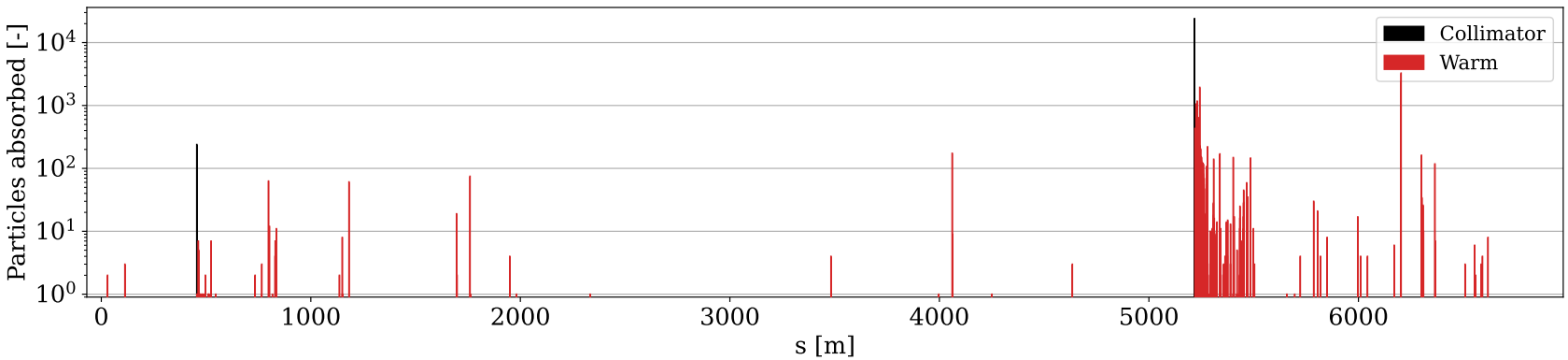
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 120°



Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 150°



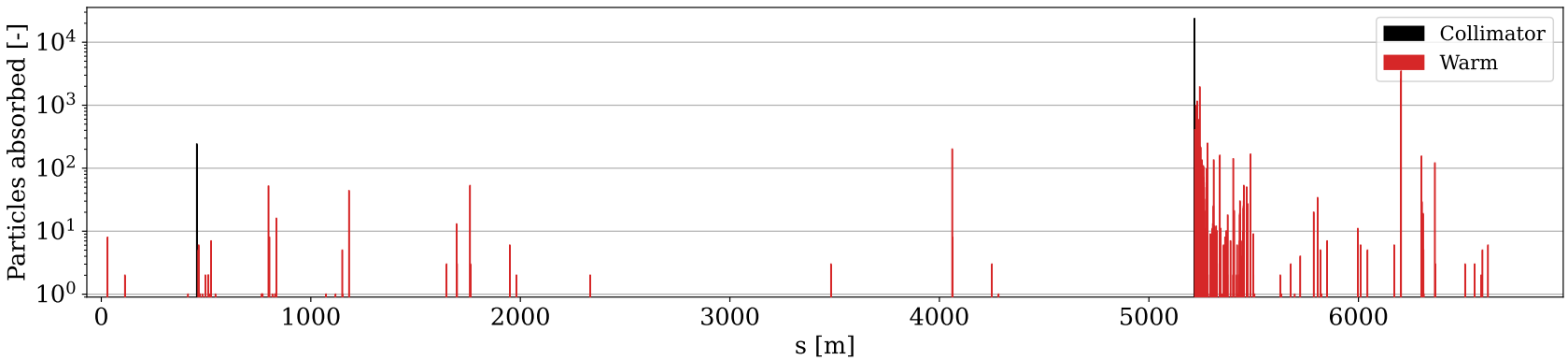
Vertical crabs — Betatron loss maps — Horizontal — Without errors — Phase 179°



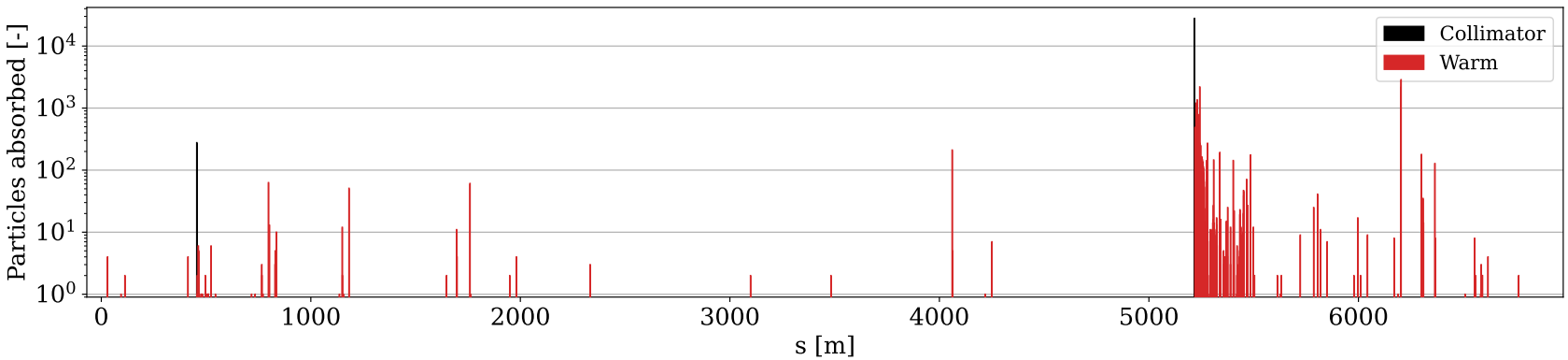
Vertical crabs — Betatron loss maps — Horizontal — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

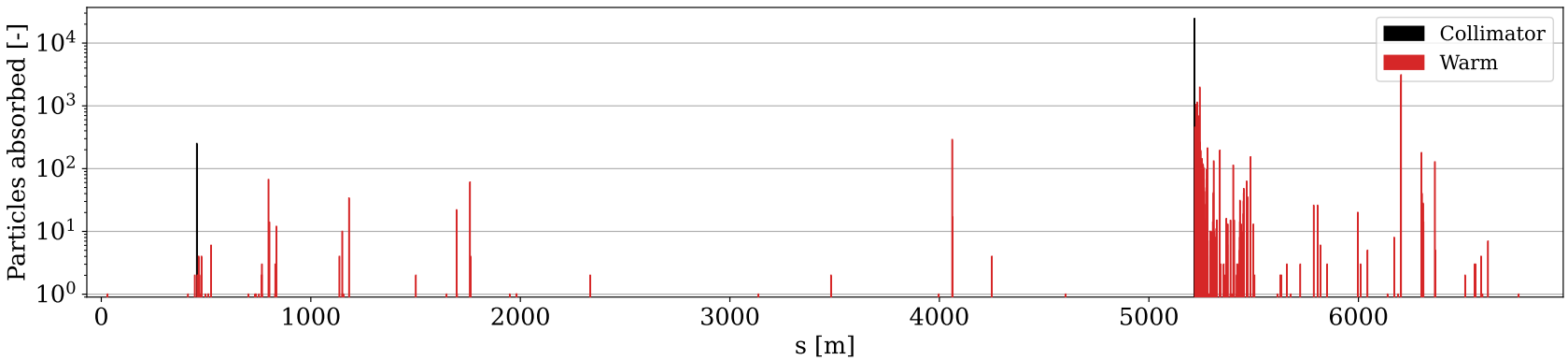
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase −179°



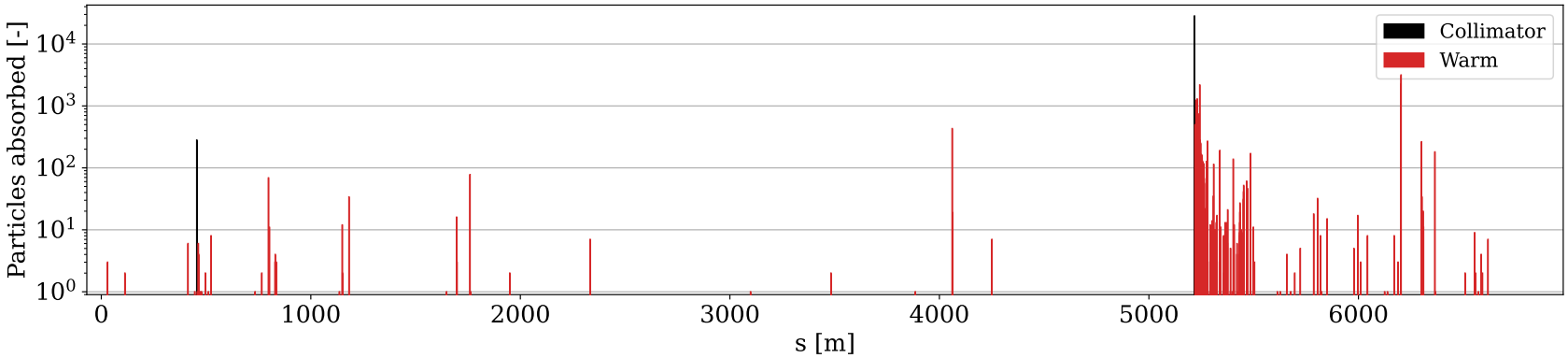
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $-150°$



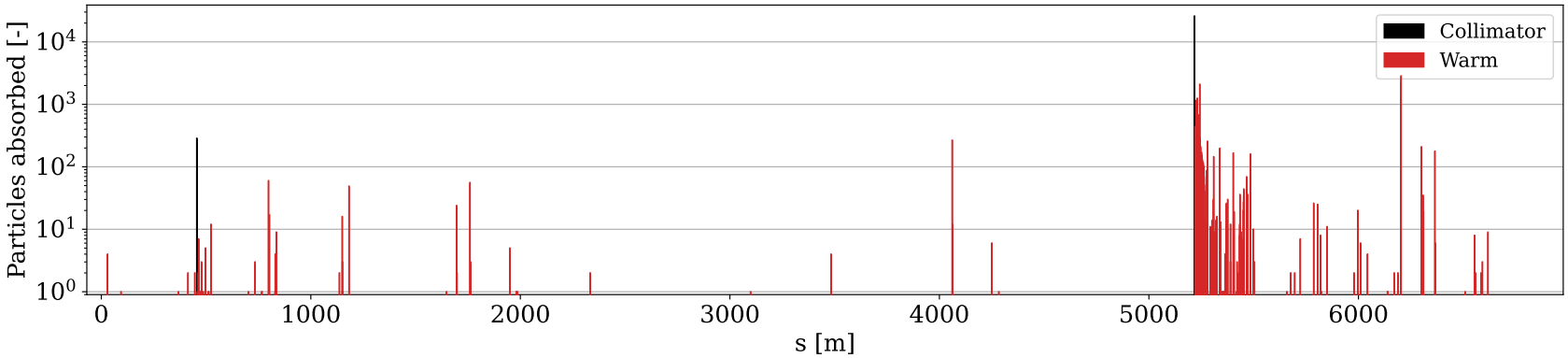
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase −120°



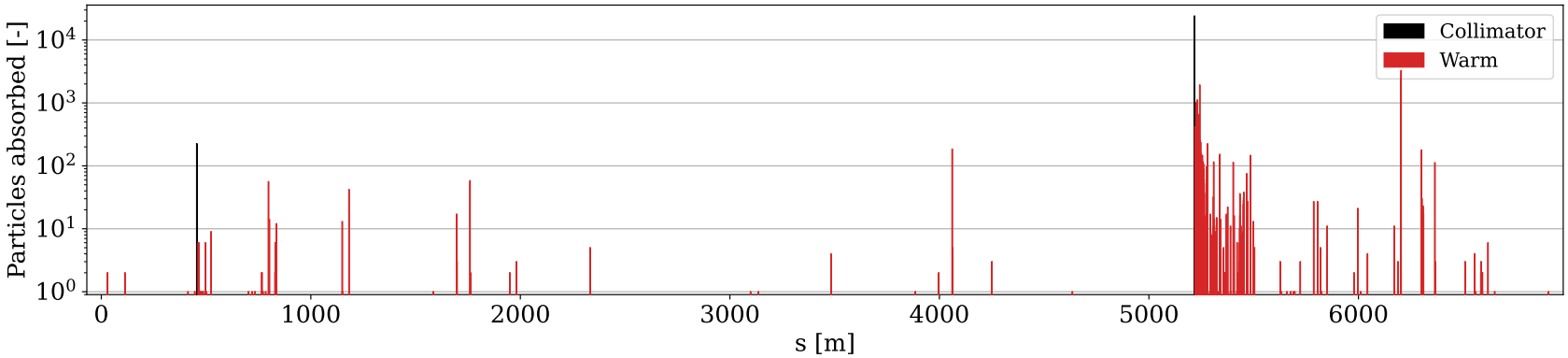
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $-90°$



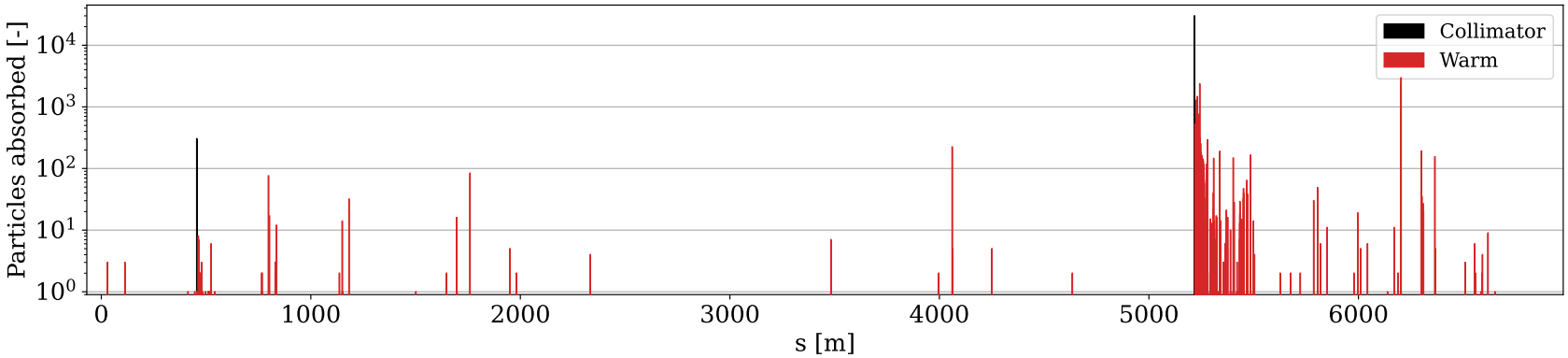
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $-60°$



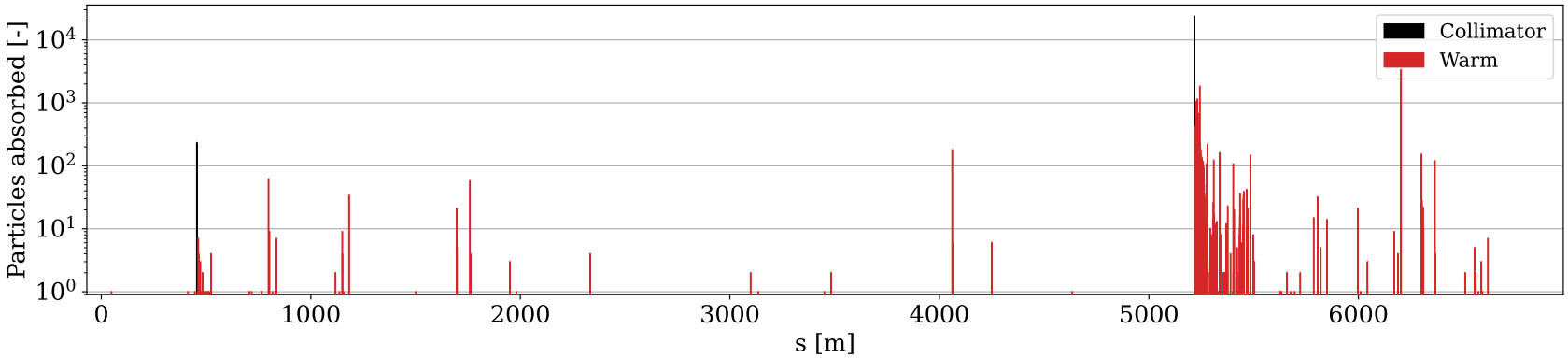
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $-30°$



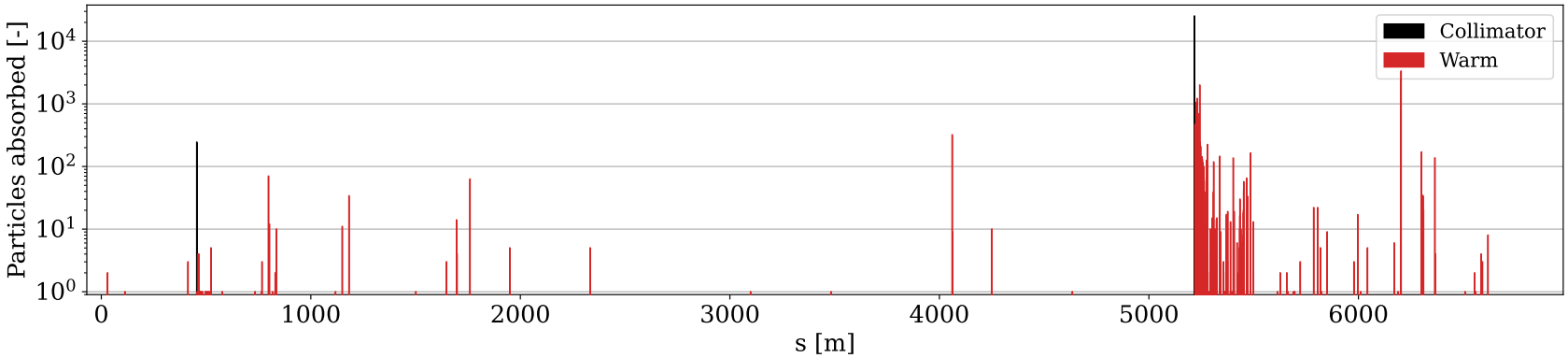
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $0°$



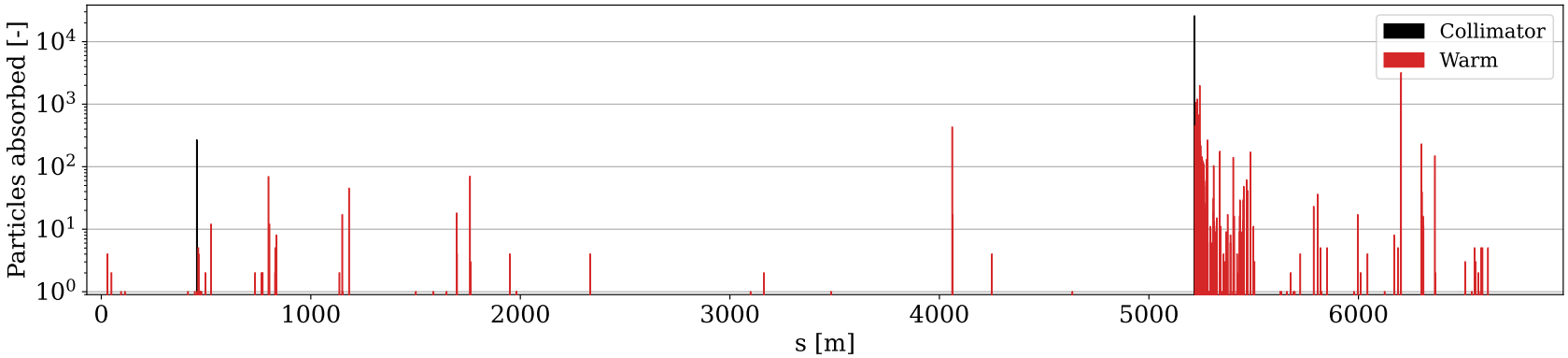
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $30°$



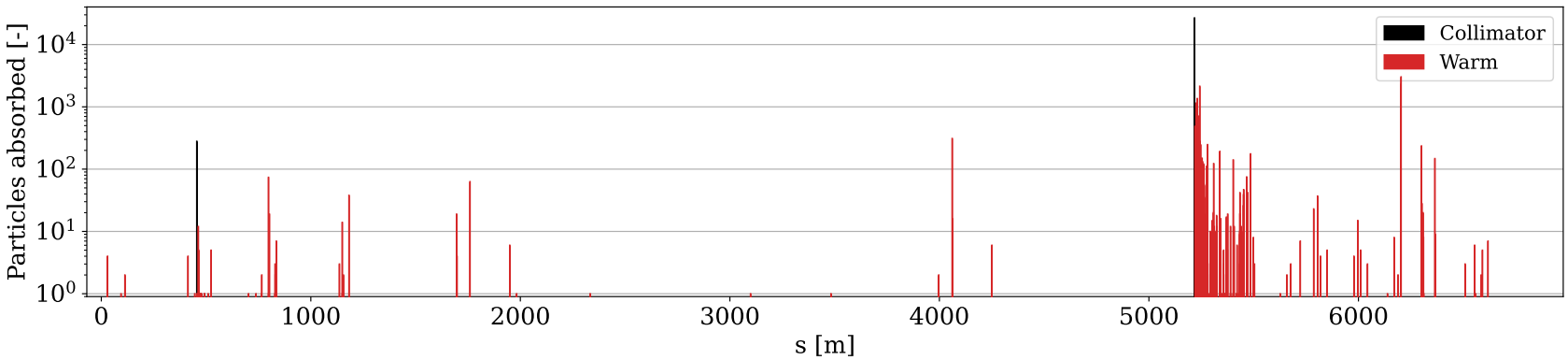
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $60°$



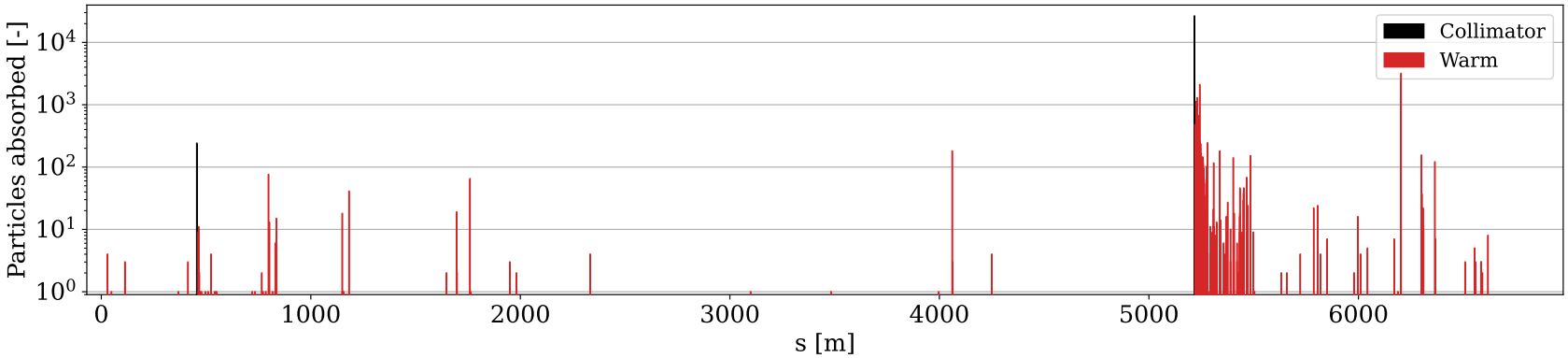
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $90°$



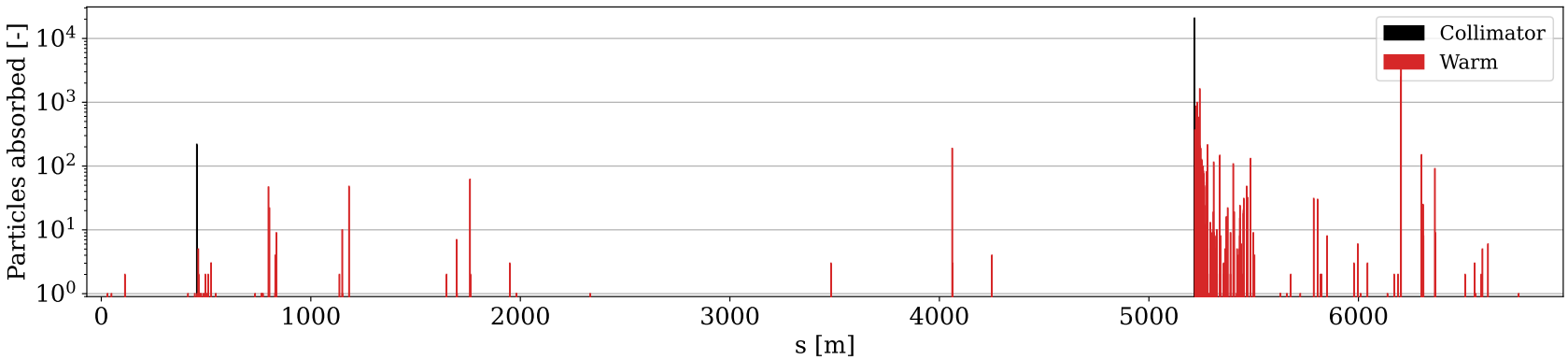
Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $120°$



Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase $150°$



Vertical crabs — Betatron loss maps — Horizontal — With errors — Phase 179°



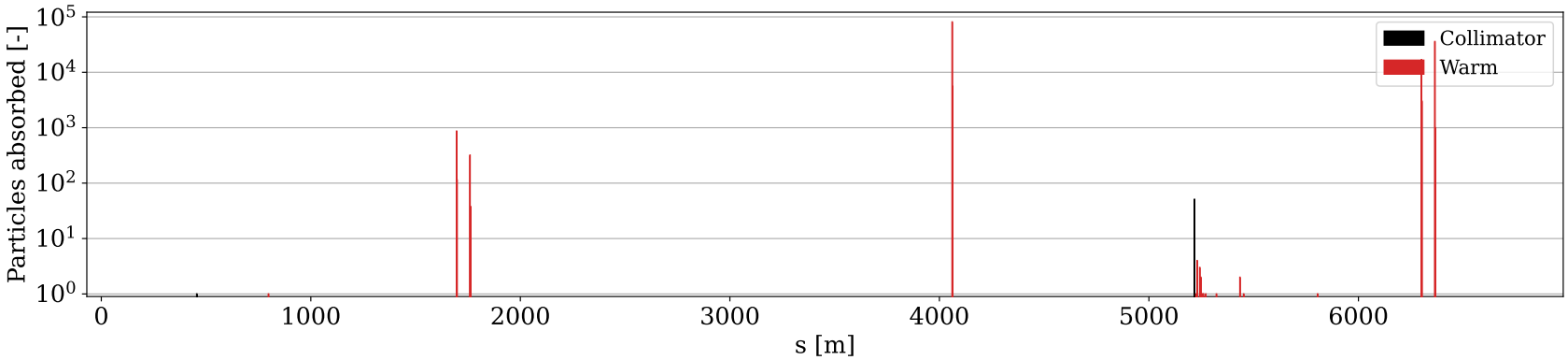
Vertical crabs — Betatron loss maps — Vertical

- Without errors
- With errors

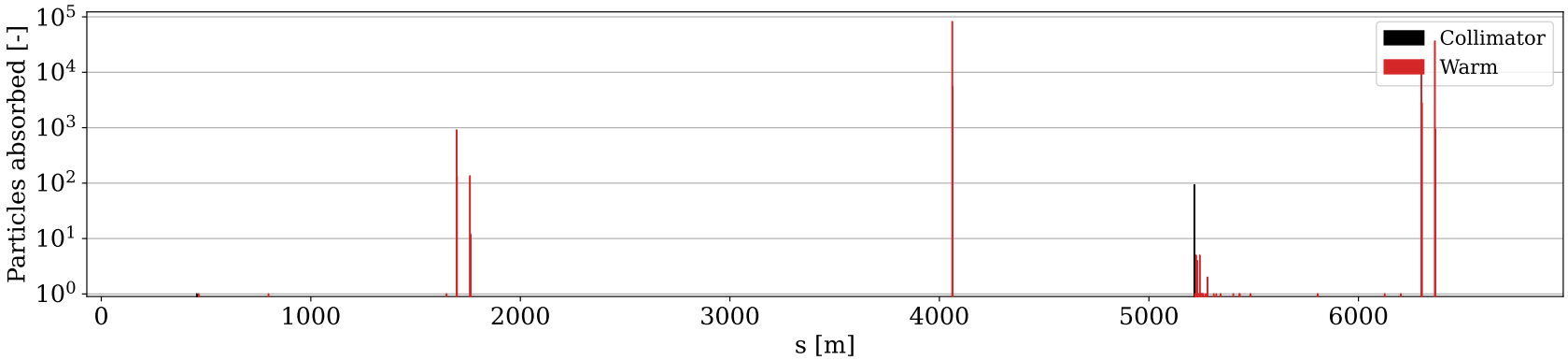
Vertical crabs — Betatron loss maps — Vertical — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

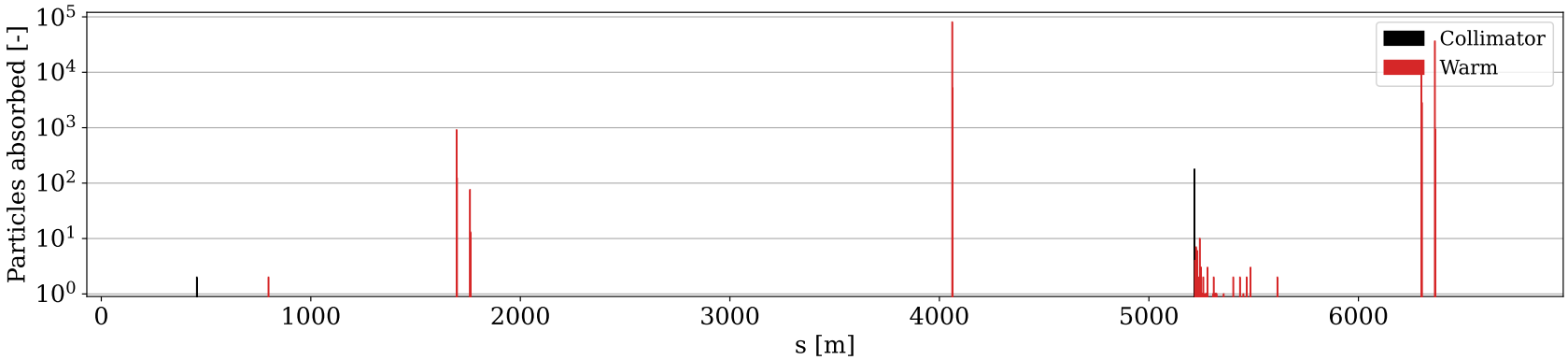
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $-179°$



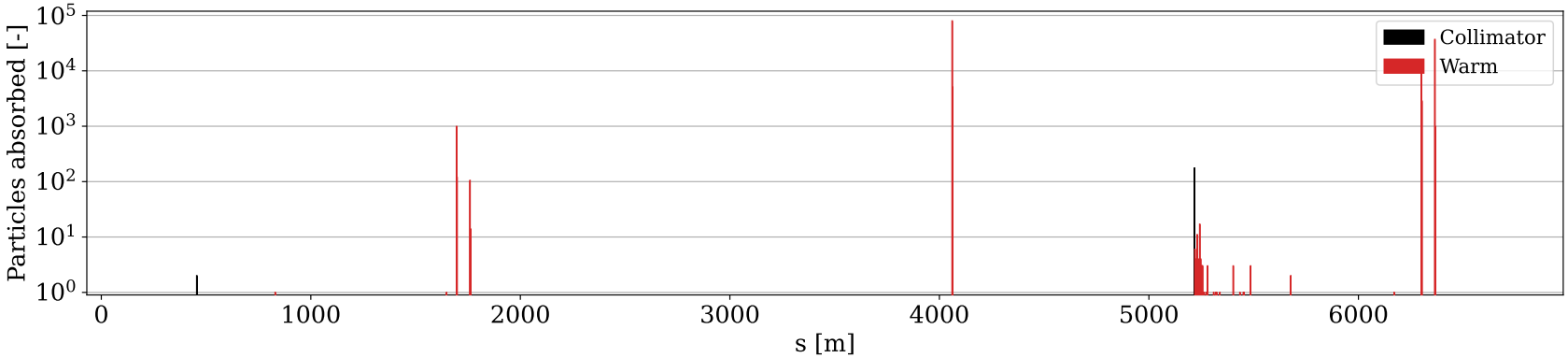
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $-150°$



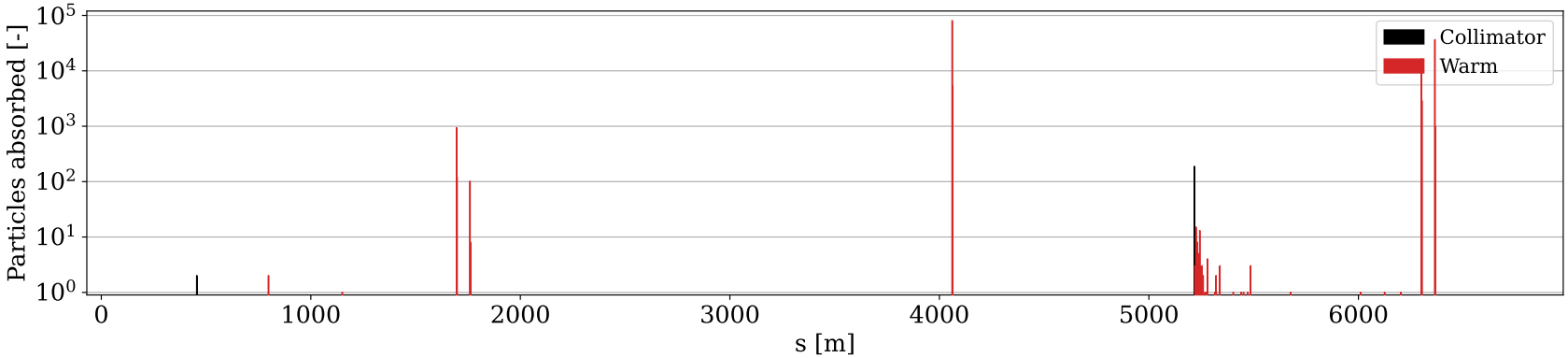
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase −120°



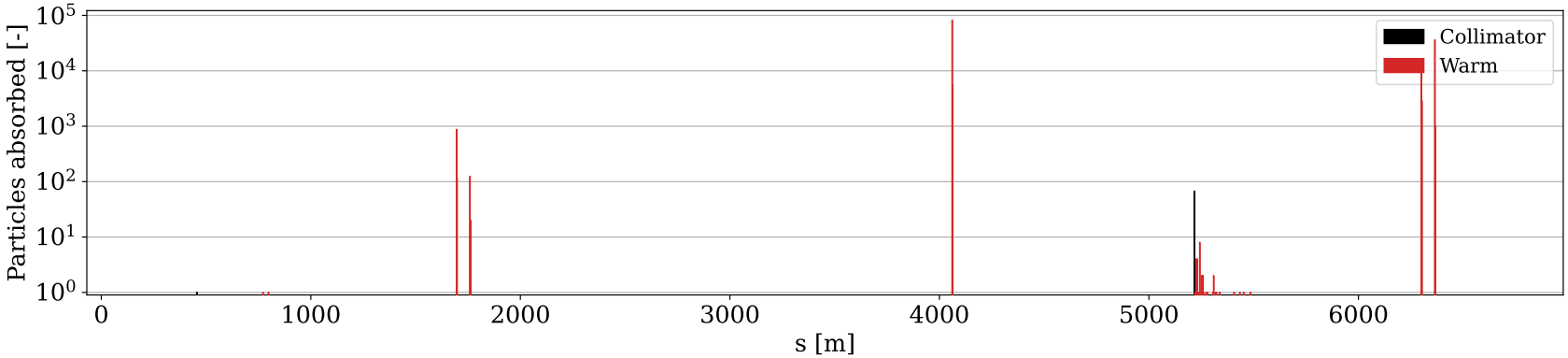
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $-90°$



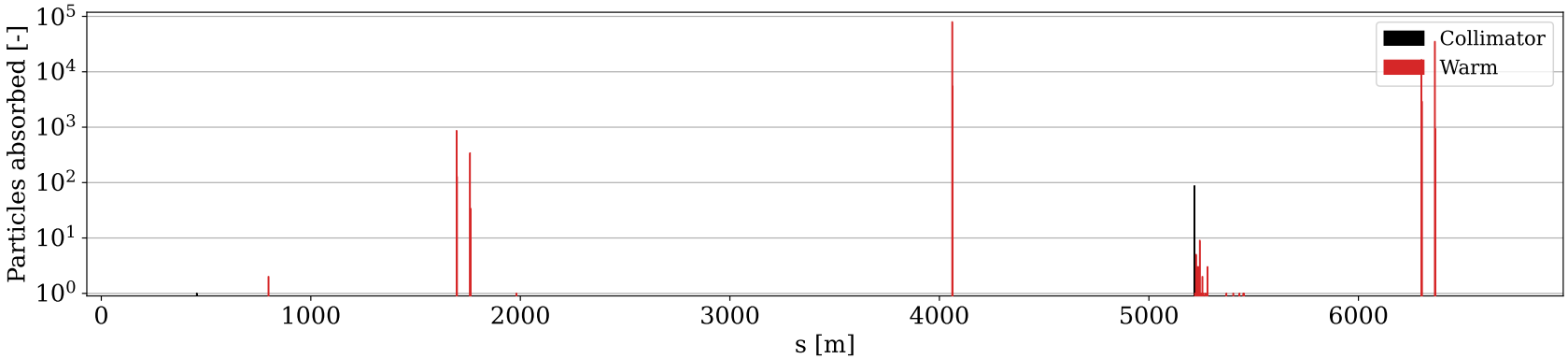
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $-60°$



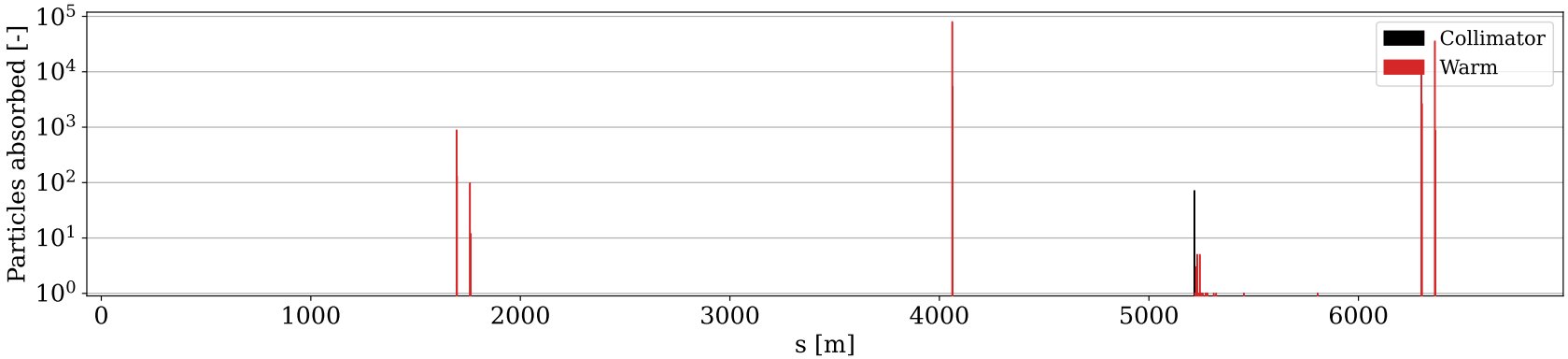
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $-30°$



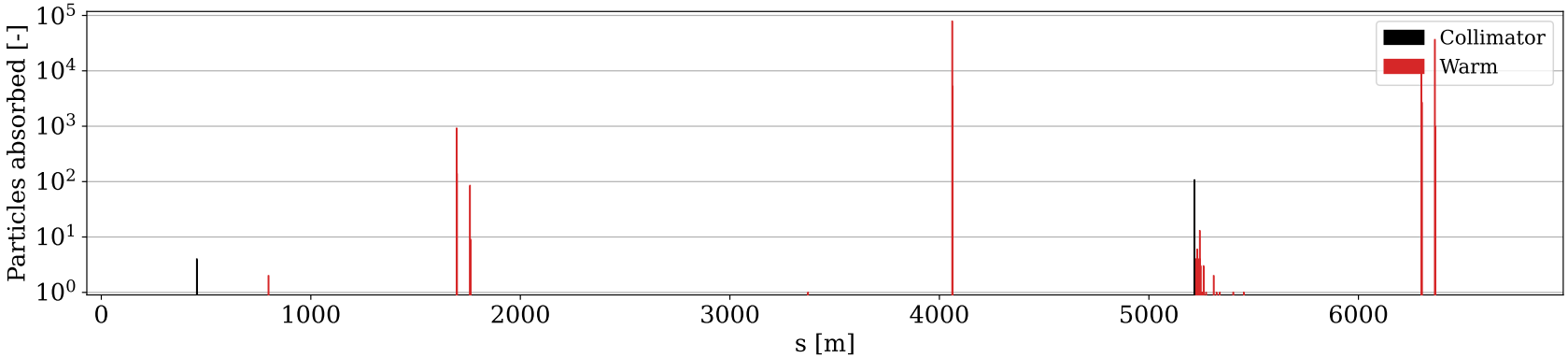
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $0°$



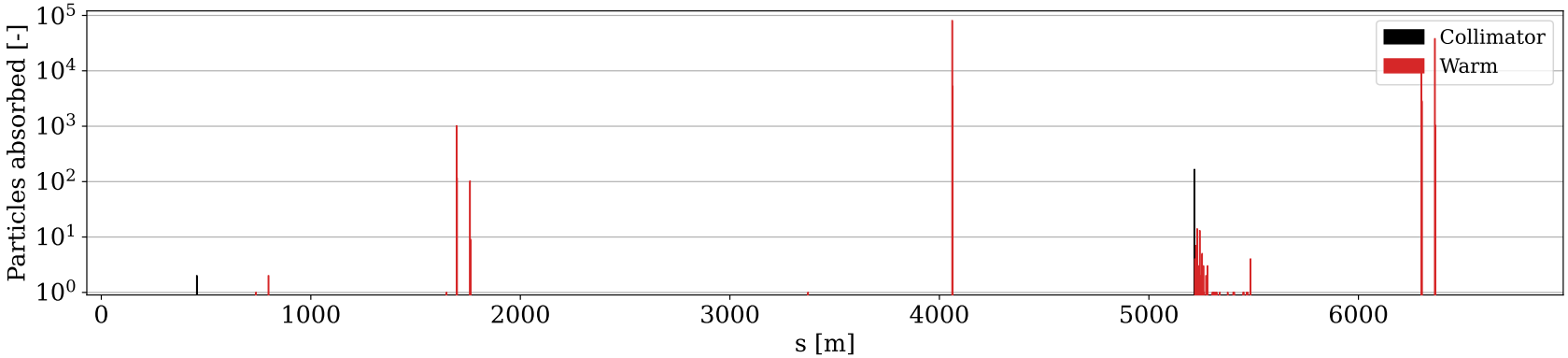
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase 30°



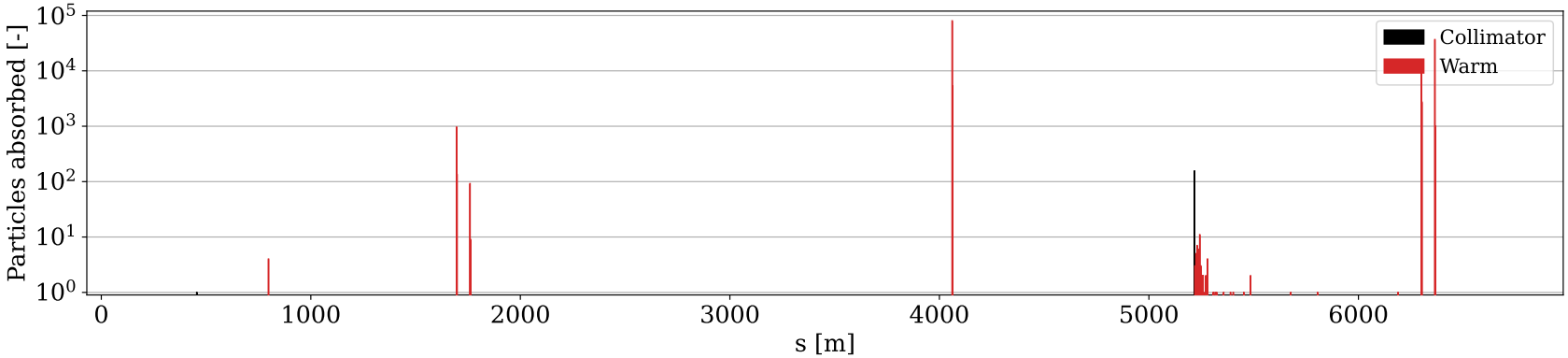
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase $60°$



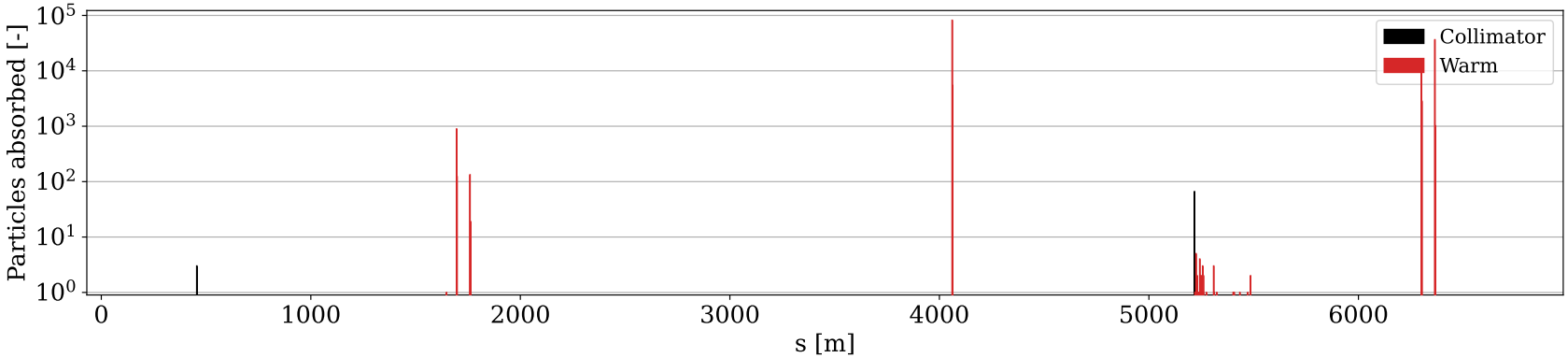
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase 90°



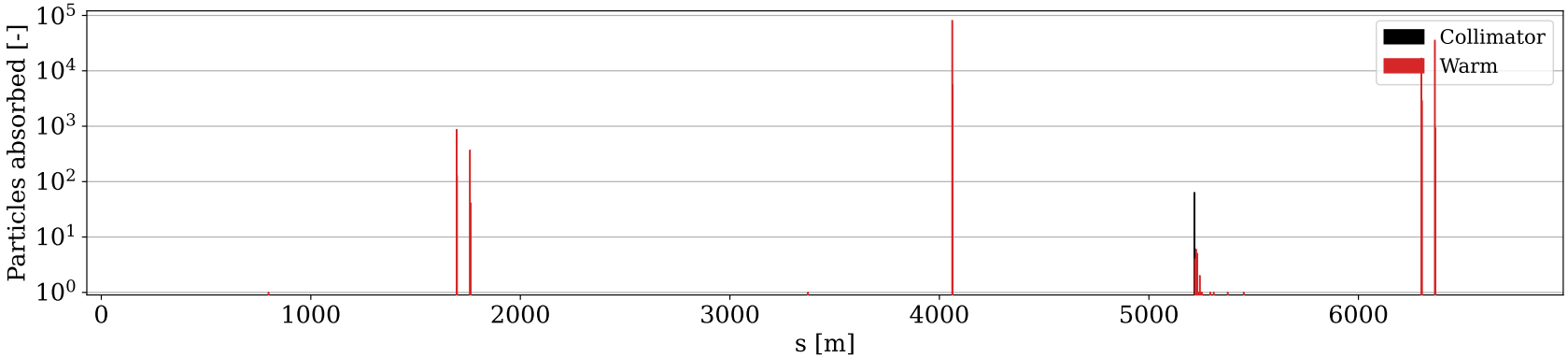
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase 120°



Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase 150°



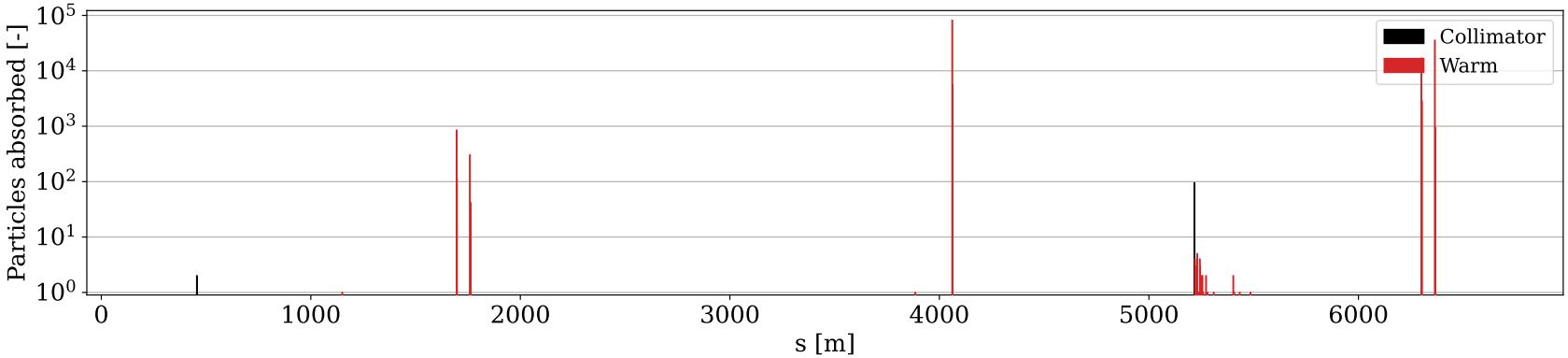
Vertical crabs — Betatron loss maps — Vertical — Without errors — Phase 179°



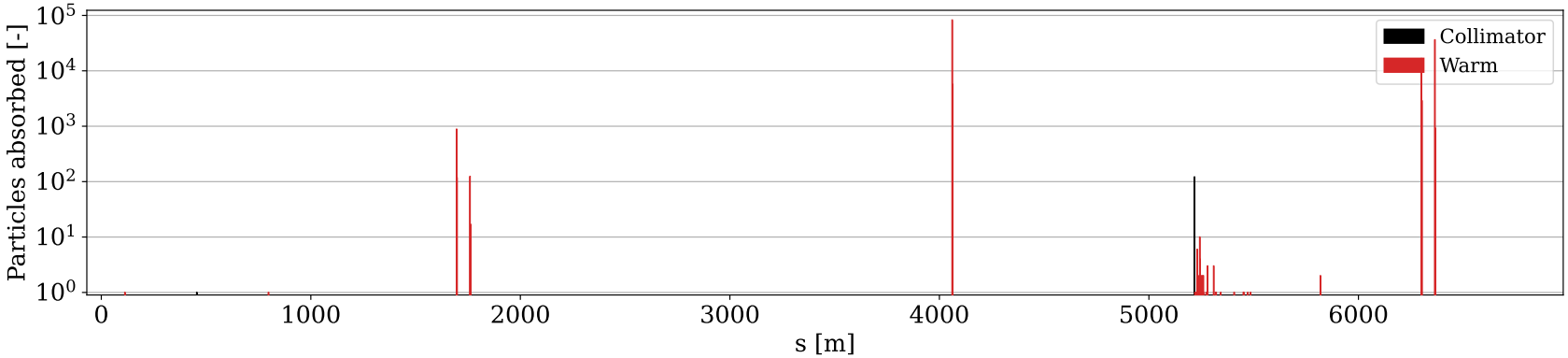
Vertical crabs — Betatron loss maps — Vertical — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

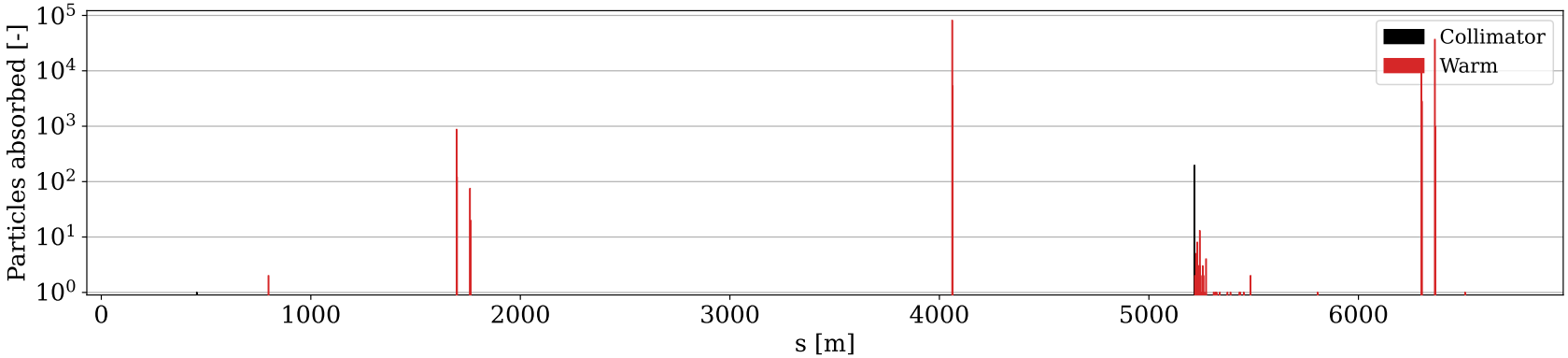
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-179°$



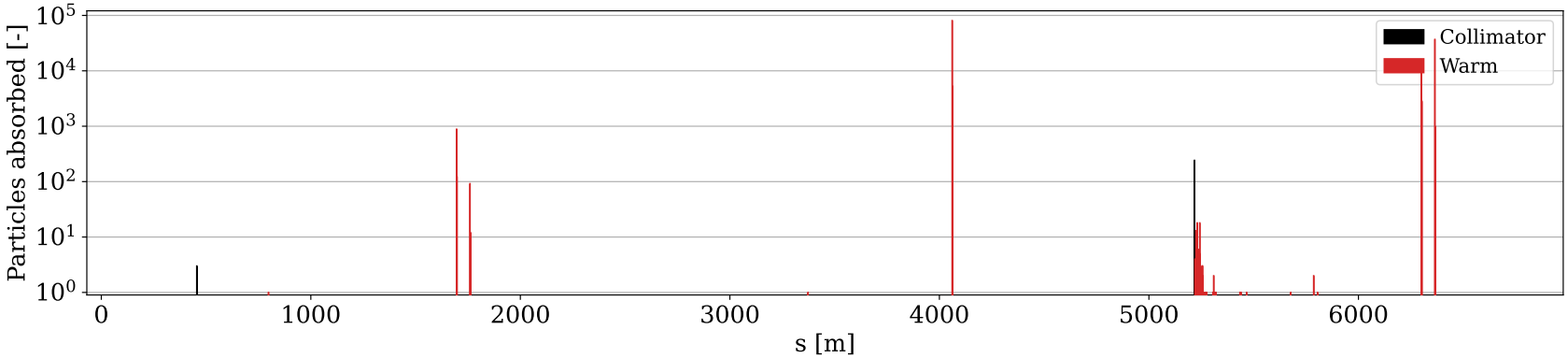
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-150°$



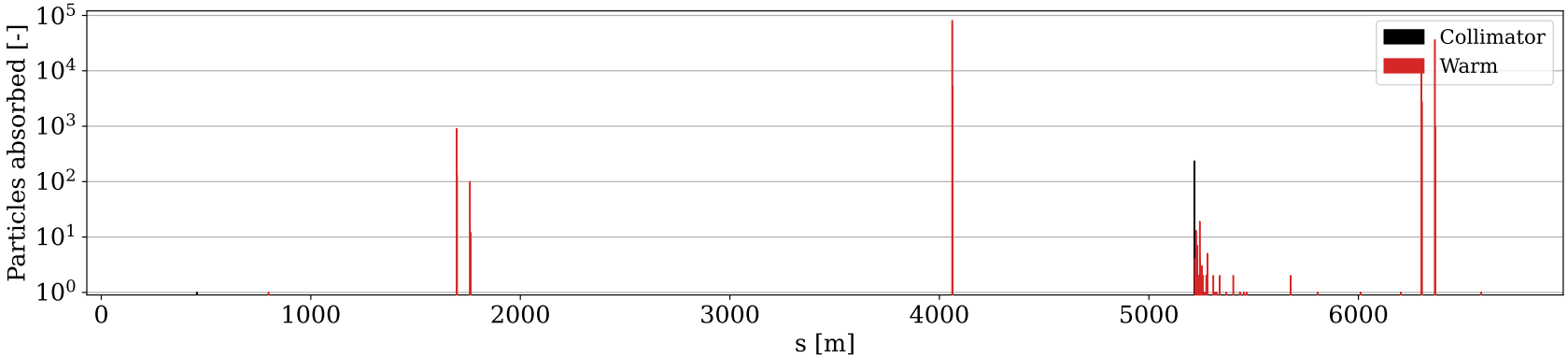
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-120°$



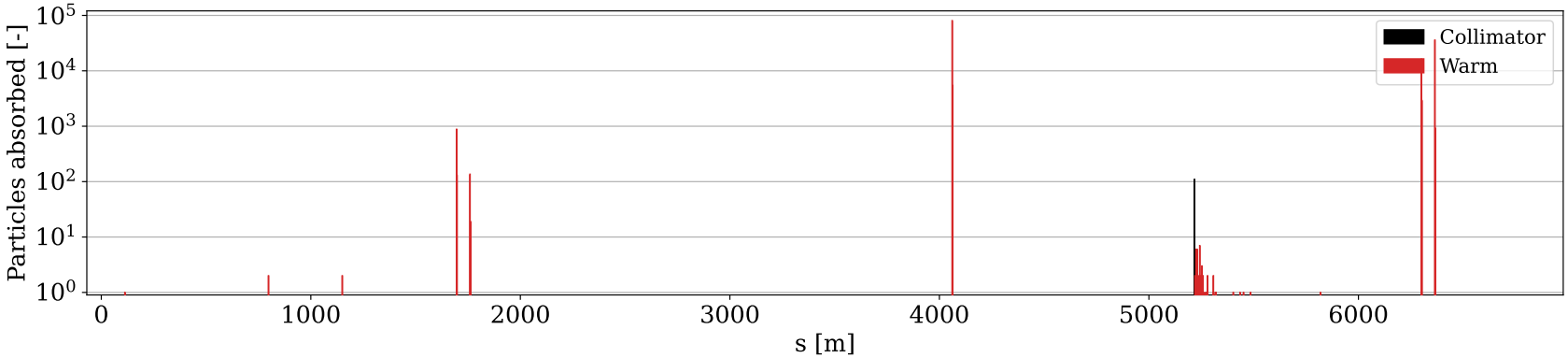
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-90°$



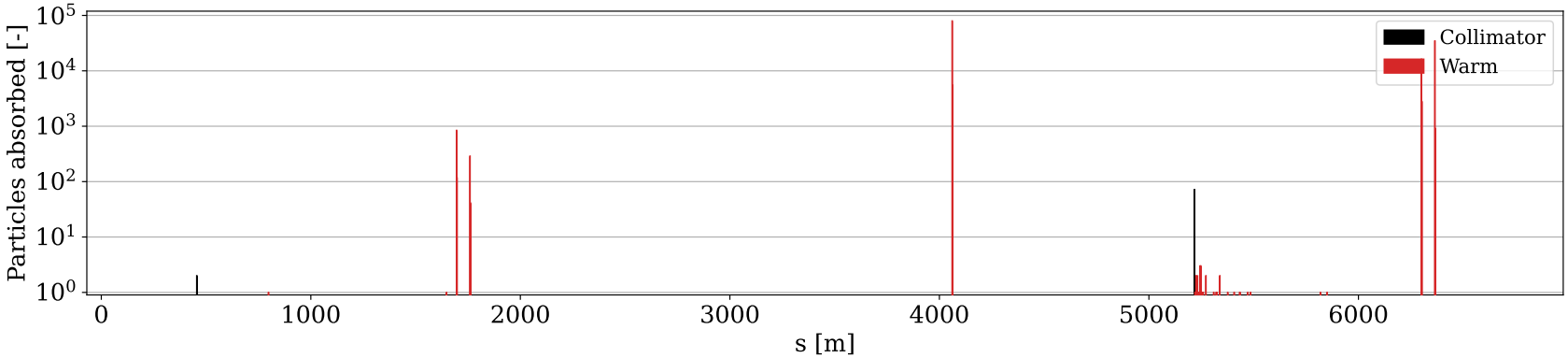
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-60°$



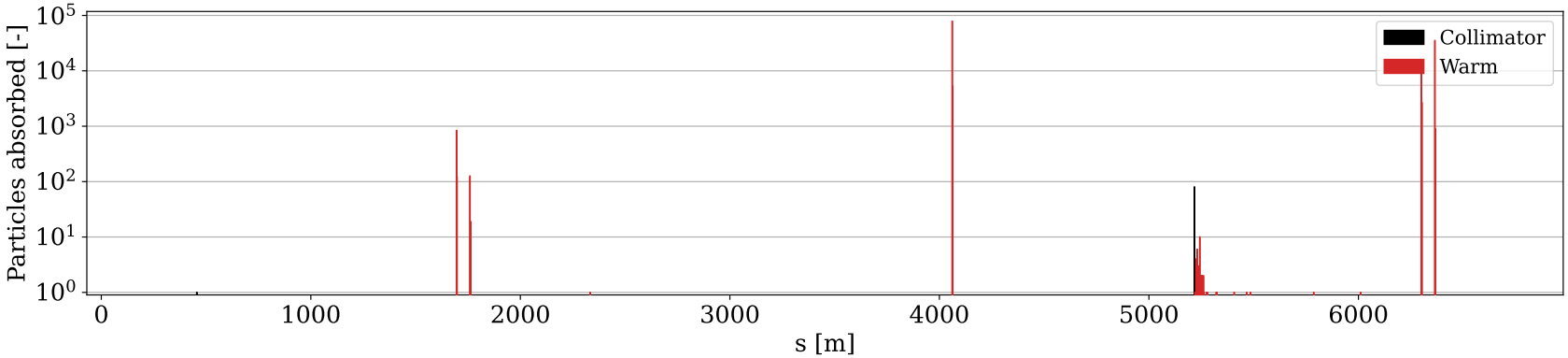
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $-30°$



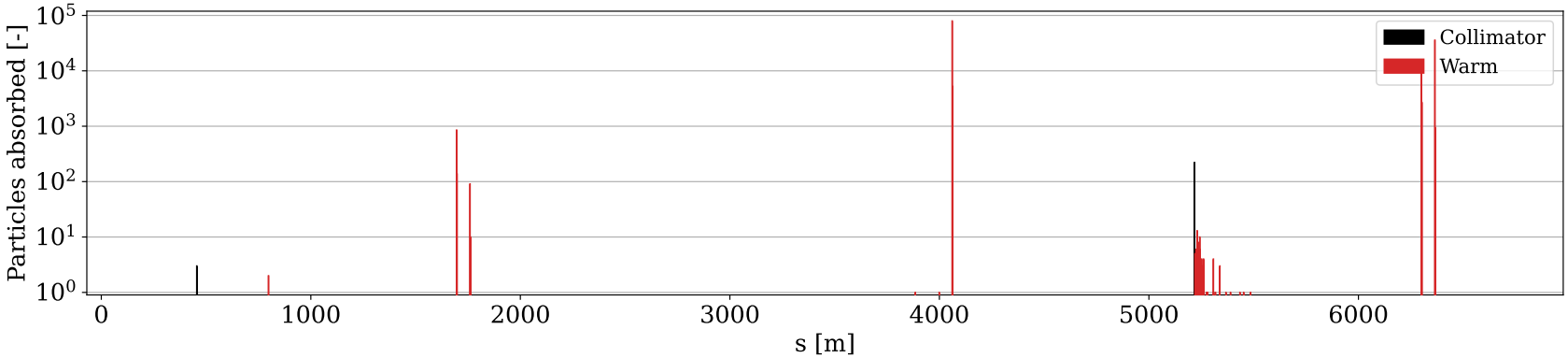
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $0°$



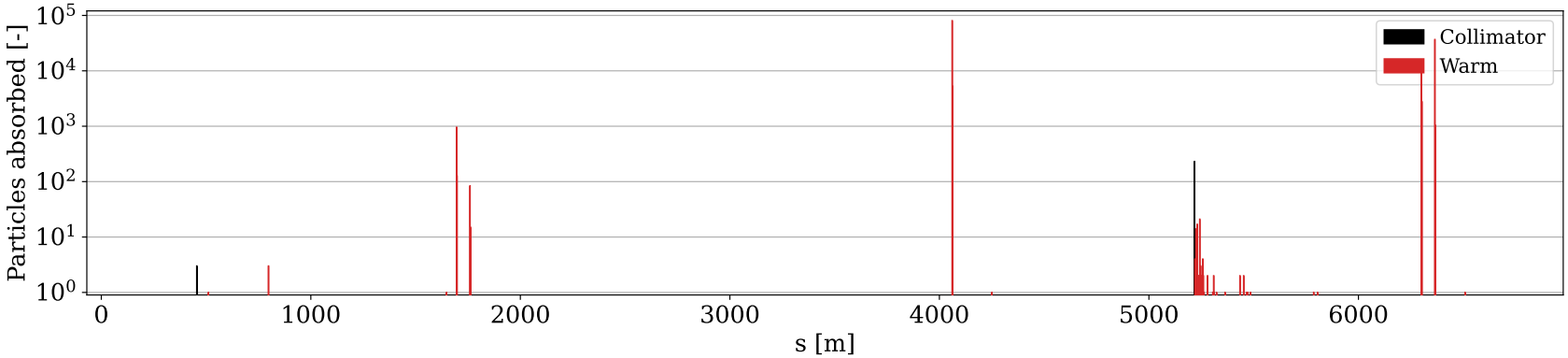
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $30°$



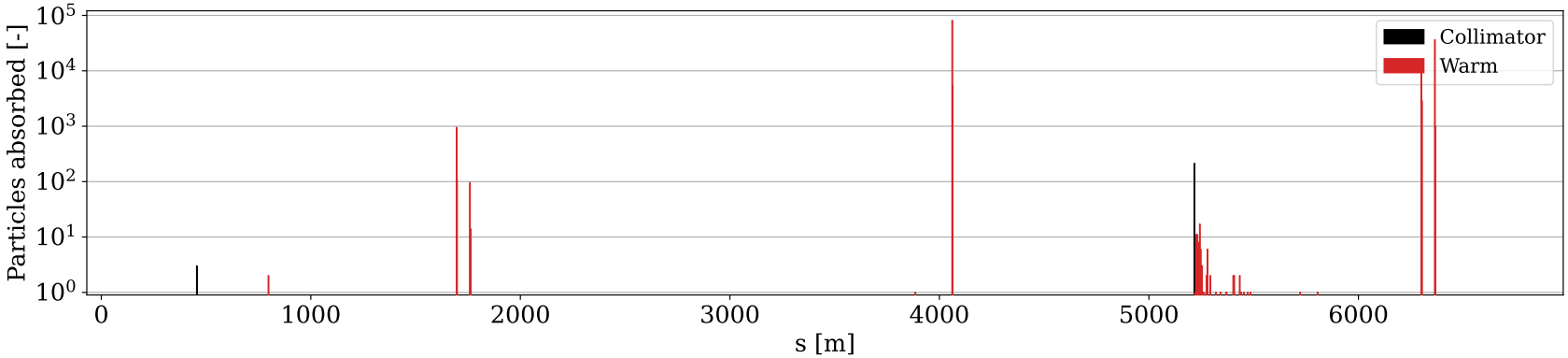
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $60°$



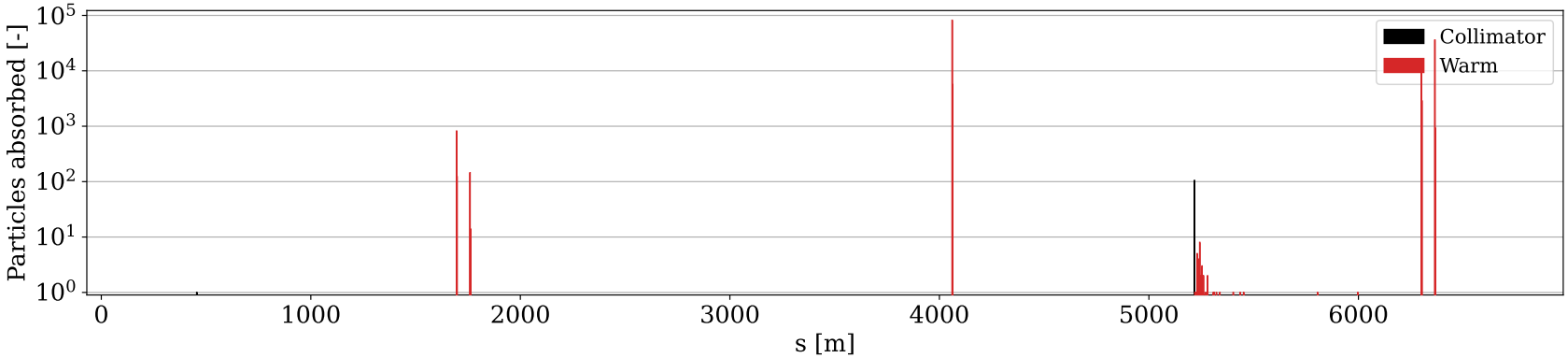
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $90°$



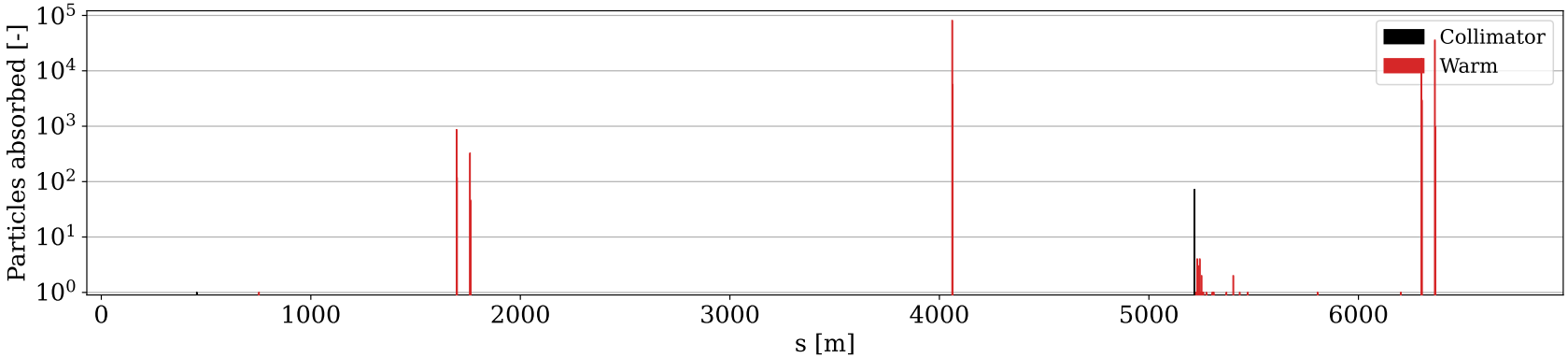
Vertical crabs — Betatron loss maps — Vertical — With errors — Phase 120°



Vertical crabs — Betatron loss maps — Vertical — With errors — Phase $150°$



Vertical crabs — Betatron loss maps — Vertical — With errors — Phase 179°



Outline

1. Overview
2. Horizontal crabs
   - Betatron loss maps
   - Off-momentum loss maps
3. Vertical crabs
   - Betatron loss maps
   - Off-momentum loss maps

Vertical crabs — Off-momentum loss maps

- DPpos
- DPneg

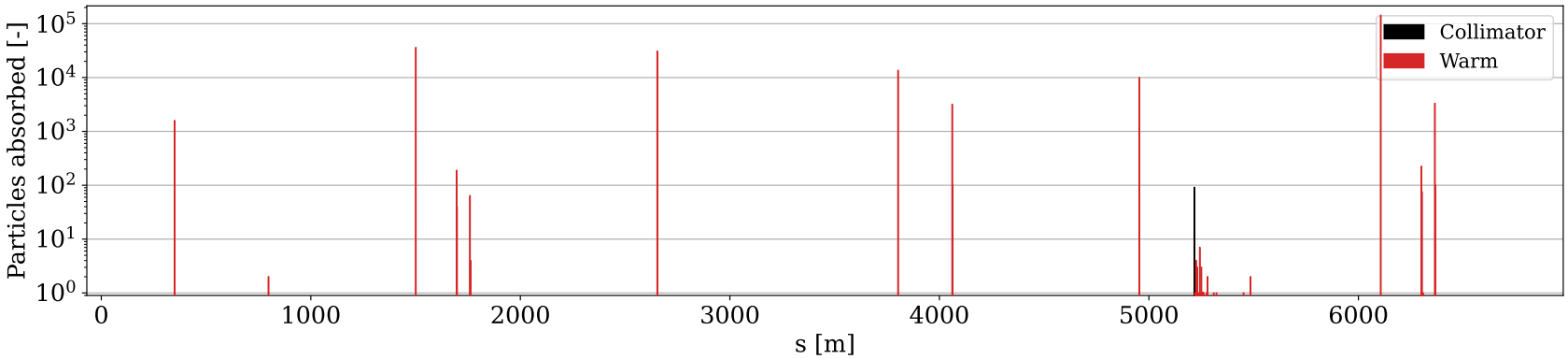
Vertical crabs — Off-momentum loss maps — DPpos

- Without errors
- With errors

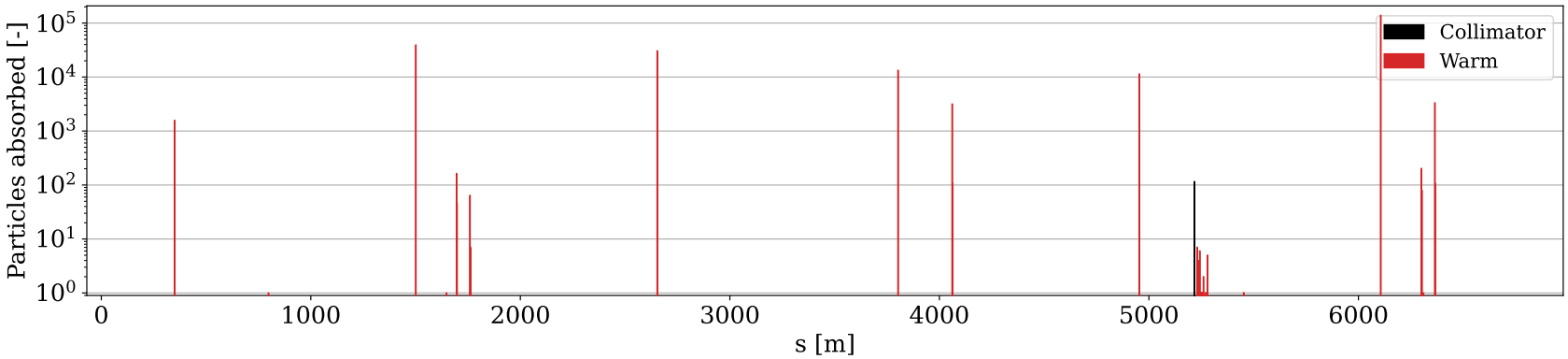
Vertical crabs — Off-momentum loss maps — DPpos — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

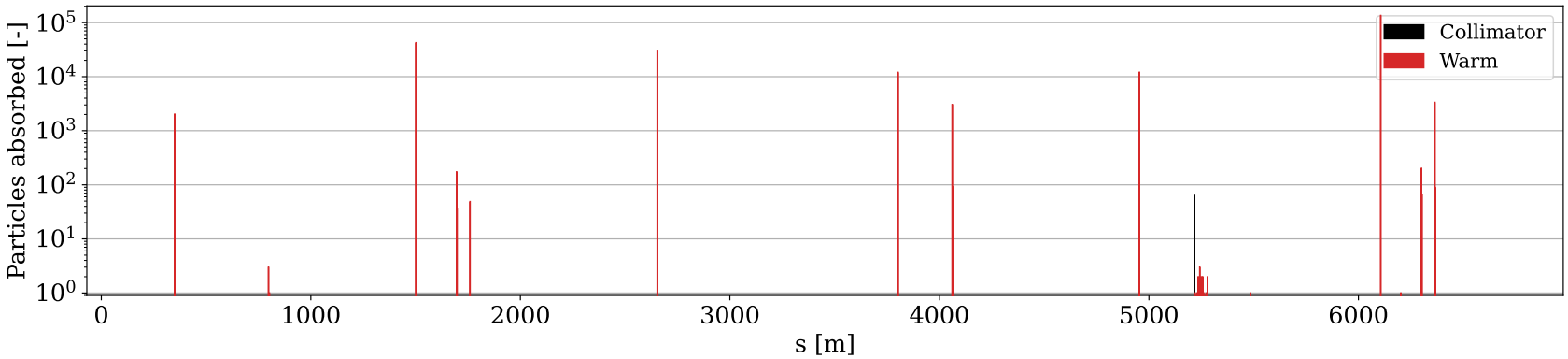
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-179°$



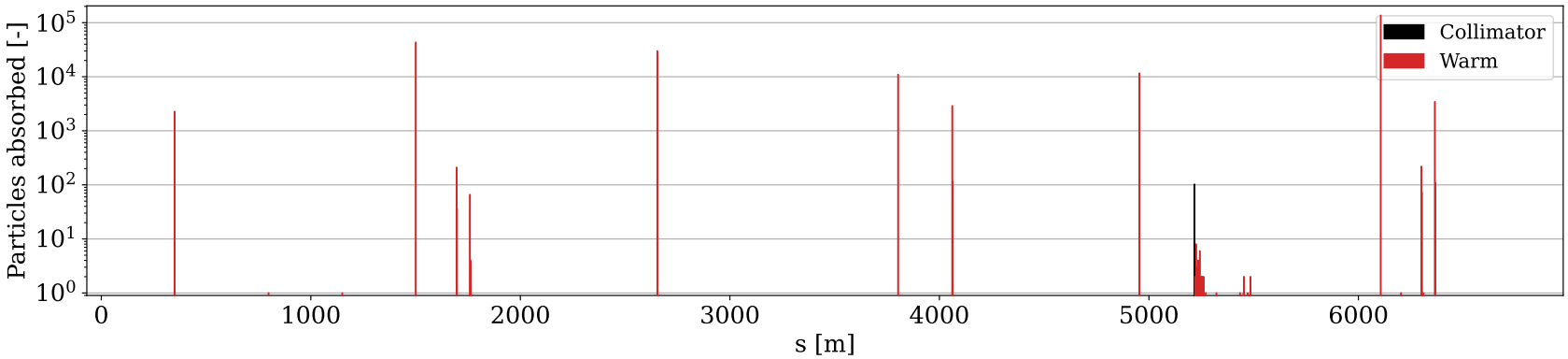
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-150°$



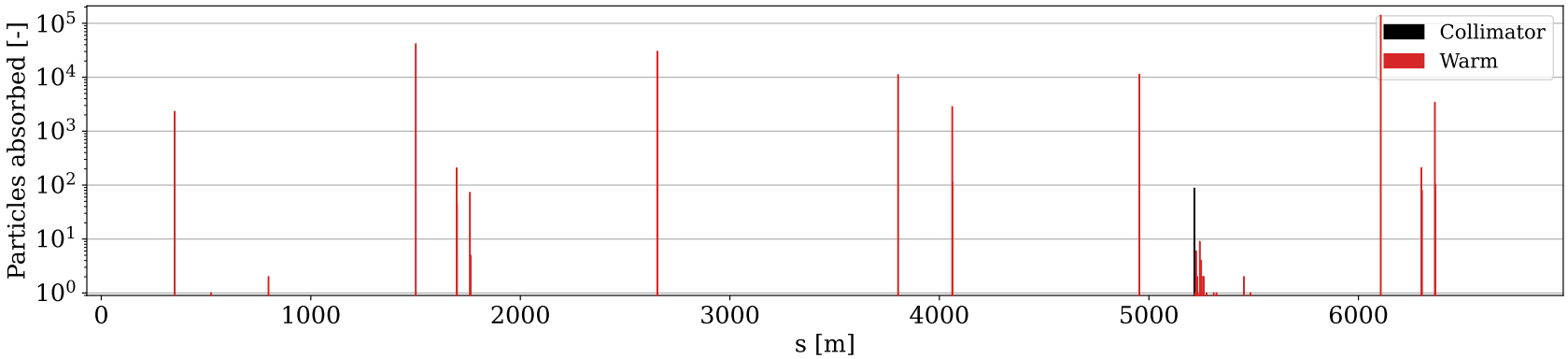
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-120°$



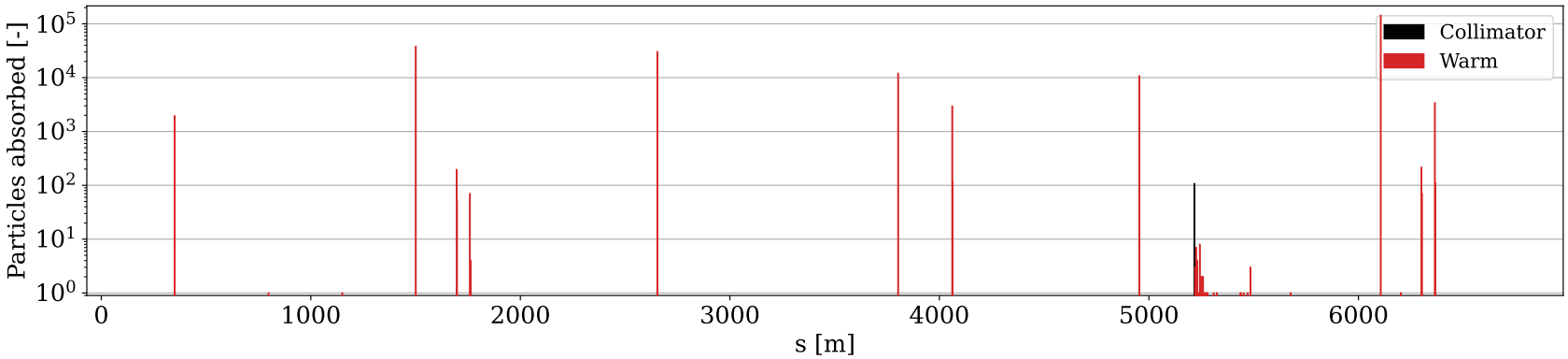
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-90°$



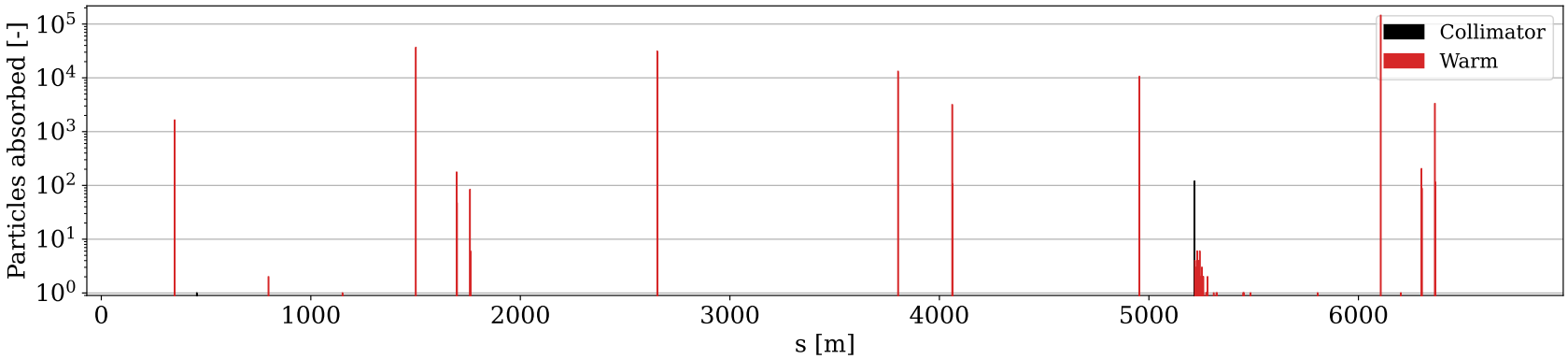
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-60°$



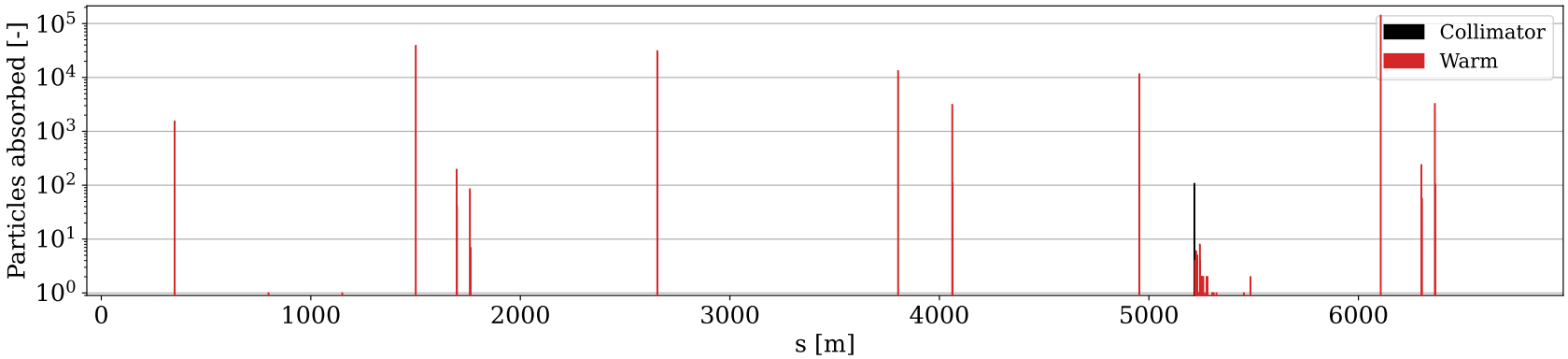
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $-30°$



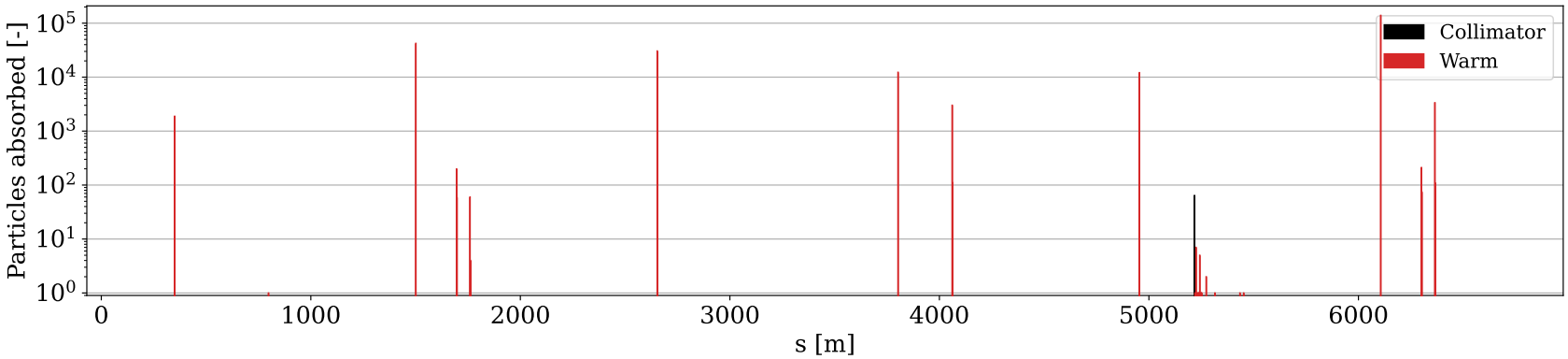
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $0°$



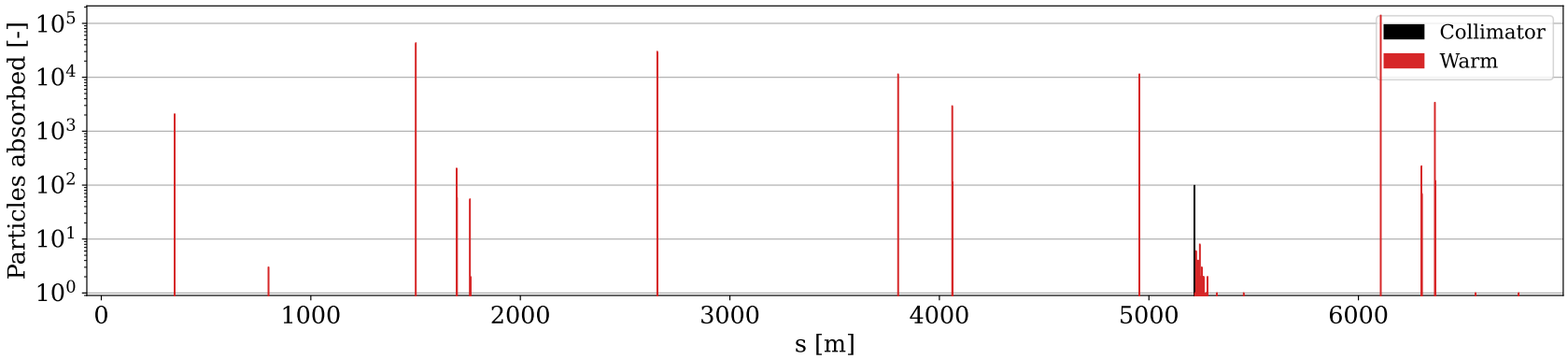
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase 30°



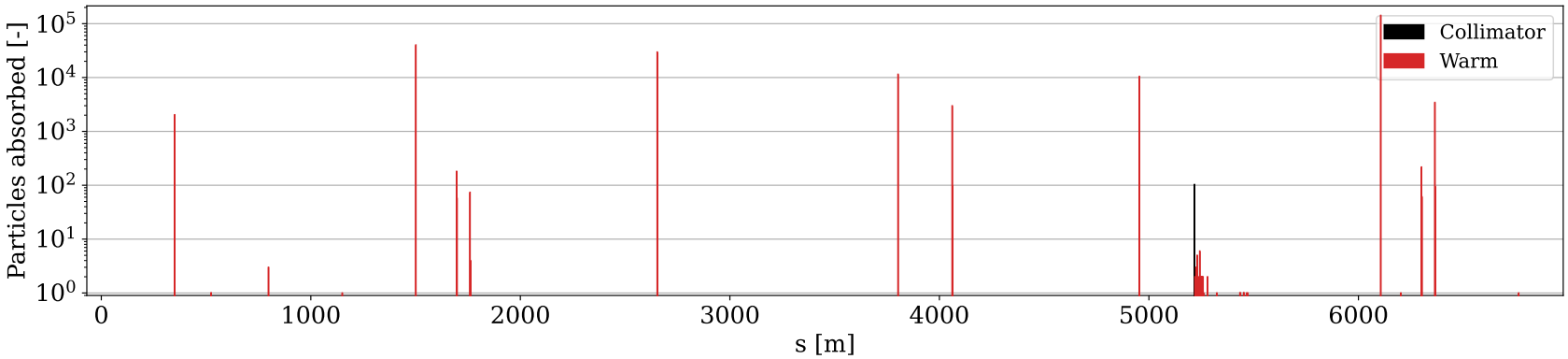
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase $60°$



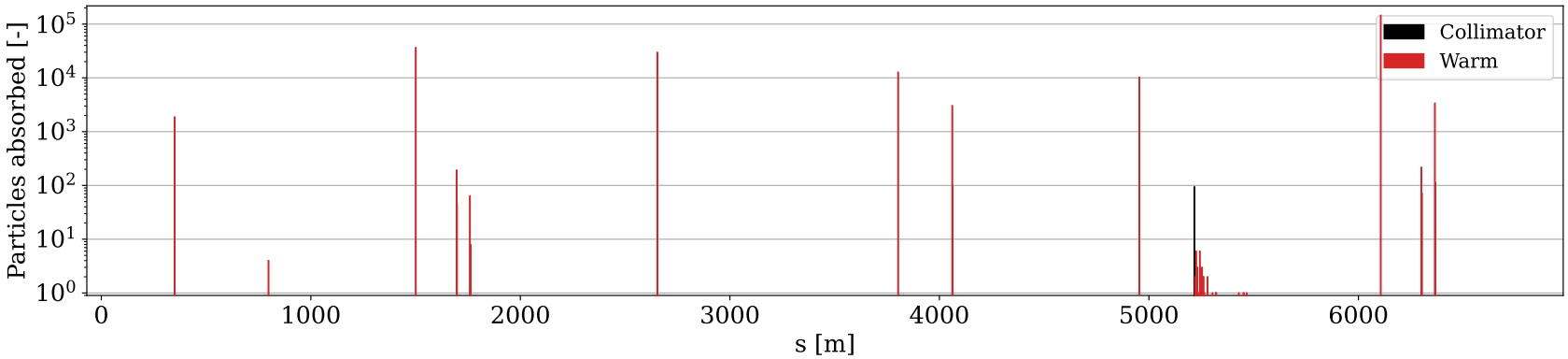
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase 90°



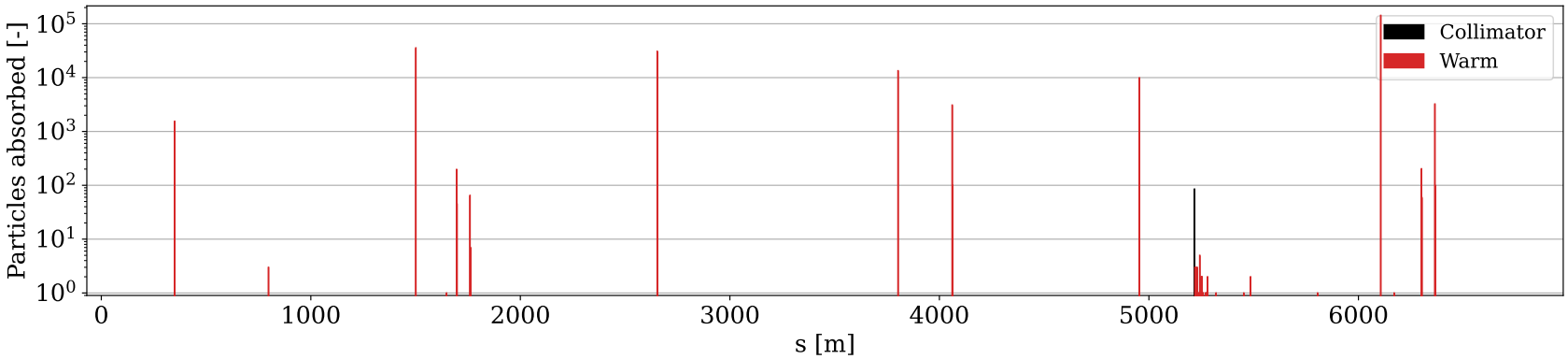
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase 120°



Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase 150°



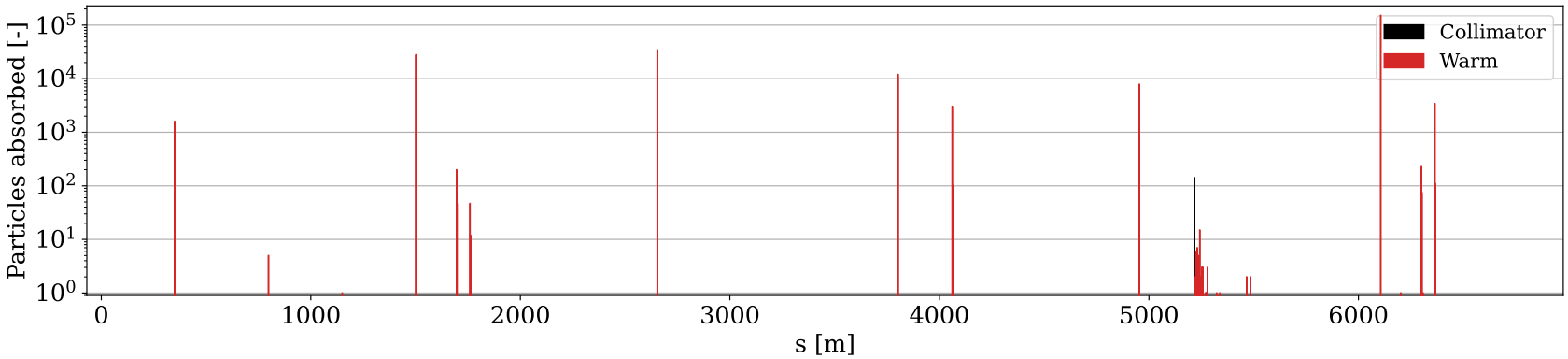
Vertical crabs — Off-momentum loss maps — DPpos — Without errors — Phase 179°



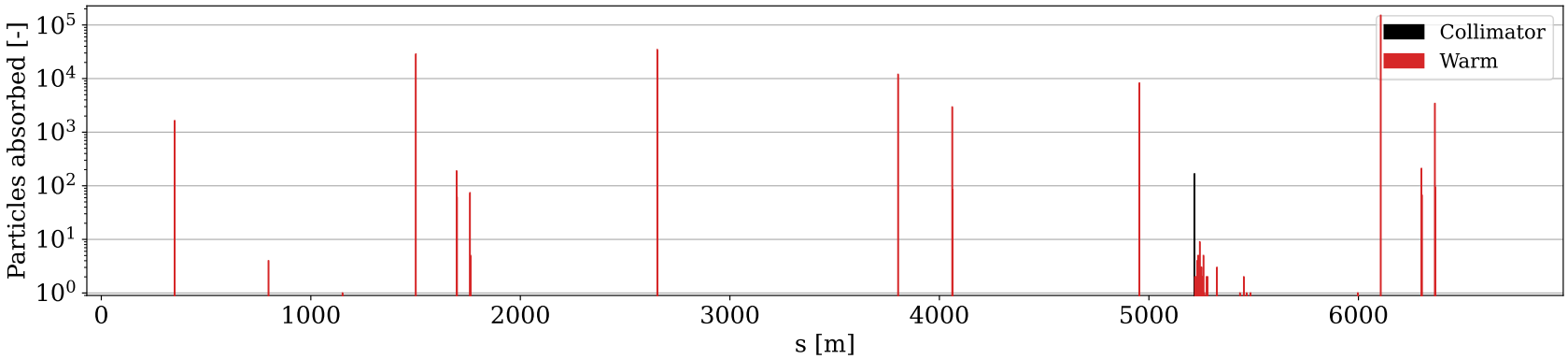
Vertical crabs — Off-momentum loss maps — DPpos — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

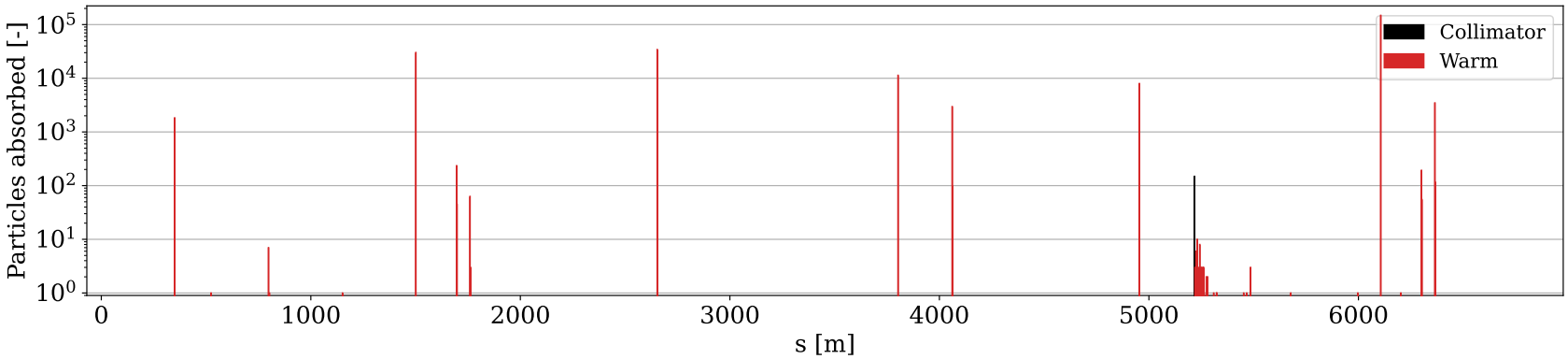
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase −179°



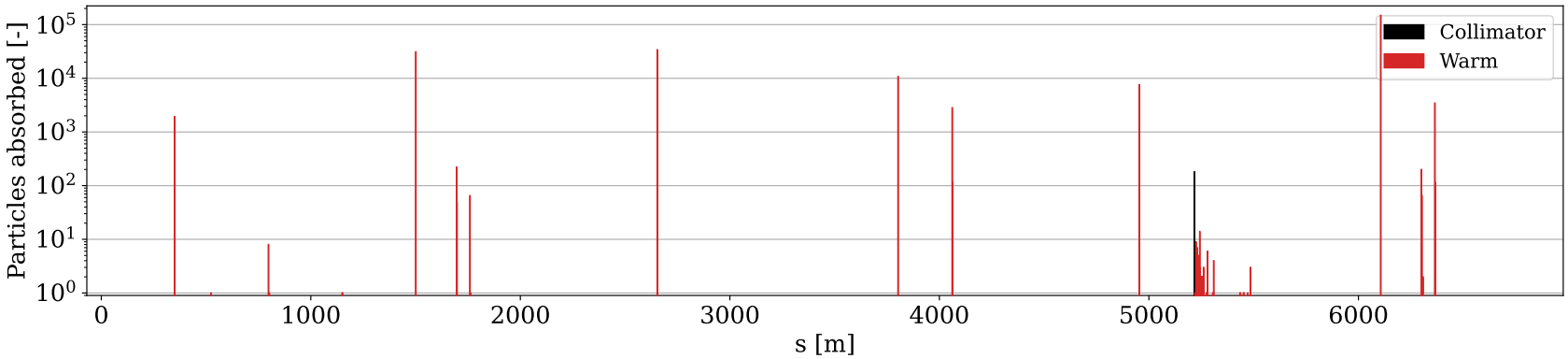
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase $-150°$



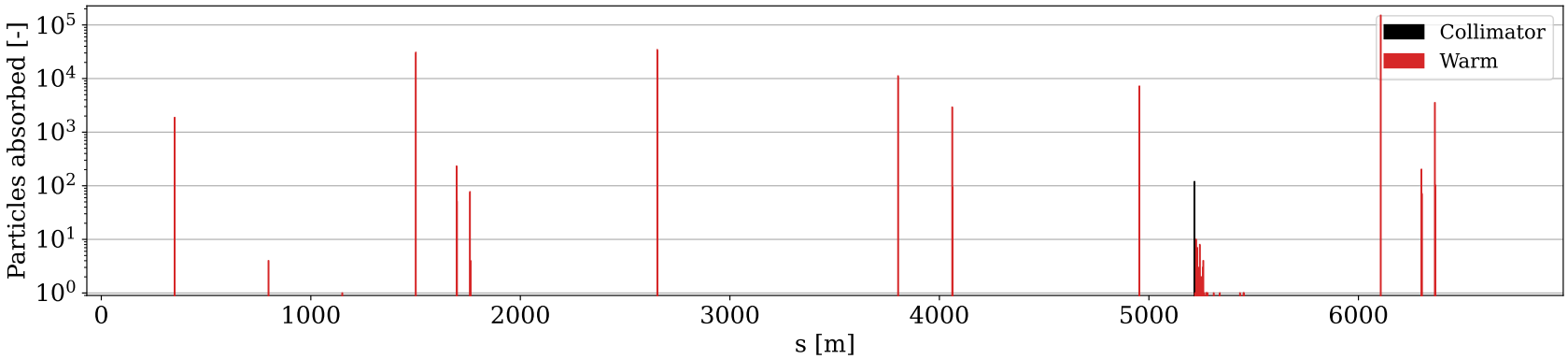
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase $-120°$



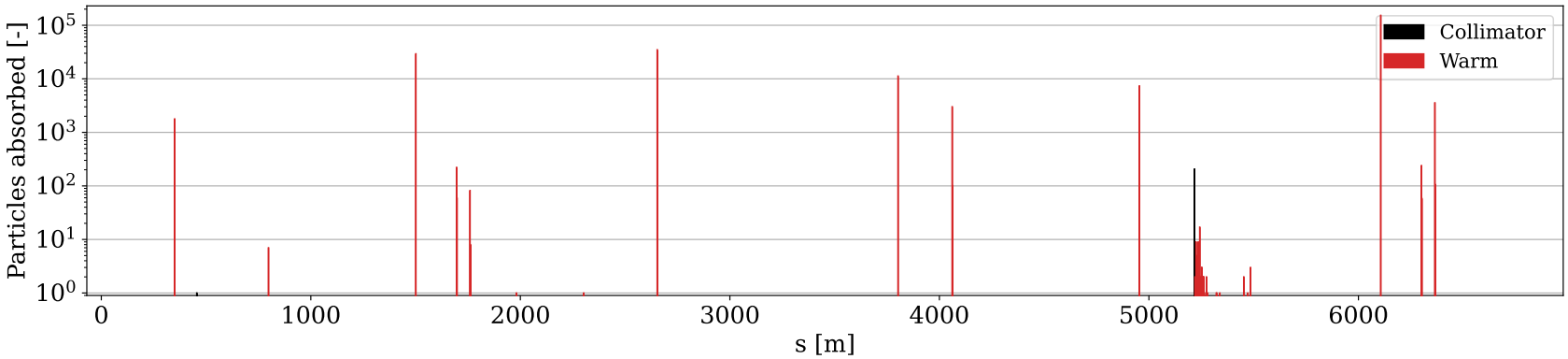
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase −90°



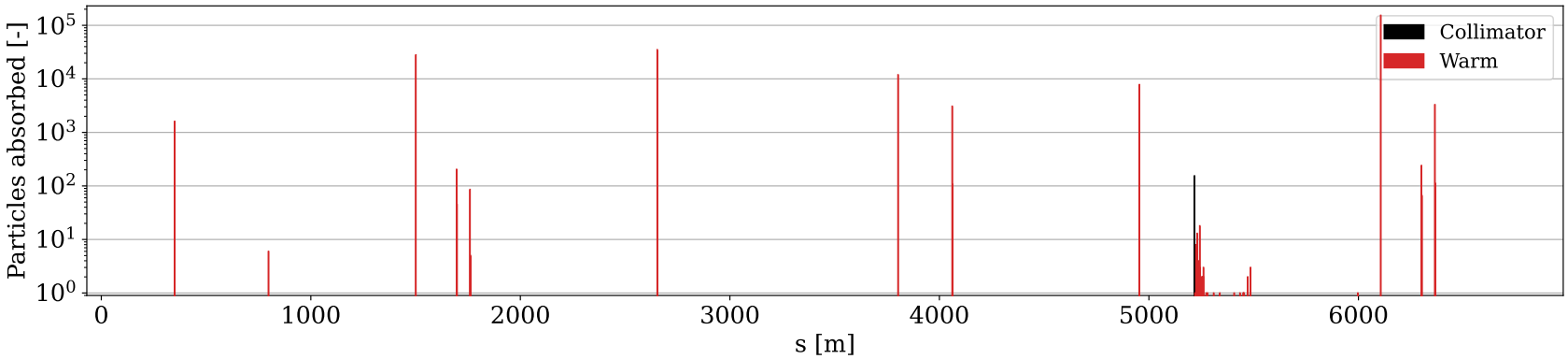
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase −60°



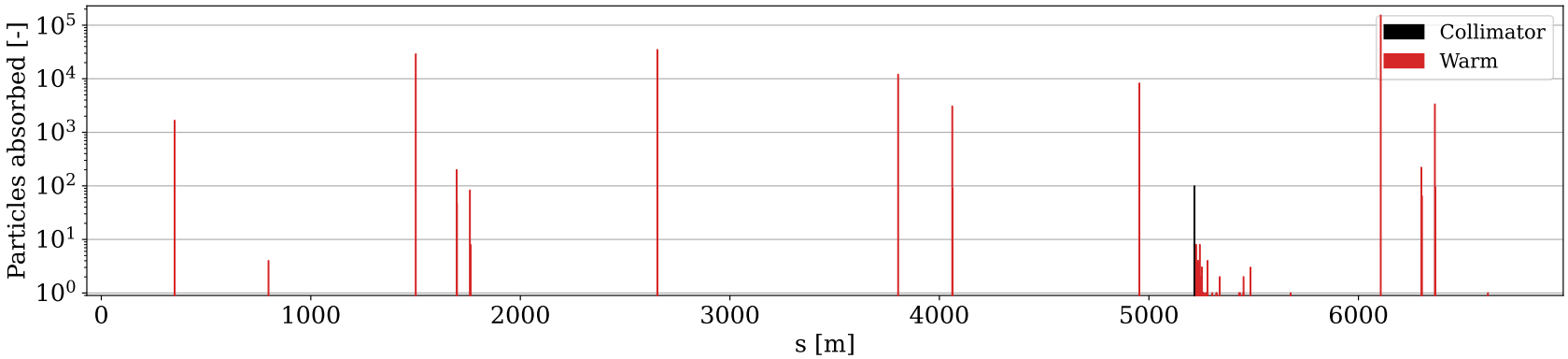
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase $-30°$



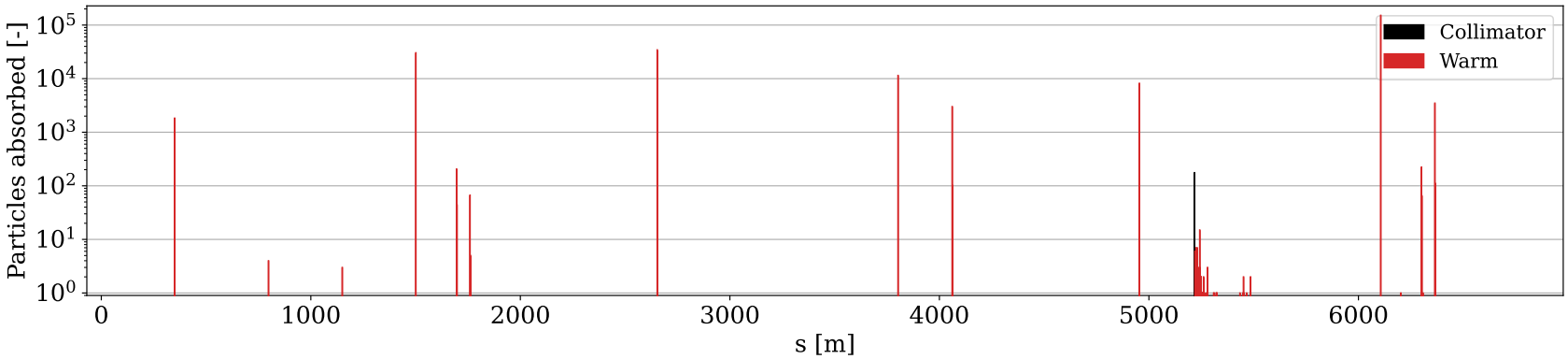
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 0°



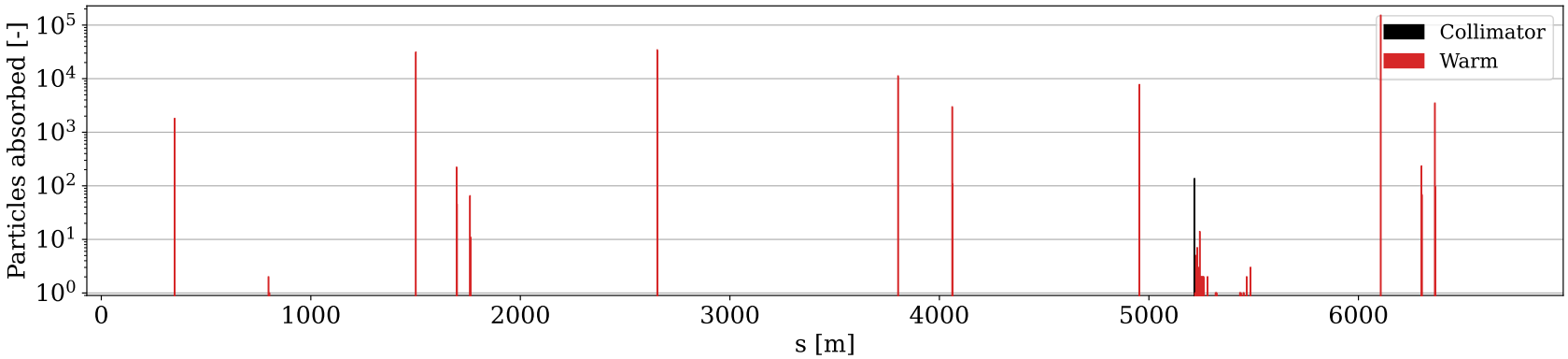
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 30°



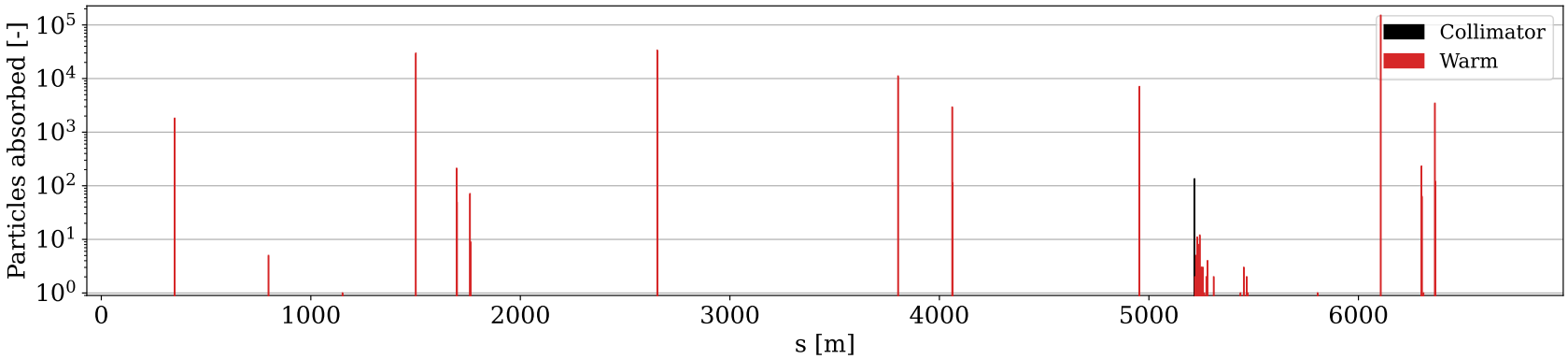
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 60°



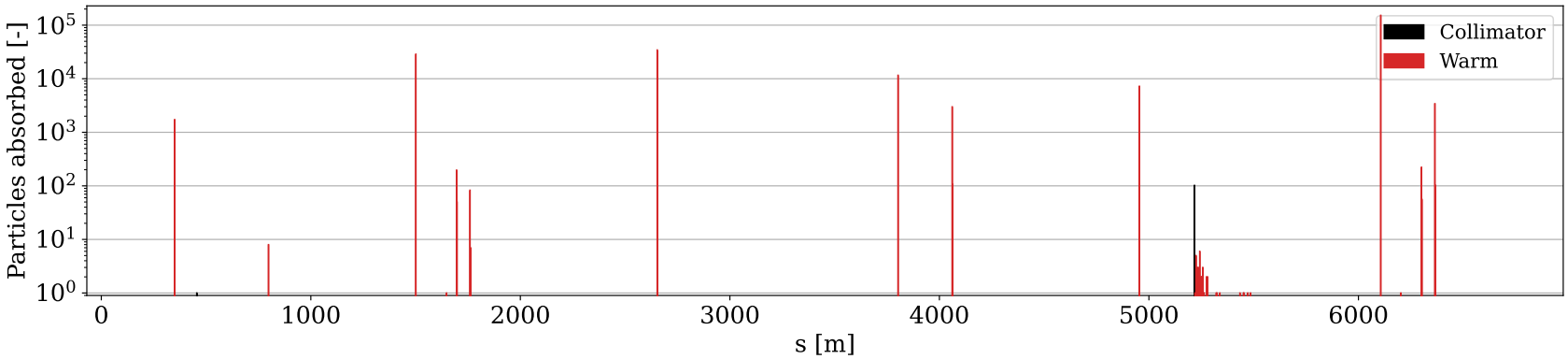
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 90°



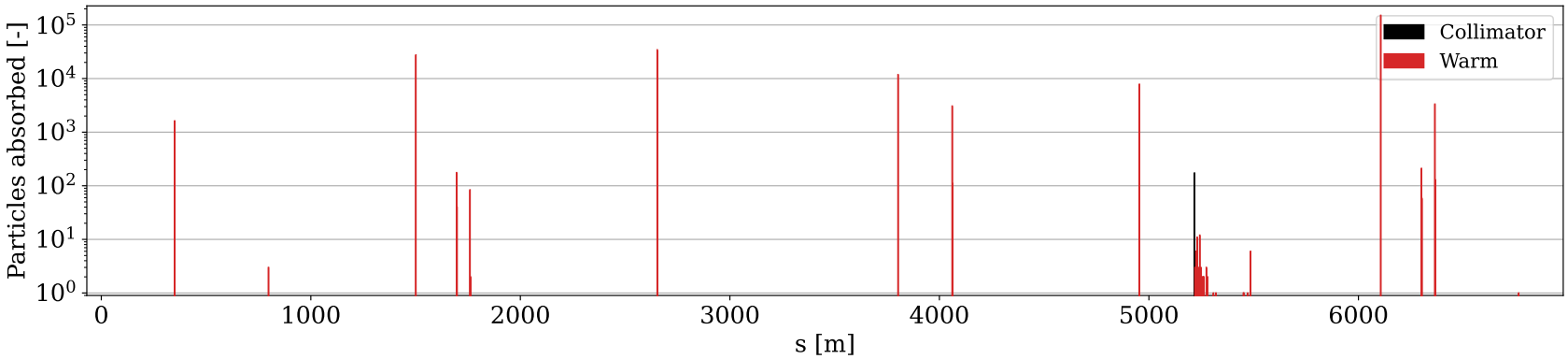
Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 120°



Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 150°



Vertical crabs — Off-momentum loss maps — DPpos — With errors — Phase 179°



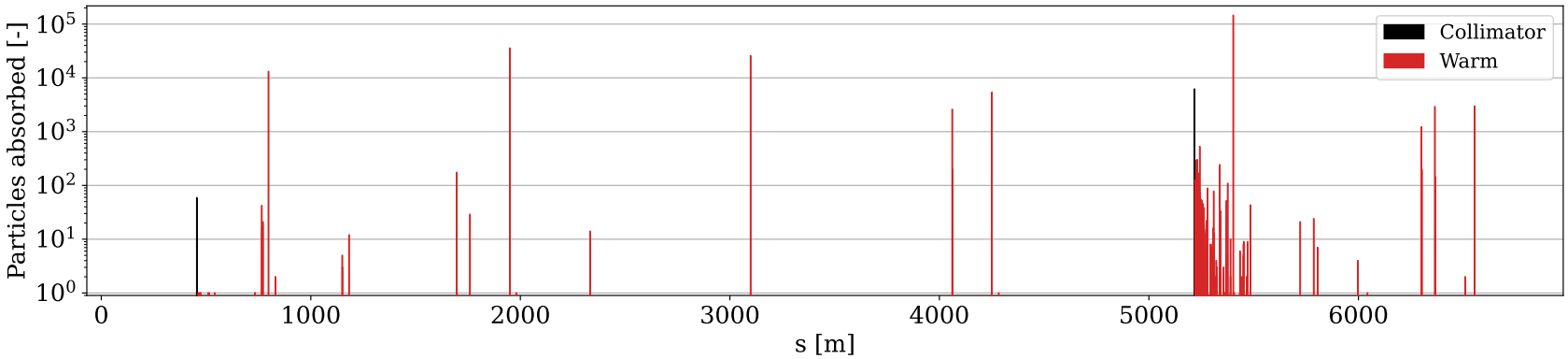
Vertical crabs — Off-momentum loss maps — DPneg

- Without errors
- With errors

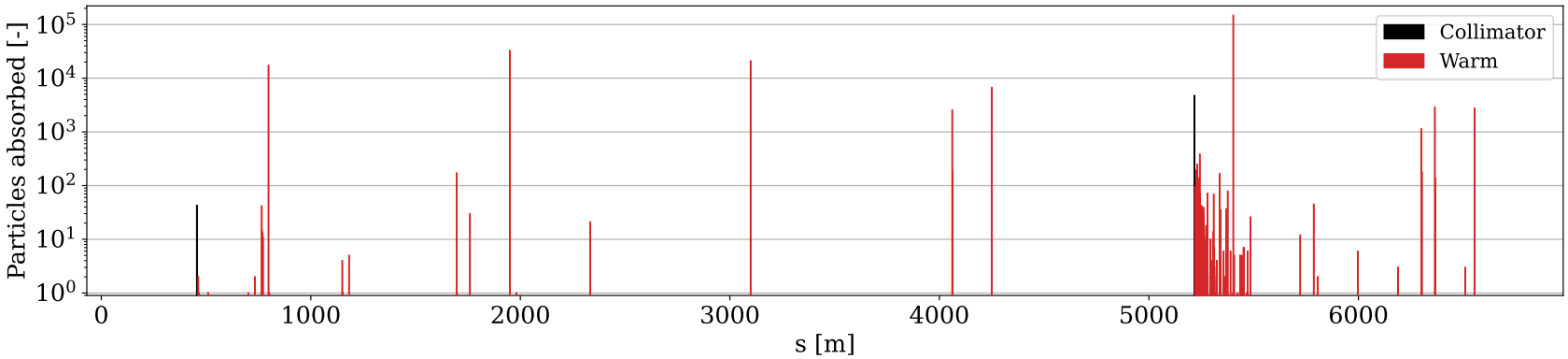
Vertical crabs — Off-momentum loss maps — DPneg — Without errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

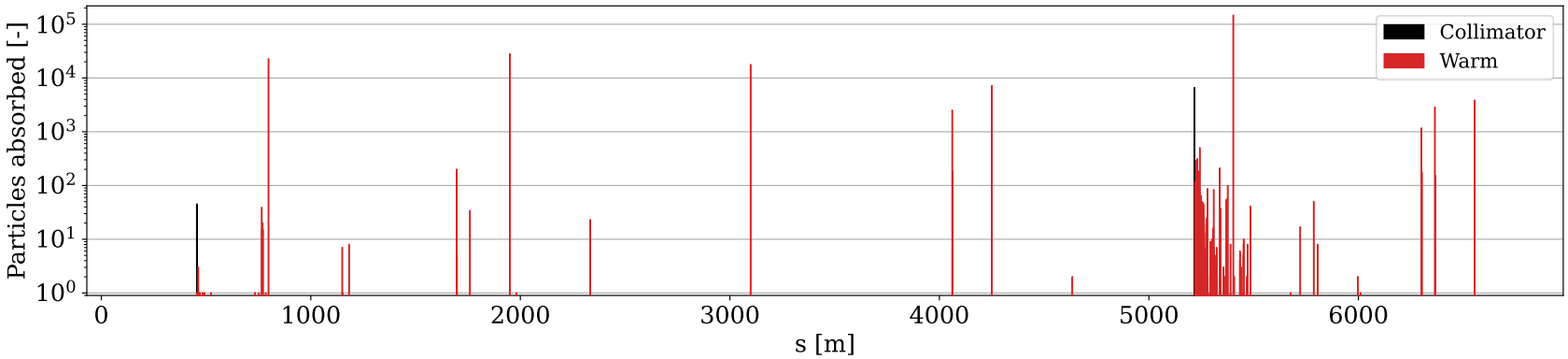
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-179°$



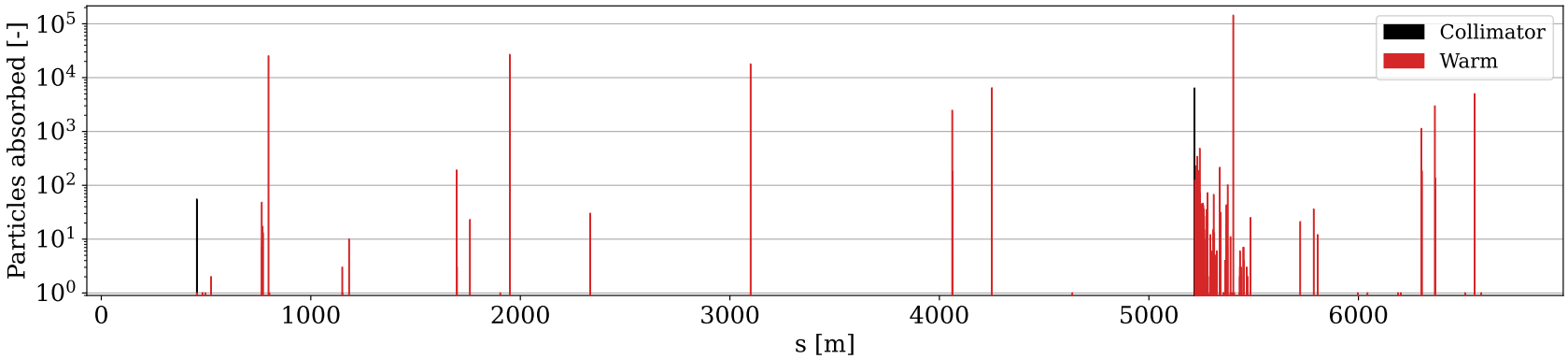
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-150°$



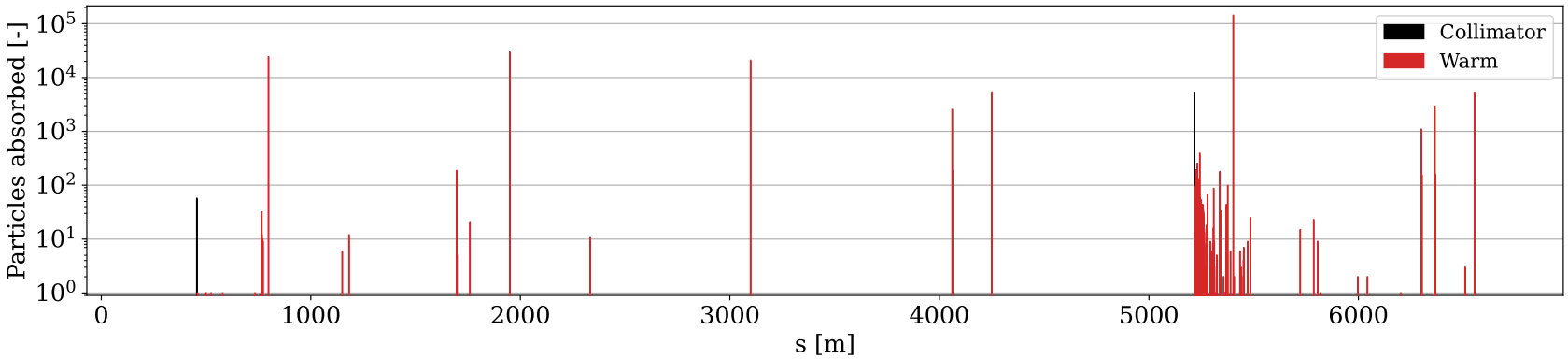
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-120°$



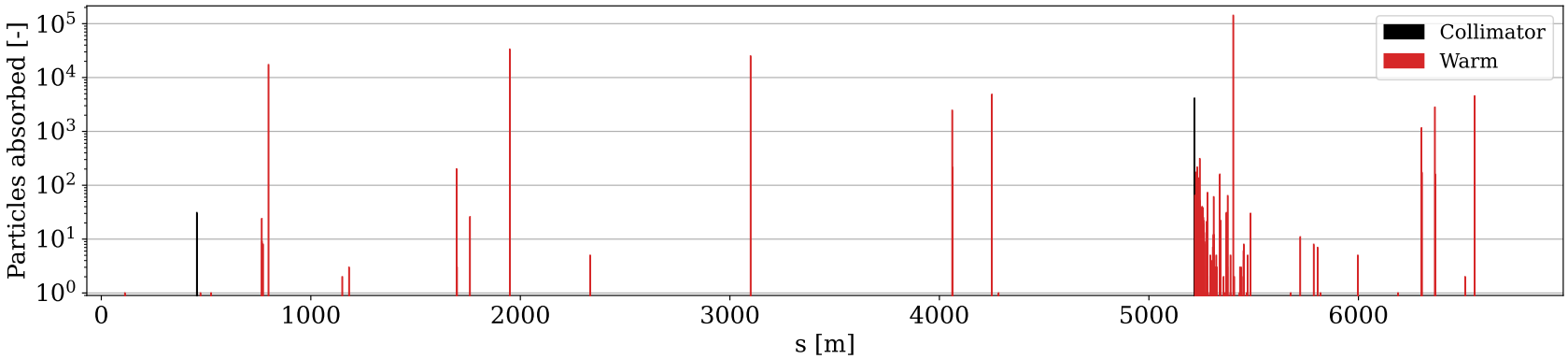
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-90°$



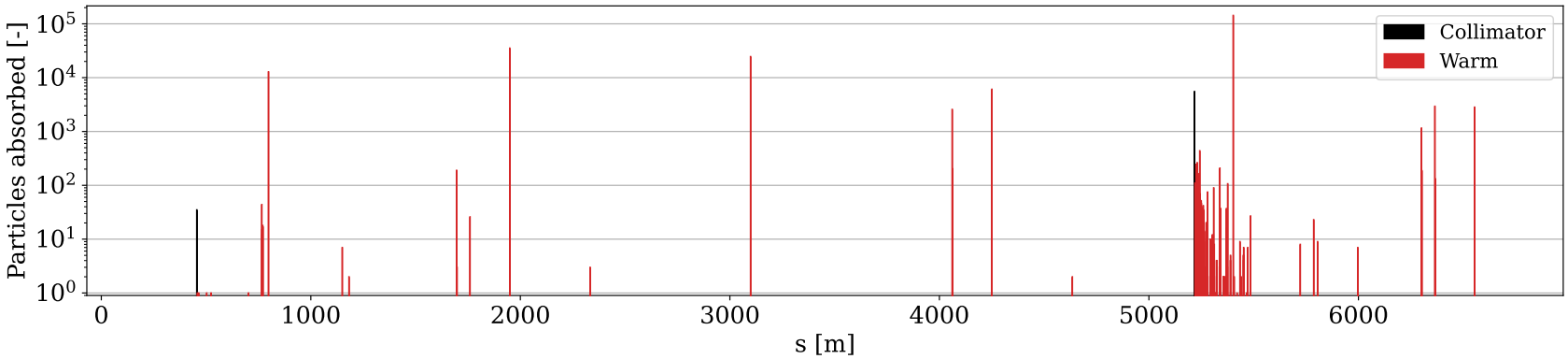
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-60°$



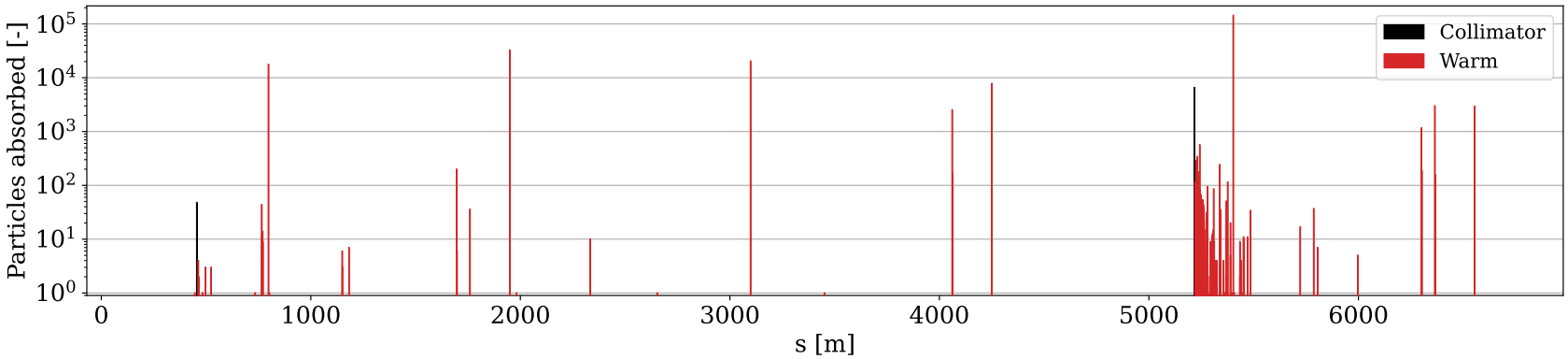
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $-30°$



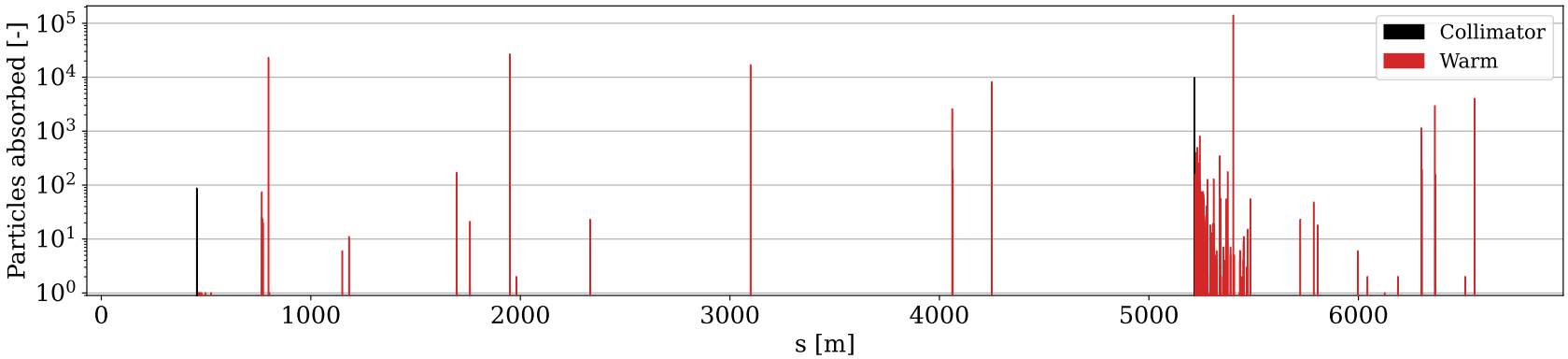
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $0°$



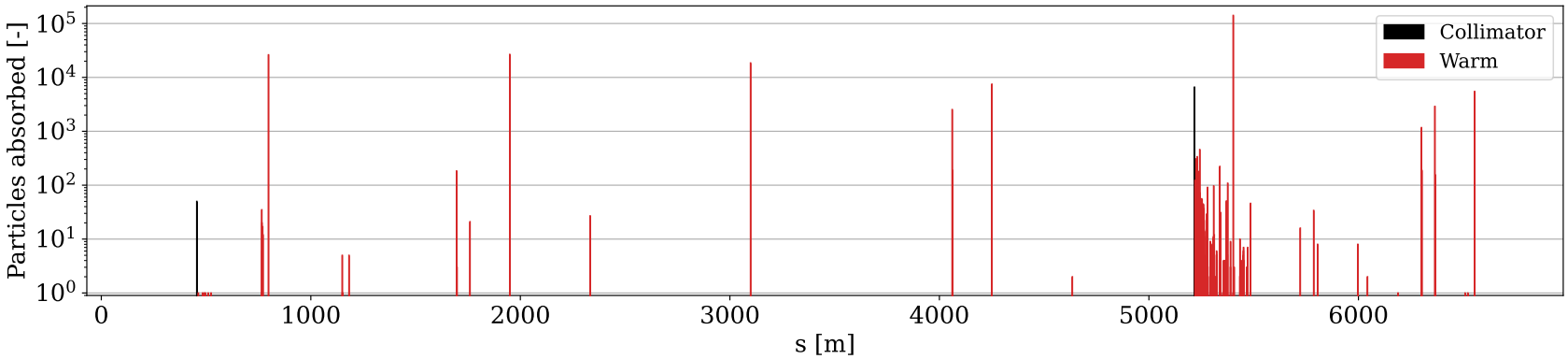
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase 30°



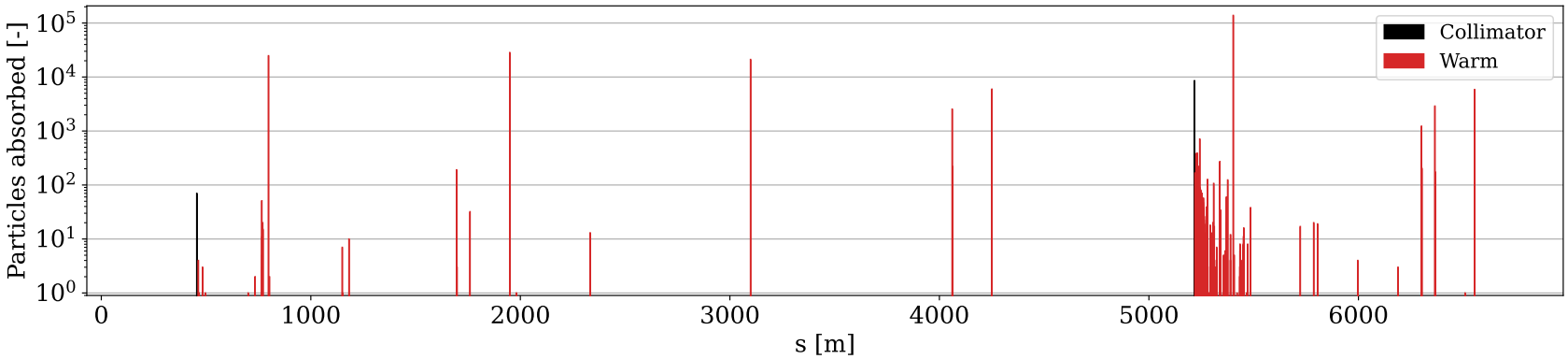
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase $60°$



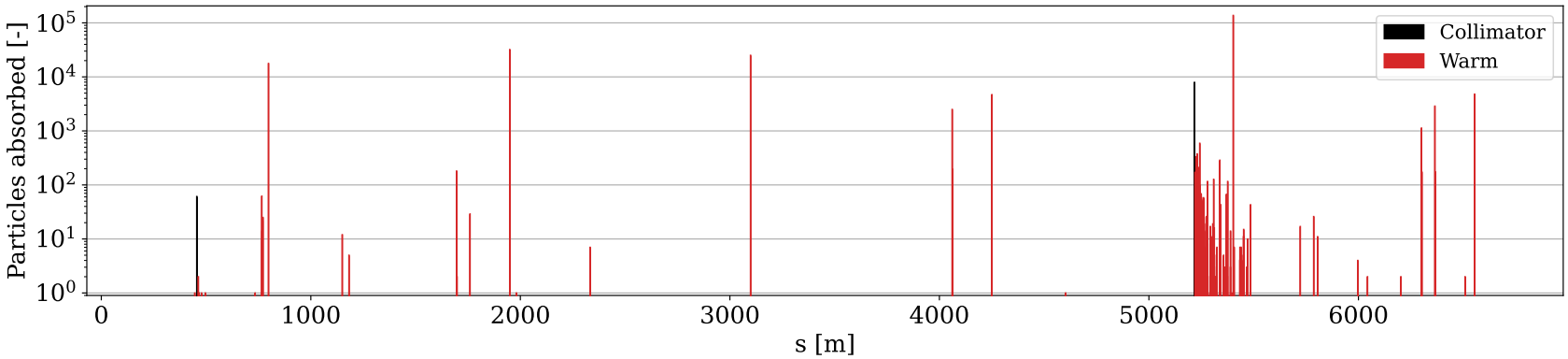
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase 90°



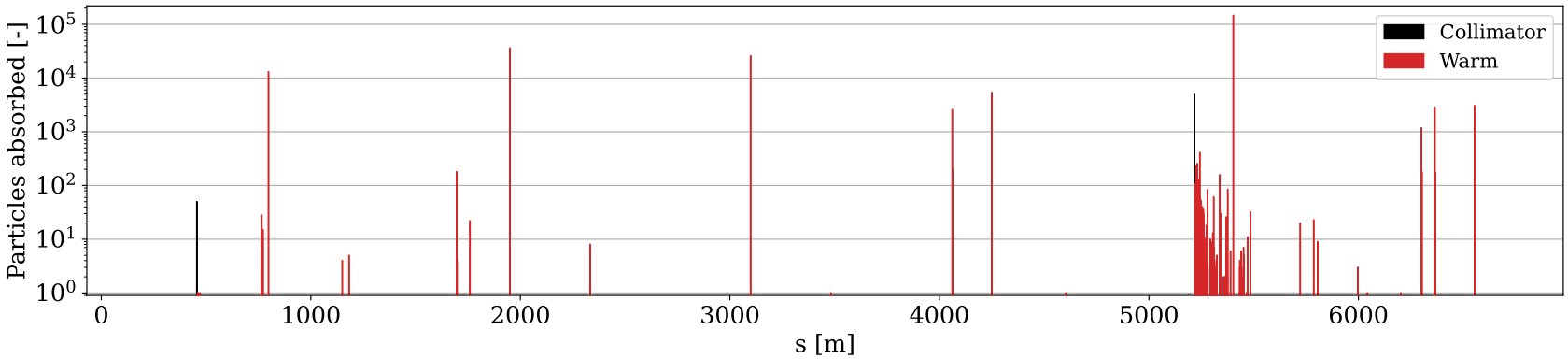
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase 120°



Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase 150°



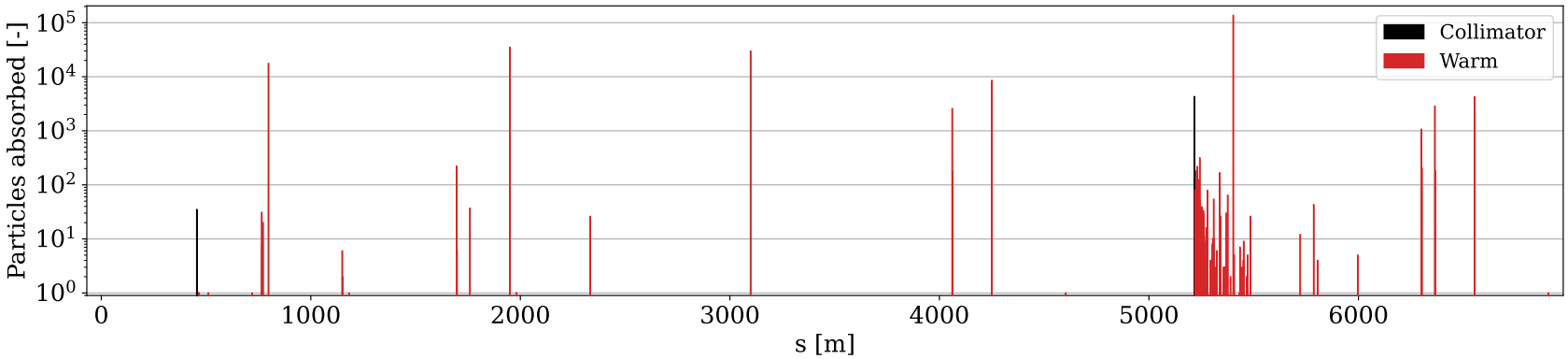
Vertical crabs — Off-momentum loss maps — DPneg — Without errors — Phase 179°



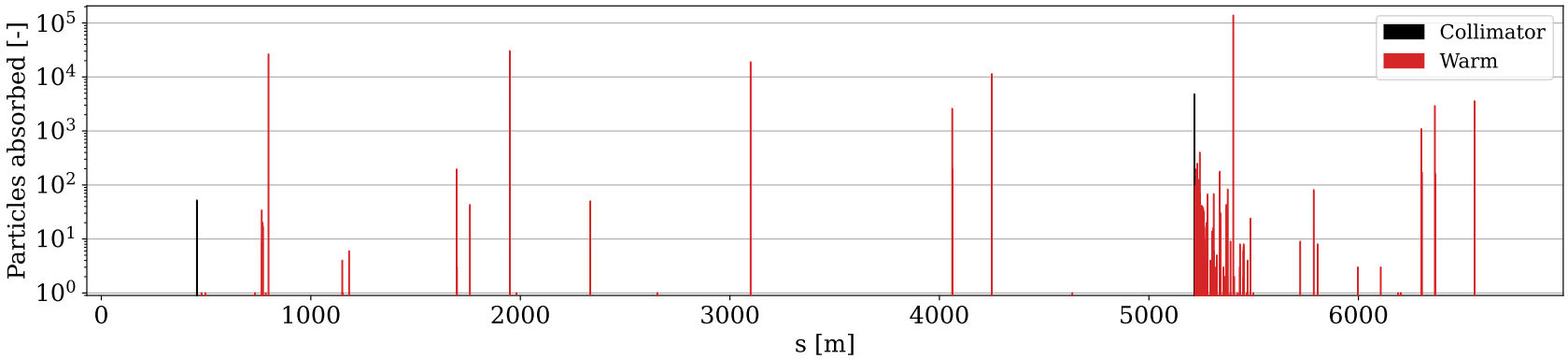
Vertical crabs — Off-momentum loss maps — DPneg — With errors

- $-179°$
- $-150°$
- $-120°$
- $-90°$
- $-60°$
- $-30°$
- $0°$
- $30°$
- $60°$
- $90°$
- $120°$
- $150°$
- $179°$

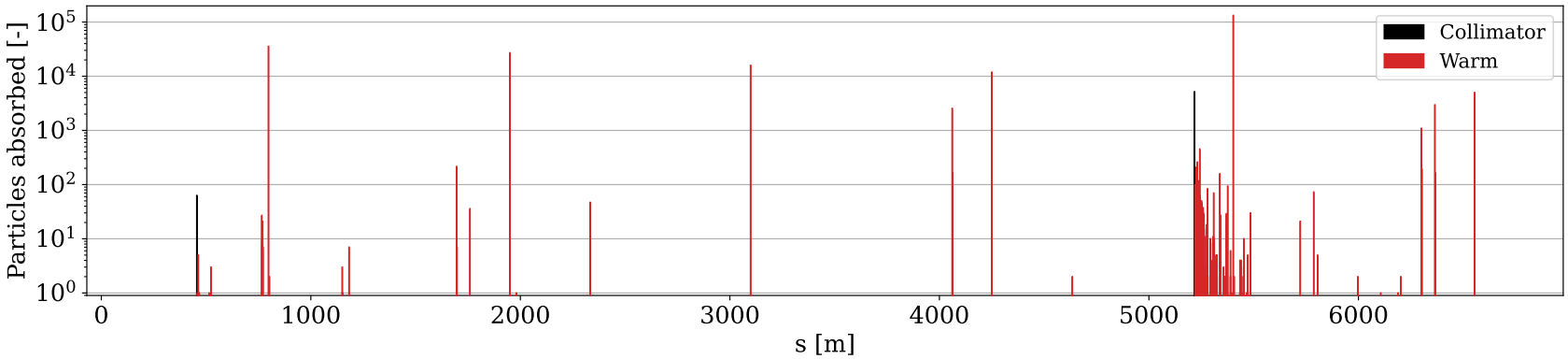
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase −179°



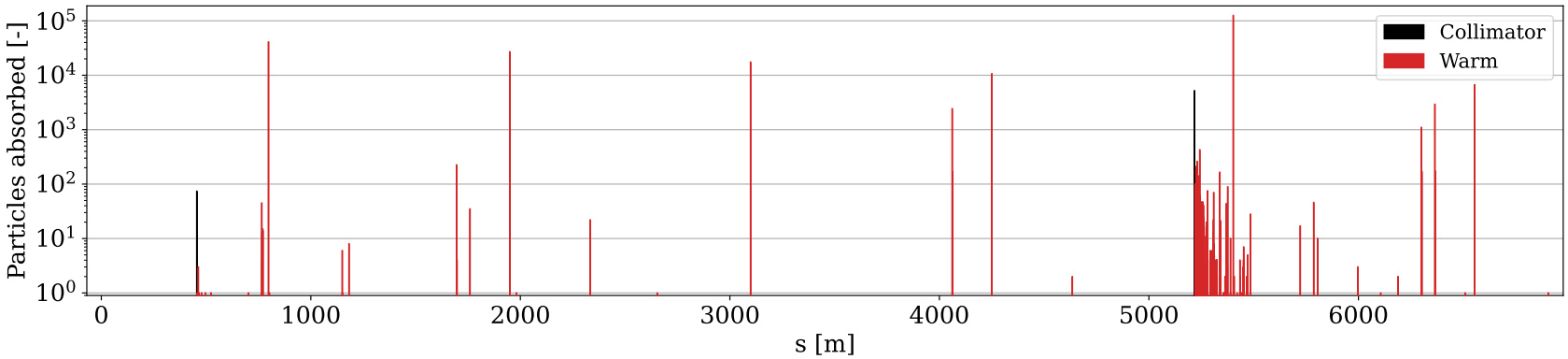
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase $-150°$



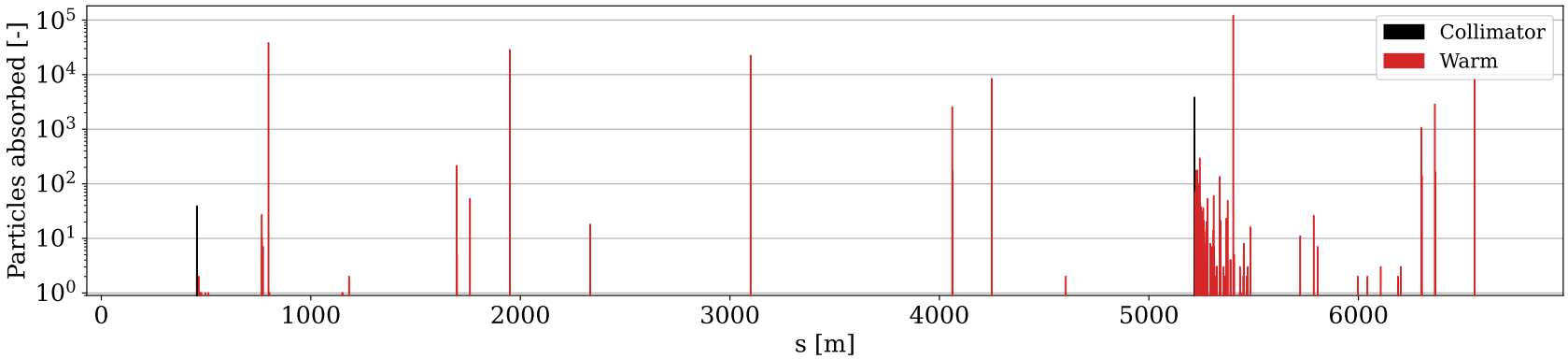
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase −120°



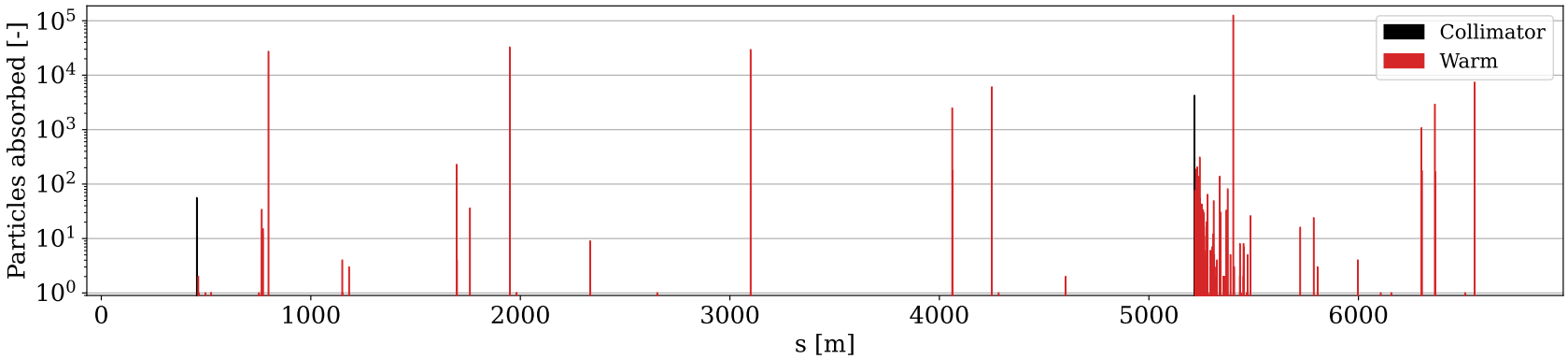
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase −90°



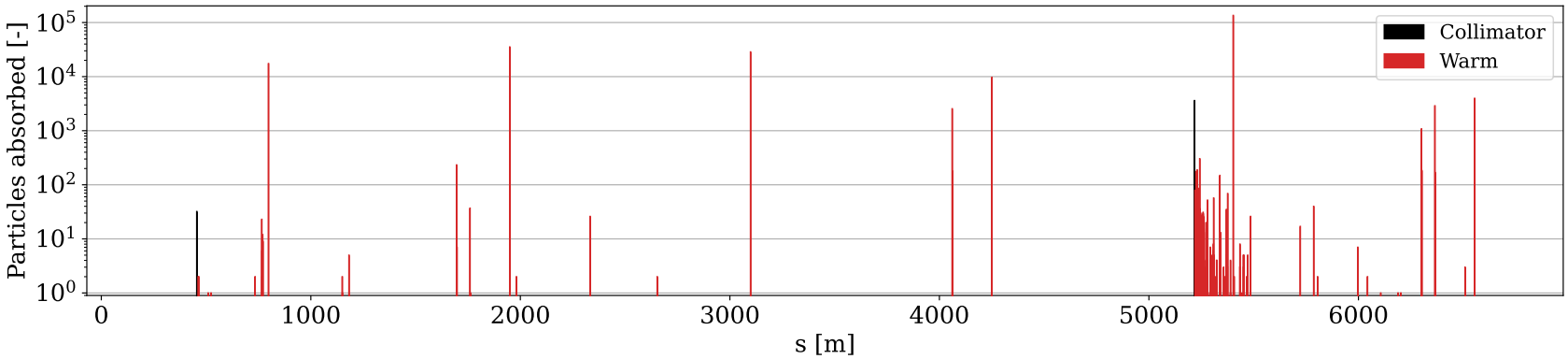
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase −60°



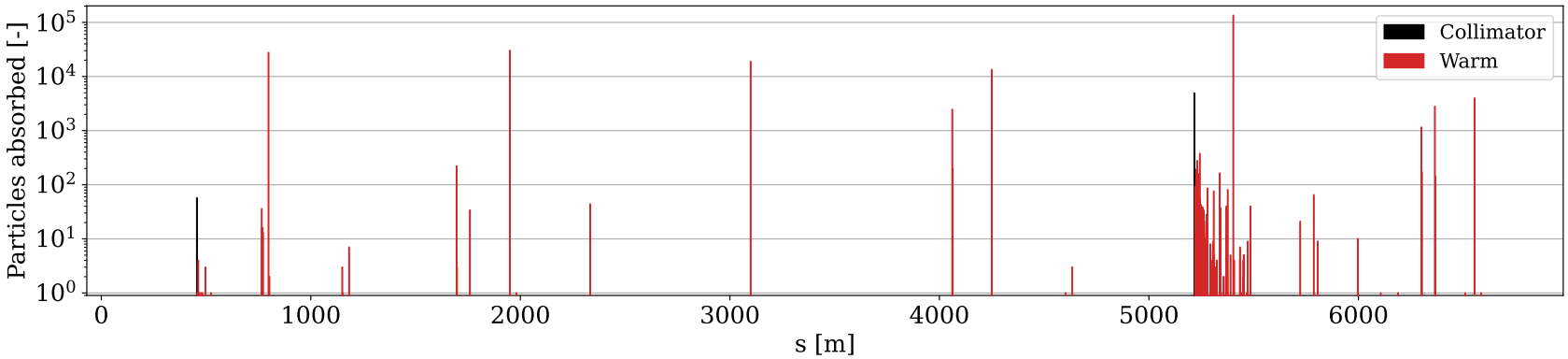
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase $-30°$



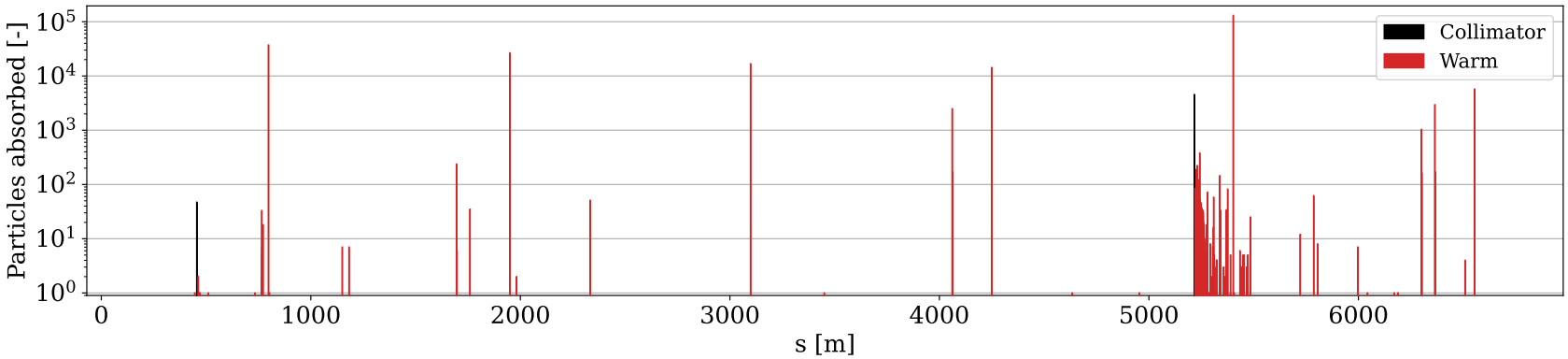
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 0°



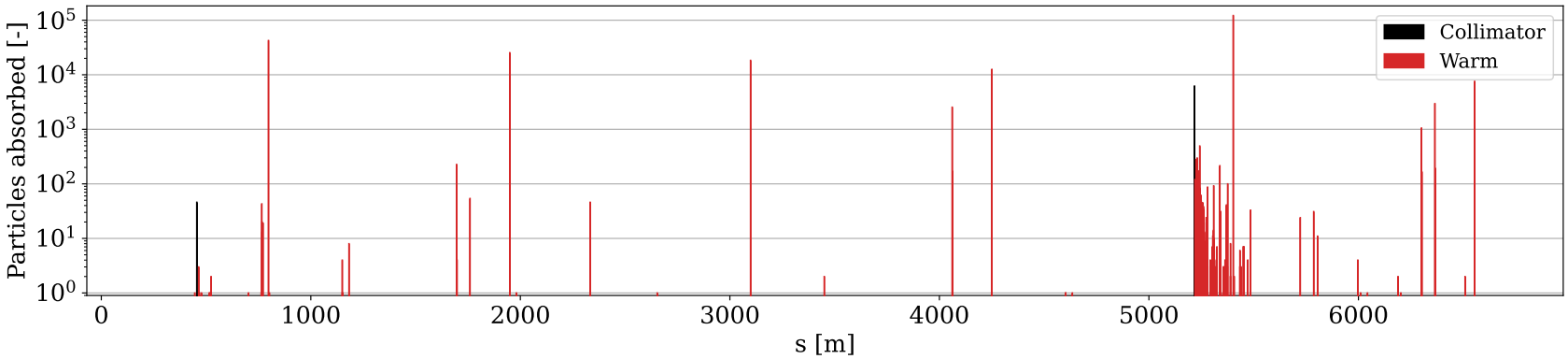
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 30°



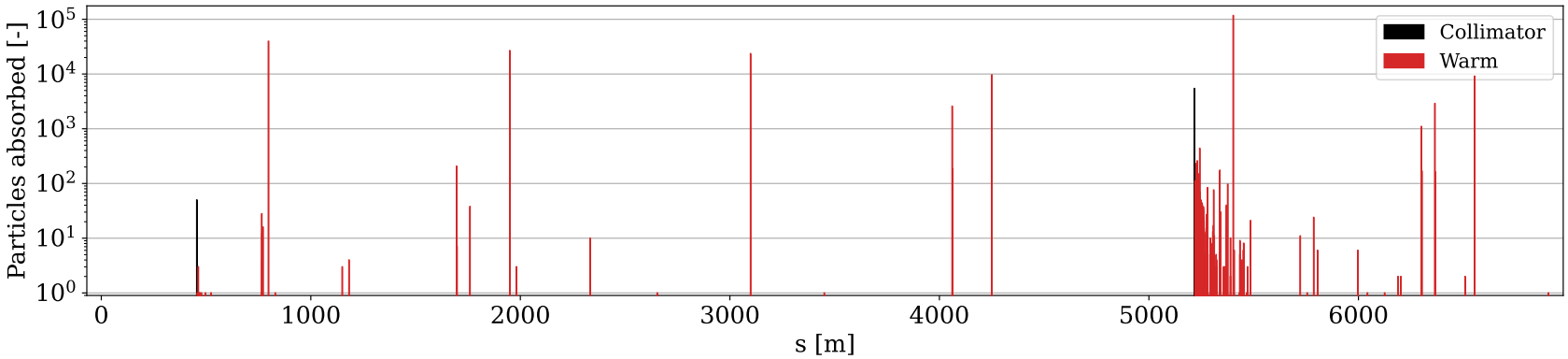
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 60°



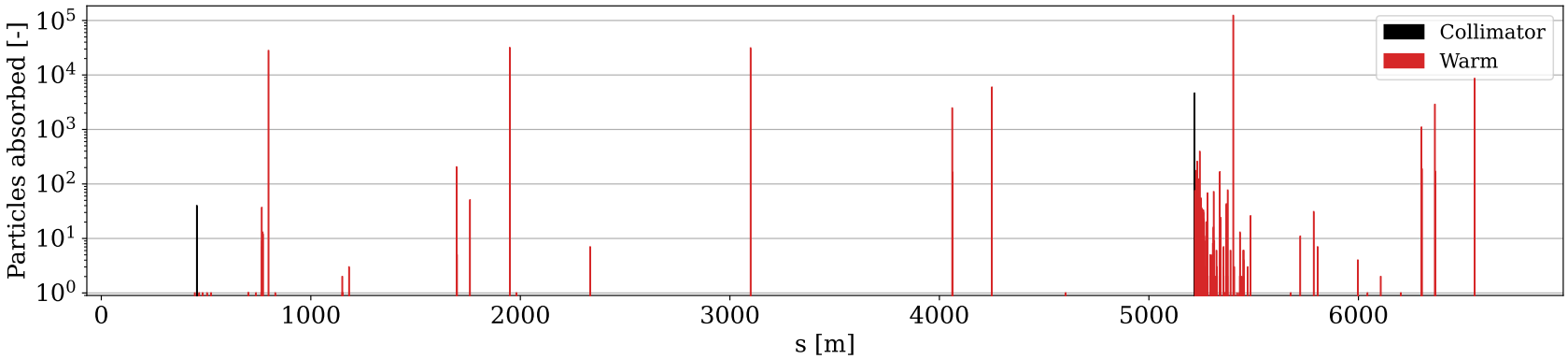
Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 90°



Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 120°



Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 150°



Vertical crabs — Off-momentum loss maps — DPneg — With errors — Phase 179°

